

Guidance on the Application of the CLP Criteria

Guidance to Regulation (EC) No 1272/2008 on classification, labelling and packaging (CLP) of substances and mixtures

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1 Document History

Version	Comment	Date
n.a.	First edition	August 2009
n.a.	Please note that change between the version published in August 2009 and that of April 2011 are not recorded in this document history.	April 2011
Version 2.0	Revision of the Guidance addressing content in relation to the environmental criteria chapters and Annexes following the 2nd Adaptation to Technical Progress to the CLP Regulation (Commission Regulation (EU) No 286/2011). The ECHA Secretariat revised the Guidance <i>Part 4 – Environmental hazards</i> and <i>Annexes</i> of the guidance document referring to the revised criteria for the long-term aquatic hazard for substances and mixtures and added new <i>Part 5 – Additional hazards</i> referring to the hazard class 'hazardous to the ozone layer'. As well, a number of examples have been included in the respective Parts and Annexes to illustrate the revisions performed. Further to this, a range of editorial corrections were proposed for <i>Part 1- General principles for classification and labelling</i>. The update includes the following: • Revision of Part 1, by eliminating and amending out of date information and restructuring the text in order to reflect the Guidance update. • All green boxes in Part 4 that are impacted by the 2nd ATP were updated. As the CLP legal text uses commas instead of dots to define numbers smaller than 1, the green boxes now show commas as well. • Revision of Part 4, by providing guidance on the application of the new long-term aquatic hazard criteria for substances and mixtures. • Section 4.1.3 Classification of substances hazardous to the aquatic environment and section 4.1.4 Classification of mixtures hazardous to the aquatic environment were substantially revised, for example by addition of new references, as well as the new/ revised examples to illustrate relevant topics in the Part 4. • New Part 5 - Additional hazards was added (please note that Part 5: Labelling was deleted from the Guidance in previous non recorded versions and covered via a new Guidance on Labelling and Packaging in accordance with Regulation (EC) No 1272/2008 published in April 2011). • Most of the I.3 sub-sections in Annex I – Aquatic toxicity were revised.	April 2012

	• In Annex II – Rapid degradation the terminology was modified. • Most of the Annex IV – Metals and Inorganic Metal Compounds was substantially modified and revised, as well as in sub-section IV.7 new examples were added.	
Version 3.0	Revision of Guidance Part 3 Health Hazards, relating to specific concentration limits (SCLs) for 4 hazard classes and the inclusion of a new Annex. The update includes the following: • Revision of Part 3, by providing guidance on the setting of lower and higher SCLs for 4 health hazard classes in section 3.2.2.5 Skin Corrosion/Irritation; section 3.3.2.5 Serious Eye Damage/Eye Irritation; section 3.7.2.5 Reproductive Toxicity and section 3.8.2.6 STOT-SE, in accordance with CLP Article 10(7); • Inclusion of a new Annex (Annex VI) providing guidance on setting SCLs for the reproductive toxicity hazard class based on potency considerations.	November 2012
Version 4.0	(i) Revision of the CLP Guidance addressing content in relation to the Part 2: Physical hazards, Part 3: Health hazards and Annex VI following the 2nd and the 4th Adaptation to Technical Progress to the CLP Regulation (Commission Regulation (EU) No 286/2011 of 10 March 2011 and Commission Regulation (EU) No 487/2013 of 8 May 2013). The revision includes: • Numbering of chapters within CLP Guidance, Parts 2 & 3 were synchronised with corresponding chapter numbering of CLP, Annex I. • Changes in the legal text due the 2nd and 4th ATPs. • Changes in the legal text due to the 4th ATP were highlighted in orange within all relevant green boxes. All changes are preceded by a note highlighting the changes. (To note: a corrigendum will change the colour of relative legal text boxes from orange to green when the 4th ATP applies). In addition, the revisions to Part 2: Physical hazards include the following: • Chapters 'Pyrophoric liquids and solids' and 'Oxidising liquids and solids' were divided into four chapters: 'Pyrophoric liquids', 'Pyrophoric solids', 'Oxidising liquids' and 'Oxidising solids' respectively. • Based on the 4th ATP the CLP Guidance Chapter 2.2 Flammable gases was extended to take into account the scope of CLP, Annex I, section 2.2 to include chemically unstable gases.	November 2013

- Further, the 4th ATP amended the criteria in CLP Annex I, Section 2.3 Flammable aerosols and renamed it into 2.3 Aerosols. Hence, the CLP Guidance was amended accordingly.
- All chapters were rechecked and redundant and/or outdated information were deleted, reorganised and/or revised. For example, 'Introduction' chapters were significantly shortened, however several "examples" sections (i.e. 'Example for classification...') were further elaborated.
- Where missing, a new sub-chapter 'Relation to other physical hazards' was added.
- Sub-chapter 2.0.4 'Physical state' was extended with additional information about substance/mixture form and some examples.
- In sub-chapter 2.1.5.2 'Additional labelling provisions' within chapter 2.1 'Explosives' further guidance about hazard communication was provided.
- In sub-chapter 2.5.6.1 a new recommendation for shot hazard codes to identify the classification of gasses under pressure was added.
- Footnotes with references to endorsed or on-going revisions of the GHS which have not yet been implemented into the CLP via a respective ATP were included in relevant sub-chapters of this guidance for information only.

In addition, the major revisions to Part 3: Health hazards include the following:

- All sections: revisions to legal text for the 4th ATP, including revisions to Precautionary Statements in the Tables with labelling information
- Section 3.1: the introduction of new guidance for the 4th ATP in section 3.1.4.1
- Sections 3.22.5 and 3.3.2.5: clarification to the recently published text (Version 3.0) for the setting of SCLs.
- Section 3.4 (sensitisation) has been significantly re-organised to present all the information on respiratory sensitisation together, followed by the information on skin sensitisation. This is in line with how the sections are presented in the CLP Regulation and in GHS documents.
- Section 3.4: integration of subcategories for respiratory and skin sensitisation based on potency of a substance; clarification of semi-quantitative terms like 'low to moderate sensitisation rate' and 'high or low exposure';

	elaboration of evaluation of human data for skin sensitisation and the addition of new examples. • Section 3.7 the introduction of new guidance for the 4th ATP in section 3.7.4.1 and section 3.7.5.1. (ii) Corrigendum of Part 1: General principles for classification and labelling and Part 4: Environmental hazards and its related Annexes I-V. The corrigendum includes the following: • The list of abbreviations was updated. • Update or deletion of outdated references to Guidance on information requirements and chemical safety assessment, Endpoint specific guidance (Chapter R.7a) within Annexes I-V. • A footnote informing the reader that with effect from 1 September 2013, Directive 98/8/EC had been repealed by Biocidal Products Regulation (EU) No 528/2012 was added. • In Part 1, Part 4 and Annexes modal verbs 'shall' were replaced with 'must' where appropriate. • A footnote related to respiratory sensitisation and skin sensitisation in Table 1.5.1-a was removed. • A correction to Example D, sub-chapter 4.1.4.7.5 was applied, namely a reference to CLP, Annex I, point (b) (ii) of Table 4.1.0 was introduced. In addition the result of a summation method calculation was corrected.	
Version 4.1	Corrigendum to take account of the end of the transition period of the 4th ATP (as foreseen in version 4.0 above): • change the colour of relative legal text boxes from orange to green; • in Part 2, to delete section 2.2.1 Flammable gases and section 2.3.1 Flammable Aerosols (outdated text) and renumber sections 2.2.2 Flammable gases (including chemically unstable gases) and 2.3.2 Aerosols accordingly; • in Part 3, to delete the "outdated text" in sections 3.7.4.1 and 3.7.5.1 in Reproductive Toxicity. In addition, minor editorial errors were corrected and minor reformatting was made.	June 2015
Version 5.0	<i>[See draft updated PART 1]</i>	Xxxx 2017

Preface

[See draft updated text in PART 1]

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List of Abbreviations

1. PART 1: GENERAL PRINCIPLES FOR CLASSIFICATION AND LABELLING

[See separate document for the text of Part 1 under consultation]

1.1. INTRODUCTION

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2. PART 2: PHYSICAL HAZARDS

2.0. INTRODUCTION

2.0.1 General remarks about the prerequisites of classification and testing

The purpose of this chapter is to give some general guidance with respect to the classification of physical hazards, the generation of test data and their interpretation. The intention of CLP is to identify hazards of chemical substances and mixtures and to provide a systematic approach – using classification – to communicate them based on harmonized criteria. The classification process involves three steps:

1. gathering of relevant information regarding the hazards of a substance or mixture (Articles 5 – 8 of CLP);
2. evaluation of hazard information to ascertain the hazards associated with the substance or mixture (Article 9 of CLP); and
3. a decision on whether the substance or mixture will be classified as hazardous and the degree of hazard, where appropriate, by comparison of the data with agreed hazard classification criteria (Article 13 of CLP).

Generally for both, substances and mixtures, the tests required in Annex I of CLP must be performed, unless there is adequate and reliable information already available. Testing is required to determine physical hazards including the physico-chemical properties necessary for the respective classification unless alternative methods are specifically permitted. Before undertaking testing of the substance or mixture, enquiries should be made to ascertain the availability of data, e.g. flash points, on the substance or mixture.

2.0.2 Safety

In most cases, the classification is based on data derived from testing. Special care is required when new or unknown substances or mixtures are tested. If possible, preliminary tests should be carried out before larger quantities are handled. Appendix 6 of the UN Recommendations on the transport of dangerous goods Manual of Tests and Criteria (UN-MTC) 'Screening procedures' allows gathering valuable information about physico-chemical properties based on small-scale tests. Further aspects of safety are given in the general introduction, Section 1.4 of the UN-MTC or within the individual test procedures.

2.0.3 General conditions for testing

Samples offered for testing must in all aspects be representative of the substance or mixture to be classified. Therefore, it is helpful to characterise or specify the sample for the purposes of documentation (i.e. batch number, production code, impurities etc.). Further characterisation (i.e. analysis) is highly recommended in cases where the presence of diluents, activators, stabilisers or moisture may influence the outcome of the test.

In some cases, additional parameters like (e.g.) physical condition, particle size and shape, specific surface area, density, crystal structure, may influence the test result. Therefore, these properties should be mentioned in the test report.

The tests must be performed on the substance or mixture in the appropriate physical form where changes in that form may influence the outcome of the test (see also Articles 5 and 6 of CLP).

2.0.4 Physical state

The physical state determines which hazard classes should be considered for testing. As the CLP states¹, hazard classification is based on intrinsic properties of the substance or mixture which are determined not only by its physical state but also its form.

As mentioned in the Chapter 1.2 of this guidance the same solid substance or mixture may have different forms such as flakes, prills, or powder. Furthermore, e.g. a powder may contain particles of different size, and particles of the same size may have different shapes, crystallinity or allotropy etc. These differences may result in different intrinsic properties, and consequently, different physical hazards of the powder. Particle size is crucial for several classes such as explosives, flammable solids, self-reactive substances, pyrophoric solids, self-heating substances, solid organic peroxides and substances which, in contact with water, emit flammable gases. Therefore not only the physical appearance, but also other parameters should be considered when identifying the form, since it may trigger different classifications of the same substance or mixture.

An example of different classification due to different intrinsic properties of forms is red phosphorus (flammable solid) and white phosphorus (pyrophoric solid) (different allotropes). It is therefore important to evaluate case by case whether available information on the physical properties of the substance and mixture placed on the market, is applicable to the examined form, and whether additional testing should be performed.

The form of a substance or mixture as placed on the market might be such that it is not possible to test it in this form, e.g. if it is in the form of tablets or pellets. In such circumstances, the physical hazards of the substance or mixture must be considered for classification especially if they are friable and produce secondary effects due to abrasion or crushing during supply and use. If phase separation does occur, the hazardous properties of the most hazardous phase of the substance or mixture must be communicated.

If further testing is required, the choice of the test method should be done after thorough evaluation of its suitability for the substance or mixture, as the properties of the form (e.g. for powders especially size and shape of the particle) may have a significant effect on the test results.

The definitions for gases, liquids and solids are given in Annex I, Part 1 of CLP:

Annex I: Part 1, 1.0. Definitions

Gas means a substance which:

- (i) at 50 °C has a vapour pressure greater than 300 kPa (absolute); or*
- (ii) is completely gaseous at 20 °C at a standard pressure of 101.3 kPa;*

Liquid means a substance or mixture which:

- (i) at 50 °C has a vapour pressure of not more than 300 kPa (3 bar);*
- (ii) is not completely gaseous at 20 °C and at a standard pressure of 101,3 kPa; and*
- (iii) which has a melting point or initial melting point of 20 °C or less at a standard pressure of 101,3 kPa;*

Solid means a substance or mixture which does not meet the definitions of liquid or gas.

In some cases (i.e. viscous substances or mixtures), a specific melting point cannot be determined. Such a substance or mixture must be regarded as a liquid if either the result of the

¹ CLP Article 5(1), 6(1) and 8(6).

ASTM D 4359-90 test as amended (standard test method for determining whether a material is a liquid or a solid) indicates 'liquid' or the result of the test for determining fluidity (penetrometer test) prescribed in Section 2.3.4 of Annex A of ADR indicates 'not pasty'.

2.0.5 Quality

The determination of data must be based on the methods named in Annex I, Part 2 of CLP. For most hazard classes in Annex I, Part 2 of CLP there is reference made to the UN-MTC which gives very detailed descriptions of the test methods. For the classification of flammable gases, oxidising gases and for the determination of the flash point there are references to international standards in Annex I, Part 2 of CLP. Whenever possible, the laboratory should validate the performance of the methods used e.g. by participating in inter-laboratory testing or by using reference materials. Any deviation from the test procedure or standard should be documented and, if necessary, justified.

The reliability of all test results used for the classification of hazardous substances and mixtures is important and therefore their transparency and comparability must be ensured.

For these purposes, CLP requires in Article 8 the following:

Article 8 (5)

[...]

Where new tests for physical hazards are carried out for the purposes of this Regulation, they shall be carried out, at the latest from 1 January 2014, in compliance with a relevant recognised quality system or by laboratories complying with a relevant recognised standard.

[...]

In general, the following alternative strategies can be pursued:

1. compliance with the principles of good laboratory practice (GLP) (as formerly required by the DSD);
2. application of EN ISO/IEC 17025 General requirements for the competence of testing and calibration laboratories as amended as a relevant recognised standard;
3. other internationally recognised standards of comparable scope.

Any laboratory that carries out physical hazard tests for classification purposes can therefore choose how to fulfil the quality requirements of CLP.

2.1. EXPLOSIVES

2.1.1. Introduction

The requirements in Chapter 2.1 'Explosives' of Annex I of CLP are identical to those in Chapter 2.1 of GHS.

The classification of explosives according to the GHS is almost entirely adopted based on the UN Recommendations on the Transport of Dangerous Goods – Model Regulations (UN RTDG Model Regulations), which are appropriate for transport and also storage of packaged explosives.

The classification of substances, mixtures and articles in the class of explosives and further allocation to a division is a very complex procedure. References to Part I of the UN-MTC and related expertise are necessary.

2.1.2. Definitions and general considerations for the classification of explosives

The following definition is given in CLP for the class of explosives.

Annex I: 2.1.1.1. The class of explosives comprises

- (a) *explosive substances and mixtures;*
- (b) *explosive articles, except devices containing explosive substances or mixtures in such quantity or of such a character that their inadvertent or accidental ignition or initiation shall not cause any effect external to the device either by projection, fire, smoke, heat or loud noise; and*
- (c) *substances, mixtures and articles not mentioned in points (a) and (b) which are manufactured with a view to producing a practical, explosive or pyrotechnic effect.*

Additional remark related to the applicability of 2.1.1.1 (a) (see also UN RTDG Model Regulations, 2.1.1.1 (a)):

- a substance or mixture which is not itself an explosive but which can form an explosive atmosphere of gas, vapour or dust is not included in this class;
- explosive behaviour related to the thermal decomposition of organic peroxides and of self-reactive substances and mixtures is covered by those specific hazard classes and therefore not included in the hazard class explosives.

In addition the following definitions apply for explosives:

Annex I: 2.1.1.2.

[...]

An explosive substance or mixture is a solid or liquid substance or mixture of substances which is in itself capable by chemical reaction of producing gas at such a temperature and pressure and at such a speed as to cause damage to the surroundings. Pyrotechnic substances are included even when they do not evolve gases.

A pyrotechnic substance or mixture is a substance or mixture of substances designed to produce an effect by heat, light, sound, gas or smoke or a combination of these as the result of non-detonative self-sustaining exothermic chemical reactions.

An unstable explosive is an explosive which is thermally unstable and/or too sensitive for normal handling, transport and use.

An explosive article is an article containing one or more explosive substances or mixtures.

A pyrotechnic article is an article containing one or more pyrotechnic substances or mixtures.

An intentional explosive is a substance, mixture or article which is manufactured with a view to produce a practical explosive or pyrotechnic effect.

Certain physical hazards (due to explosive properties) are altered by dilution, as is the case for desensitized explosives, by inclusion in a mixture or article, packaging or other factors.

Explosive substances and mixtures wetted with water or alcohols, or diluted with other substances to suppress their explosive properties, may be treated differently to their non-wetted or non-diluted counterparts i.e. different hazard classes may apply, depending on the physical properties of the wetted/diluted substance or mixture.

2.1.3. Relation to other physical hazards

For safety reasons, substances, mixtures or articles which have already been classified as Explosives (Class 1 according to the UN RTDG Model Regulations) should not be considered for classification in any other physical hazard classes. Since the explosion hazard is more severe than other physical hazards there is no need to further perform classification tests for other potential physical hazards.

When considering substances and mixtures for classification within the hazard class explosives, the following checks should be performed with respect to other hazard classes:

Substances, mixtures and articles that have been manufactured with a view to producing a practical explosive or pyrotechnic effect, are classified as explosives by definition according to 2.1.1.1(c) of Annex I of the CLP. It should be checked whether such a substance or mixture is an unstable explosive.

Thermally unstable substances or mixtures that are not classified as explosives should be considered for classification as self-reactive substances and mixtures.

Mixtures of oxidising substances and mixtures with combustible material that are not classified as explosives should be considered for classification as self-reactive substances and mixtures, oxidising liquids or oxidising solids.

Due to the complexity of these issues, expert advice should always be sought when dealing with classification of substances and mixtures with potentially explosive properties.

2.1.4. Classification of substances, mixtures or articles as explosives

2.1.4.1. Identification of hazard information

Information on the following types of hazards is relevant for the evaluation of substances, mixtures and articles for the class of explosives:

- sensitivity to shock;
- effects of heating and ignition under confinement;
- thermal stability;
- sensitiveness to impact and friction;
- mass explosion hazard;
- projection hazard;
- fire and radiant heat hazard.

2.1.4.2. Screening procedures and waiving of testing

The screening procedure is described in:

CLP, Annex I, Part 2, paragraphs 2.1.4.2 and 2.1.4.3; Appendix 6 of the UN-MTC.

The screening procedure may be used for new substances or mixtures which are suspected of having explosive properties. It should not be used for substances and mixtures manufactured with the intention of producing a practical explosive or pyrotechnic effect.

Explosive properties are associated with the presence of certain chemical groups in a molecule which can react to produce very rapid increases in temperature and/or pressure. The screening procedure is aimed at identifying the presence of such reactive groups and the potential for rapid energy release.

Examples of groups which may indicate explosive properties in organic materials are:

- C-C unsaturation (e.g. acetylenes, acetylides, 1, 2-dienes);
- C-Metal, N-Metal (e.g. Grignard reagents, organo-lithium compounds);

- Contiguous nitrogen atoms (e.g. azides, aliphatic azo compounds, diazonium salts, hydrazines, sulphonylhydrazides);
- Contiguous oxygen atoms (e.g. peroxides, ozonides);
- N-O (e.g. hydroxyl amines, nitrates, nitro compounds, nitroso compounds, N-oxides, 1,2-oxazoles);
- N-halogen (e.g. chloramines, fluoroamines);
- O-halogen (e.g. chlorates, perchlorates, iodosyl compounds).

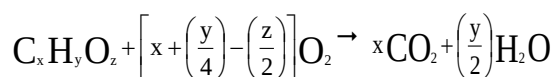
A substance or mixture is not classified as explosive:

- a. when there are no chemical groups associated with explosive properties present in the molecule;

or

- b. when the substance or mixture contains chemical groups associated with explosive properties which include oxygen and the calculated oxygen balance is less than -200;

The oxygen balance is calculated for the chemical reaction:



Using the formula:

$$\text{Oxygen balance} = -1600 \times \frac{[2x + (y/2) - z]}{\text{molecular weight}}$$

or

- c. when the organic substance or a homogenous mixture of organic substances contains chemical groups associated with explosive properties but the exothermic decomposition energy is less than 500 J/g and the onset of exothermic decomposition is below 500 °C. (The temperature limit is to prevent the procedure being applied to a large number of organic materials which are not explosive but which will decompose slowly above 500 °C to release more than 500 J/g.) The exothermic decomposition energy may be determined using a suitable calorimetric technique;

or

- d. for mixtures of inorganic oxidising substances with organic material(s), the concentration of the inorganic oxidising substance is:
 - less than 15 % by mass, if the oxidising substance is assigned to Categories 1 or 2;
 - less than 30 % by mass, if the oxidising substance is assigned to Category 3.

If the screening procedure identifies the substance or mixture to be a potential explosive or if it is a mixture containing any known explosives, the classification (acceptance) procedure for the class of explosives (see Section [2.1.4.5.1](#)) has to be applied. If the exothermic decomposition energy of organic materials is less than 800 J/g, a UN gap test is not required, neither according to Series 1 Type (a) nor according to Series 2 Type (a).

2.1.4.3. Classification criteria

The criteria for the classification of explosives are given in the following tables.

Annex I: 2.1.2.1. *Substances, mixtures and articles of this class are classified as an unstable explosive on the basis of the flowchart in Figure 2.1.2. The test methods are described in Part I of the UN RTDG, Manual of Tests and Criteria.*

2.1.2.2. *Substances, mixtures and articles of this class, which are not classified as an unstable explosive, shall be assigned to one of the following six divisions depending on the type of hazard they present:*

- (a) *Division 1.1 Substances, mixtures and articles which have a mass explosion hazard (a mass explosion is one which affects almost the entire quantity present virtually instantaneously);*
- (b) *Division 1.2 Substances, mixtures and articles which have a projection hazard but not a mass explosion hazard;*
- (c) *Division 1.3 Substances, mixtures and articles which have a fire hazard and either a minor blast hazard or a minor projection hazard or both, but not a mass explosion hazard:*
 - (i) *combustion of which gives rise to considerable radiant heat; or*
 - (ii) *which burn one after another, producing minor blast or projection effects or both;*
- (d) *Division 1.4 Substances, mixtures and articles which present no significant hazard:*
 - *substances, mixtures and articles which present only a small hazard in the event of ignition or initiation. The effects are largely confined to the package and no projection of fragments of appreciable size or range is to be expected. An external fire shall not cause virtually instantaneous explosion of almost the entire contents of the package;*
- (e) *Division 1.5 Very insensitive substances or mixtures which have a mass explosion hazard:*
 - *substances and mixtures which have a mass explosion hazard but are so insensitive that there is very little probability of initiation or of transition from burning to detonation under normal conditions;*
- (f) *Division 1.6 Extremely insensitive articles which do not have a mass explosion hazard:*
 - *articles which contain only extremely insensitive substances or mixtures and which demonstrate a negligible probability of accidental initiation or propagation.*

2.1.2.3. *Explosives, which are not classified as an unstable explosive, shall be classified in one of the six divisions referred to in section 2.1.2.2 based on Test Series 2 to 8 in Part I of the UN RTDG, Manual of Tests and Criteria according to the results of the tests laid down in Table 2.1.1:*

Table 2.1.1

Criteria for explosives

Category	Criteria
Unstable explosives or explosives of Divisions 1.1 to 1.6	<i>For explosives of Divisions 1.1 to 1.6, the following are the core set of tests that need to be performed:</i> <i>Explosibility: according to UN Test Series 2 (section 12 of the UN RTDG, Manual of Tests and Criteria). Intentional explosives (¹) shall not be subject to UN Test Series 2.</i> <i>Sensitiveness: according to UN Test Series 3 (section 13 of the UN RTDG, Manual of Tests and Criteria).</i>

	<i>Thermal stability: according to UN Test 3(c) (sub-section 13.6.1 of the UN RTDG, Manual of Tests and Criteria).</i> <i>Further tests are necessary to allocate the correct Division.</i>
<i>(¹) This comprises substances, mixtures and articles which are manufactured with a view to producing a practical, explosive or pyrotechnic effect.</i>	

Where the test is conducted in the package form and the packaging is changed, a further test must be conducted where it is considered that the change in packaging will affect the outcome of the test.

Classification tests must be performed on the substance or mixture as supplied. If the same chemical is to be presented in a physical form different from that which was tested and which is considered likely to materially alter its performance in a classification test, the substance or mixture must also be tested in the new form.

2.1.4.4. Testing and evaluation of hazard information

Where test data are available, these must be evaluated against the set criteria for classification.

When the screening procedure indicates that a substance or mixture may possess explosive properties, a cautious approach when performing the tests is necessary to ensure safe handling.

For information on the test procedures see the following Section [2.1.4.5](#) where the individual test series are described in context with the respective decision logic.

The test procedures for the classification of explosives are described in detail in the Part I of the UN-MTC.

2.1.4.5. Classification procedure and decision logics

Any substance, mixture or article having or suspected of having explosives characteristics must be considered for classification in the hazard class of explosives. Substances, mixtures and articles classified in this hazard class must be assigned to the appropriate division or must be classified as unstable explosive.

The classification is divided into two stages, the acceptance procedure and the assignment procedure.

In the acceptance procedure, intrinsic explosive properties of a substance, mixture or article are determined through tests of its sensitivity, stability and explosion effects. If the substance, mixture or article is not characterised as unstable explosive and is provisionally accepted into the class of explosives, it is then necessary to ascertain the correct division by the assignment procedure. The further subdivision into compatibility groups A to S is described in detail in the UN RTDG Model Regulations, Section 2.1.2. The compatibility groups and their recommended combination identify types of explosives which are deemed to be compatible, e.g. for combined storage or transportation and can therefore be used to distinguish technical requirements (especially) in these sectors. However, assignment of compatibility groups is not part of the classification system according to CLP.

The tests for acceptance and the further tests to determine the correct division are grouped into eight test series. Classification procedures, test methods and criteria are described in detail in Part I of the UN-MTC.



NOTE: The person responsible for the classification of explosives should be experienced in this field and be familiar with the criteria for classification.

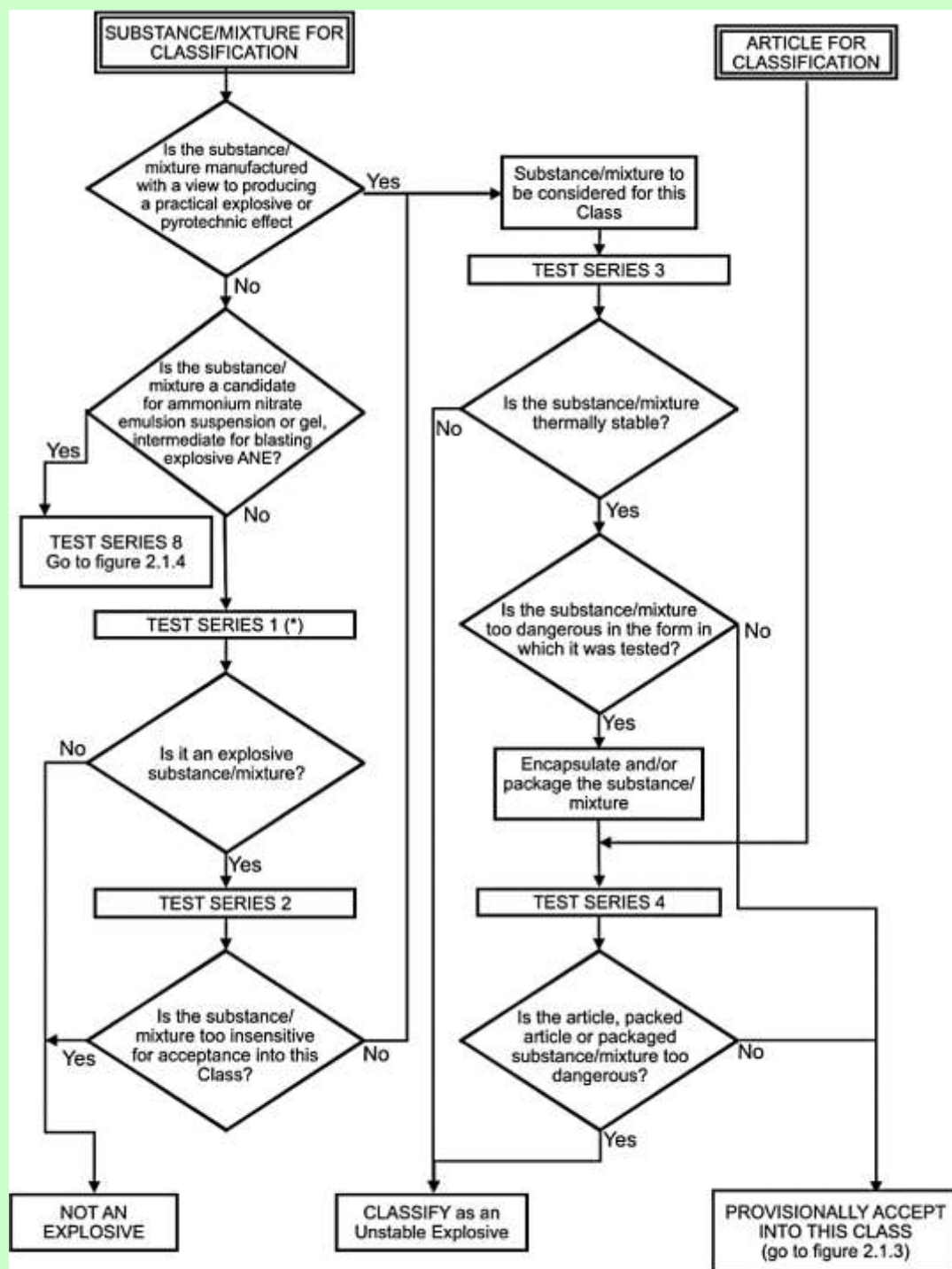
2.1.4.5.1. Acceptance procedure

The acceptance procedure is used to determine whether or not a substance, mixture or article is a candidate for the class of explosives or is an unstable explosive.

The test methods used for deciding on provisional acceptance into the class of explosives are grouped into four series, numbered 1 to 4 (see CLP Annex I, Figure 2.1.2 reported below).

Annex I: Figure 2.1.2

Procedure for provisional acceptance of a substance, mixture or article in the class of explosives (Class 1 for transport)



(*) For classification purposes, start with Test Series 2.

The numbering of Test Series 1 to 4 relates to the sequence of assessing the results rather than the order in which the tests should be conducted. **It may be important for the safety of test personnel that certain tests, using small amounts of material, be conducted first before proceeding to experiment with larger quantities.**

Starting the testing procedure with Test Series 3 is highly recommended, because these tests involve relatively small sample sizes, which reduces the risk to test personnel.

Test Series 1

Within Test Series 1 the question 'Is it an explosive substance / mixture?' is answered on the basis of the results of three types of tests to assess possible explosive effects. The question is answered 'Yes' if a '+' is obtained in any of the three types of tests. If the answer is 'No', the substance / mixture is rejected from this class; it is not an explosive. Under certain conditions the test Type 1 (a) can be replaced by certain tests of Test Series F, see UN-MTC, Section 11.3.5.

The three types of test used are (recommended test is indicated within brackets):

Type 1 (a): a shock test with defined booster and confinement to determine the ability of the substance to propagate a detonation (UN Gap test);

Type 1 (b): a test to determine the effect of heating under confinement (Koenen test); and

Type 1 (c): a test to determine the effect of ignition under confinement (time/pressure test).

Test Series 2

Series 2 tests are used to answer the question 'Is the substance / mixture too insensitive for acceptance into this Class?'. In general, the basic apparatus and method used is the same as that for Test Series 1 but with less stringent criteria, e.g. in the case of gap tests, the gap used is greater than zero. The question is answered 'No' if a '+' is obtained in any of the three types of test. If the answer is 'Yes', the substance / mixture is rejected from this class; it is not an explosive. Under certain conditions the test Type 2 (a) can be replaced by certain tests of Test Series F, see UN-MTC, Section 12.3.4.

The following three types of test are used (recommended test is indicated within brackets):

Type 2 (a): a shock test with defined initiation system and confinement to determine sensitivity to shock (UN gap test);

Type 2 (b): a test to determine the effect of heating under confinement (Koenen test); and

Type 2 (c): a test to determine the effect of ignition under confinement (Time/pressure test).

If the substance or mixture is manufactured with a view to produce a practical explosive or pyrotechnic effect, it is unnecessary to conduct Test Series 1 and 2 for purposes of classification.

Test Series 3

As stated above it is recommended to carry out Test Series 3 before Test Series 1 and 2 for safety reasons due to the small sample amount needed. It is also recommended to carry out Test Series 3 even if negative results have been obtained in Test Series 1 and/or 2 because only Test Series 3 gives information about the thermal stability and the sensitivity to mechanical stimuli (impact and friction).

Test Series 3 is used to answer the questions 'Is the substance / mixture thermally stable?' and 'Is the substance / mixture too dangerous in the form in which it was tested?' This

involves tests for determining the sensitiveness of the substance or mixture to mechanical stimuli (impact and friction), and to heat and flame.

The following four types of tests are used (recommended test is indicated within brackets):

Type 3 (a): a falling weight test to determine sensitiveness to impact (BAM Fallhammer);

Type 3 (b): a friction; or impacted friction test to determine sensitiveness to friction (BAM friction apparatus);

Type 3 (c): an elevated temperature test to determine thermal stability (thermal stability test at 75 °C); and

Type 3 (d): an ignition test to determine the response of a substance or mixture to fire (small scale burning test).

The first question is answered 'No' if a '+' is obtained in Test type 3(c). Then the substance / mixture is considered as thermally unstable and either classified as an unstable explosive or as a self-reactive substance or mixture.

The second question is answered 'Yes' if a '+' is obtained in any of the Test types 3(a), 3(b) or 3(d). If a '+' is obtained, the substance / mixture may be encapsulated or packaged to reduce its sensitiveness to external stimuli or is classified as an unstable explosive.

Furthermore, the explosive may be desensitized in order to suppress/reduce its explosive properties in which case the classification procedure has to be restarted.

Test Series 4

Series 4 tests are intended to answer the question 'Is the article, packaged article or packaged substance or mixture too dangerous?'. Conditions which may occur during supply and use include high /low temperature and high relative humidity, vibration, bumping and dropping.

The two types of test to be carried out are:

Type 4 (a): a test of thermal stability for articles; and

Type 4 (b): a test to determine the hazard from dropping.

The question is answered 'Yes' if a '+' is obtained in either Test type 4 (a) or 4 (b) and the substance or mixture or article is classified as an unstable explosive.

It is important to note that a substance / mixture which fails Test Series 2 (i.e. it is sensitive enough for acceptance into the class of explosives) may still, if properly packaged, leave the class of explosives provided that it is not designed to have an explosive effect and does not exhibit any explosive hazard in Test Series 6 of the assignment procedure (see example for musk xylene). Such an exclusion from the class of explosives is restricted to the specific type and size of package tested.

Especially for substances / mixtures, which have explosive properties according to Test Series 1 and/or 2 but can leave the class of explosives after Test Series 6 due to proper packaging, it is necessary to communicate these properties in the Safety Data Sheet (SDS). Furthermore, the results from Test types 3 (a) and 3 (b) should be documented in the SDS when they meet the criteria of EU test method A.14 in Regulation (EC) No 440/2008 (these are substances with a sensitiveness to impact, determined by UN Test Series 3 (a) (ii) of 40 J or less and/or a sensitiveness to friction, determined by Test Series 3 (b) (i) of 360 N or less).

2.1.4.5.2. Assignment procedure to a division

The assignment procedure to one of six divisions, depending on the type of hazard they present, applies to all substances, mixtures and/or articles that are candidates for class of explosives. A substance, mixture or article must be assigned to the division which corresponds to the results of the tests to which the substance, mixture or article, as offered for supply and use, has been subjected. Other test results, and data assembled from accidents which have occurred, may also be taken into account.

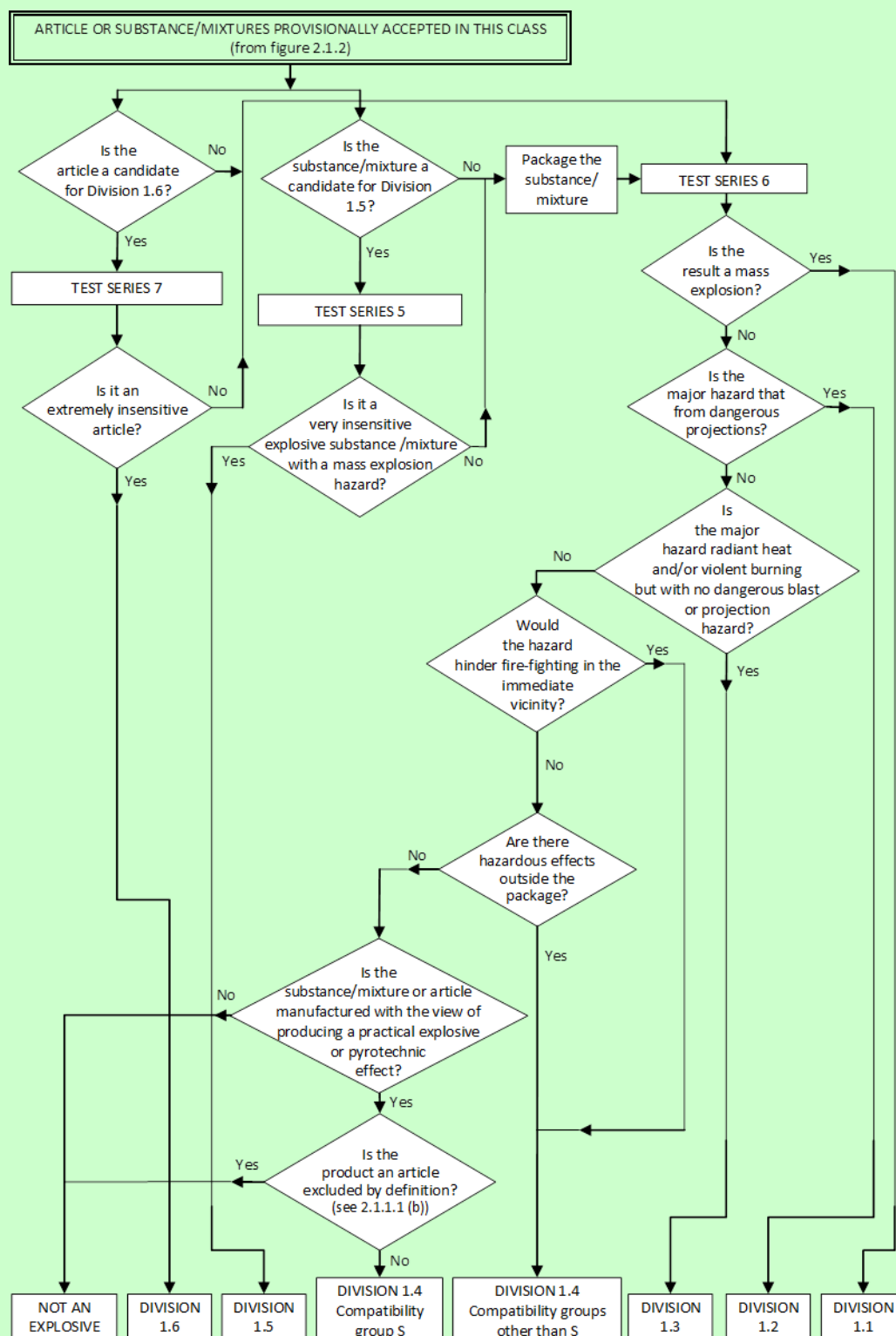
The test methods used for assignment to a division are grouped into three series – numbered 5 to 7 – designed to provide the information necessary to answer the questions in Figure 2.1.3 in CLP.



NOTE: The person responsible for the classification of explosives should be experienced in this field and be familiar with the criteria for classification.

Annex I: Figure 2.1.3

Procedure for assignment to a division in the class of explosives (Class 1 for transport)



Test Series 5

Test Series 5 is only carried out for explosive substances/mixtures which are very insensitive and therefore candidates for division 1.5. Typical substances/mixtures are blasting agents such as ANFO, slurries, and emulsion explosives.

The results from three types of series 5 tests are used to answer the question 'Is it a very insensitive explosive substance / mixture with a mass explosion hazard?'.

The test types are (recommended test is indicated within brackets):

Type 5 (a): a shock test to determine the sensitivity to intense mechanical stimulus (cap sensitivity test);

Type 5 (b): thermal tests to determine the tendency of transition from deflagration to detonation (French or USA DDT test); and

Type 5 (c): a test to determine if a substance, when in large quantities, explodes when subjected to a large fire.

The question is answered 'No' if a '+' is obtained in any of the three test types. A candidate for Division 1.5 should pass one test of each type.

Test Series 6

The results from four types of series 6 tests are used to determine which division, amongst Divisions 1.1, 1.2, 1.3 and 1.4, corresponds most closely to the behaviour of the substance, mixture or article to be classified if a load is involved in a fire resulting from internal or external sources or an explosion from internal sources. The results are also necessary to assess whether a substance, mixture or article can be assigned to Compatibility Group S of Division 1.4 and whether or not it should be excluded from this class. Test Series 6 should be applied to packages of substances, mixtures or articles in the condition and form in which they are offered for supply and use.

The four test types are (recommended test is indicated within brackets):

Type 6 (a): a test on a single package to determine if there is mass explosion of the contents (single package test);

Type 6 (b): a test on packages of an explosive substance, mixture or explosive articles, or non-packaged explosive articles, to determine whether an explosion is propagated from one package to another or from a non-packaged article to another (stack test); and

Type 6 (c): a test on packages of an explosive substance, mixture or explosive articles, or non-packaged explosive articles, to determine whether there is a mass explosion or a hazard from dangerous projections, radiant heat and/or violent burning or any other dangerous effect when involved in a fire (bonfire test);

Type 6 (d): a test on an unconfined package of explosive articles to which special provision 347 of Chapter 3.3 of the UN RTDG Model Regulations applies, to determine if there are hazardous effects outside the package arising from accidental ignition or initiation of the contents.

Test types 6 (a), 6 (b), 6 (c) and 6 (d) are performed in alphabetical order. However, it is not always necessary to conduct tests of all types. Test type 6 (a) may be waived if explosive articles are carried without packaging or when the package contains only one article. Test type 6 (b) may be waived if in each type 6 (a) test:

- the exterior of the package is undamaged by internal detonation and/or ignition; or
- the contents of the package fail to explode, or explode as feebly as would exclude propagation of the explosive effect from one package to another in test type 6(b).

Test type 6(c) may be waived if, in a type 6(b) test, there is practically instantaneous explosion of virtually the total contents of the stack. In such cases the product is assigned to Division 1.1.

Test type 6 (d) is a test used to determine whether a 1.4S classification is appropriate and is only used if Special Provision 347 of Chapter 3.3 of the UN RTDG Model Regulations applies. The results of test series 6 (c) and 6 (d) indicate if 1.4S is appropriate, otherwise the classification is 1.4 other than S.

If a substance or mixture gives a '—' result (no propagation of detonation) in the Series 1 type (a) test, the 6(a) test with a detonator may be waived.

If a substance gives a '—' result (no or slow deflagration) in a Series 2 type (c) test, the 6 (a) test with an igniter may be waived.

Test Series 7

Test Series 7 aims at military explosives (Extremely Insensitive Substance: EIS or article containing an EIS) and is generally not relevant for explosives for civil use. Therefore the individual tests are not described here. If needed, they can be found in the UN- MTC, Part I, Section 17.

Test Series 8

The question whether a candidate for ammonium nitrate emulsion or suspension or gel, intermediate for blasting explosives (ANE) is insensitive enough for classification as oxidising is answered by series 8 tests. The three test types are (recommended test is indicated within brackets):

Type 8 (a): a test to determine the thermal stability (Thermal Stability Test for ANE);

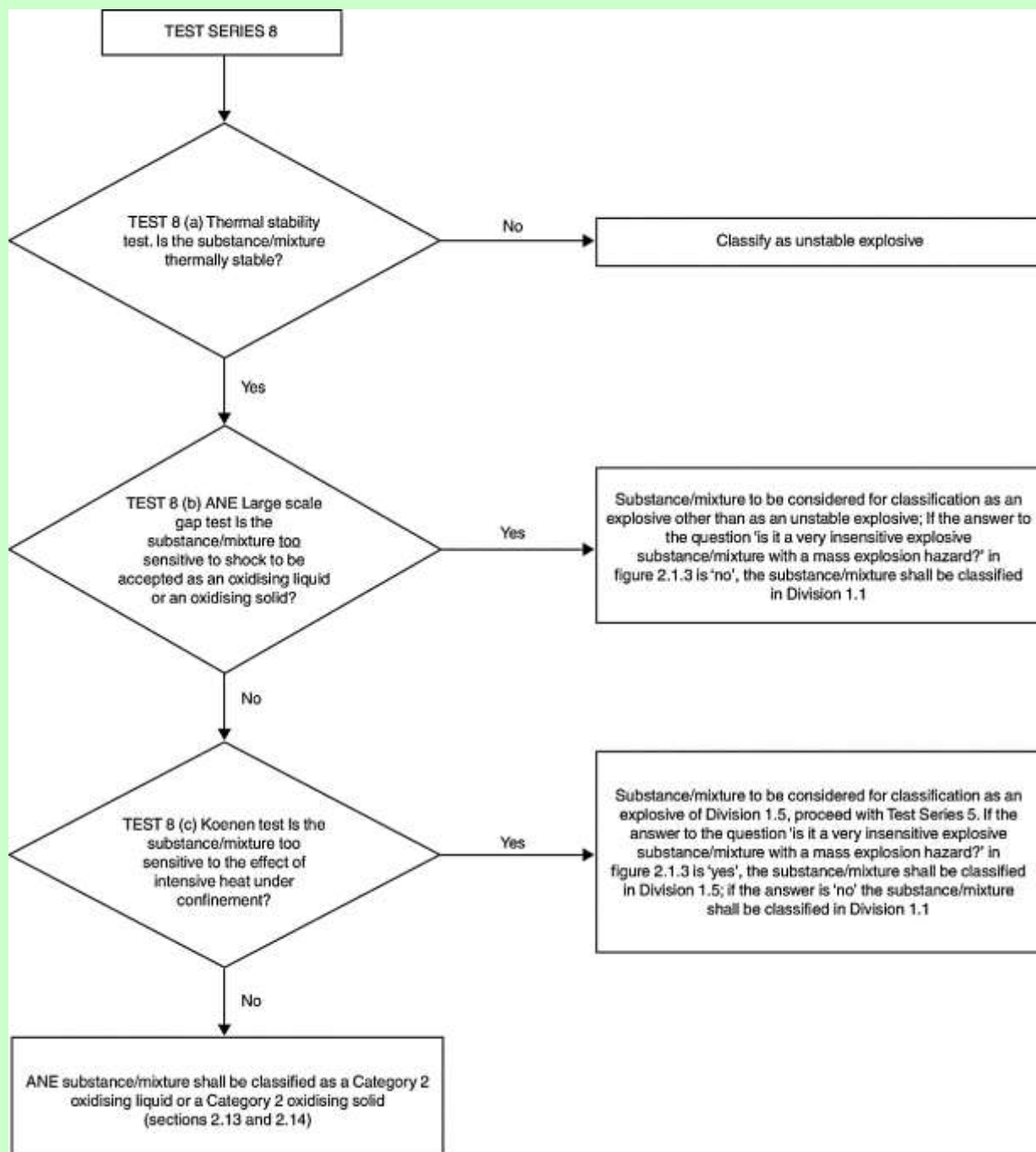
Type 8 (b): a shock test to determine sensitivity to intense shock (ANE gap test); and

Type 8 (c): a test to determine the effect of heating under confinement (Koenen test).

Test Series 8 is used to establish whether an ammonium nitrate emulsion or suspension or gel, intermediate for blasting explosives (ANE) may leave the class of explosives or not. Substances or mixtures failing any of the tests must be classified as explosives (Division 1.1. or 1.5) or as an unstable explosive in accordance with CLP Annex I, Figure 2.1.4. If they pass all three tests they are classified as an oxidising liquid or solid.

Annex I: Figure 2.1.4

Procedure for the classification of ammonium nitrate emulsion, suspension or gel (ANE)



2.1.5. Hazard communication for explosives

2.1.5.1. Pictograms, signal words, hazard statements and precautionary statements²

Annex I: Table 2.1.2							
Label elements for explosives							
Classification	Unstable Explosive	Division 1.1	Division 1.2	Division 1.3	Division 1.4	Division 1.5	Division 1.6
GHS Pictograms							
Signal Word	Danger	Danger	Danger	Danger	Warning	Danger	No signal word
Hazard Statement	H200: Unstable Explosive	H201: Explosive; mass explosion hazard	H202: Explosive; severe projection hazard	H203: Explosive; fire, blast or projection hazard	H204: Fire or projection hazard	H205: May mass explode in fire	No hazard statement
Pre-cautionary Statement Prevention	P201 P250 P280	P210 P230 P234 P240 P250 P280	P210 P230 P234 P240 P250 P280	P210 P230 P234 P240 P250 P280	P210 P234 P240 P250 P280	P210 P230 P234 P240 P250 P280	No pre-cautionary statement
Pre-cautionary Statement Response	P370 + P372 + P380+P373	P370 + P372 + P380 + P373	P370 + P372 + P380 + P373	P370 + P372 + P380 + P373	P370 + P372 + P380 + P373 P370 + P380 + P375	P370 + P372 + P380 + P373	No pre-cautionary statement
Pre-cautionary Statement Storage	P401	P401	P401	P401	P401	P401	No pre-cautionary statement

² The combination statement P370+P372+P380+P373 applies to division 1.4 except for compatibility group S in transport packaging, whereas the combination statement P370+P380+P375 applies to division 1.4 compatibility group S in transport packaging.

<i>Pre-cautionary Statement</i>	<i>P501</i>	<i>P501</i>	<i>P501</i>	<i>P501</i>	<i>P501</i>	<i>P501</i>	<i>No pre-cautionary statement</i>
<i>Disposal</i>							

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

The intrinsic explosive properties of substances and mixtures regarding their stability and sensitivity are only investigated within Test Series 1, 2 and 3 during the acceptance procedure. Subsequent tests for the assignment to the Divisions 1.1, 1.2, 1.3 and 1.4 (Test Series 6) are carried out with the packaged substances, mixtures or articles. The type of packaging may significantly influence the test outcome.

Consequently, there are some deficiencies in the hazard communication of the GHS for unpacked or repacked explosive substances and mixtures, especially for substances and mixtures, which are provisionally accepted in the class of explosives but are later rejected from this class due to their packaging in the assignment procedure (see CLP Annex I, Figure 2.1.1 and Figure 2.1.3 and Section [2.1.4.5.1](#) of this guidance). These substances and mixtures have explosive properties but there might be no hazard communication about these properties due to the subsequent classification in a hazard class other than the class of explosives. Musk xylene is an example which illustrates this issue (see Section [2.1.7.2](#)). The results of Test Series 6 for musk xylene in the specified packaging lead to the exclusion of this substance from the hazard class of explosives. But musk xylene on its own (unpacked) shows explosive properties due to heating under confinement (Koenen test). Also repacking of the substance in a packaging other than the tested one can result in a completely different outcome of Test Series 6.

This issue is not sufficiently clarified under GHS, but should be kept in mind by everyone applying the CLP criteria.

2.1.5.2. Additional labelling provisions

2.1.5.2.1. Packaging dependance

Explosives are normally classified in their transport packaging. The packaging itself may be crucial for the classification. This is clear from the Figure 2.1.3 in Section [2.1.4.5.2](#) especially when it comes to Test Series 6. The assignment of an explosive substance or mixture to a particular Division within the hazard class of explosives is thus only valid for the substance and mixture in the packaging in which it was tested, which is usually the transport packaging. Because of the package-dependence of the classification, paragraph 2.1.2.4 of the Annex I to the CLP prescribes:

Annex I: 2.1.2.4. *If explosives are unpackaged or repacked in packaging other than the original or similar packaging, they shall be retested.*

Further, according to NOTE 1 to Table 2.1.2 in Section 2.1.3 of Annex I to CLP, unpackaged explosives or explosives repacked in packaging other than the original or similar packaging must have the following label elements:

Annex I: 2.1.3. Hazard communication

[...]

NOTE 1: Unpackaged explosives or explosives repackaged in packaging other than the original or similar packaging shall include the following label elements:

(a) the pictogram: exploding bomb;

(b) the signal word: "Danger"; and

(c) the hazard statement: 'explosive; mass explosion hazard'

unless the hazard is shown to correspond to one of the hazard categories in Table 2.1.2, in which case the corresponding symbol, signal word and/or the hazard statement shall be assigned.

Normally, if explosives are unpackaged or repacked in packaging other than the original or similar packaging the classification procedure needs to be performed again in order to determine which Division the explosive belongs to in the new packaging. The label elements prescribed in the NOTE 1 to Table 2.1.2, as quoted above, are the same as those of Division 1.1 and in practice this Division constitutes the most severe classification of a repackaged explosive. (Please note that Table 2.1.2 foresees also the hazard category 'Unstable explosive', which is assigned on the basis of the intrinsic properties of a substance or mixture via Test Series 3 and it is not package dependent). Therefore, the CLP allows labelling of a repackaged explosive with labelling corresponding to Division 1.1 instead of retesting. This, however, overestimates the hazardous properties unless the explosive in fact belongs to Division 1.1.

Many explosives are supplied in inner packages which are placed together in an outer package and where the entity as a whole, i.e. the combination of inner and outer packages, constitutes the transport packaging. According to the UN RTDG Model Regulations and the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) the classification tests are performed in the transport packaging. Under Article 33(1) of the CLP where the hazard pictograms(s) required by CLP relate to the same hazard as in the rules for the transport of dangerous goods, the respective CLP hazard pictogram(s) do not need to appear on the outer packaging.

The classification in accordance with rules on the transport of dangerous goods is almost entirely identical to the corresponding classification procedure used in the CLP and hence the CLP classification will automatically be known for the transport packaging. However, the CLP classification for the inner package alone strictly speaking is not known to manufacturer, importer or downstream user as this will not have been attained through the classification of the transport packaging. On the other hand, it is normally not practicable to perform the required tests on the inner packages. Therefore, normally the same classification as for the transport packaging may be assumed for the inner packages. The labelling requirements for the inner packages are the ones foreseen in Table 2.1.2 of Annex I to the CLP. However, the following exceptions apply:

- Transport packages in which the packaging is designed such that mass explosion is prevented by the packaging, e.g. by arranging the individual inner packages crosswise (so that they are not neighbouring each other) and by separating them with specified material. This is especially the case when packing instruction P101 according to the section 4.1.5 of the ADR applies. In this case the inner package should be labelled in accordance with the Note 1 to Table 2.1.2 of Annex I to the CLP (i.e. as Division 1.1 unless tested otherwise).
- Packages in which explosives of different divisions are contained (for such cases see especially the mixed packing provisions MP 20 to MP 24 in section 4.1.10 of the ADR).
- Furthermore, it does not apply if the packaging is changed, as stated in the Note 1 to Table 2.1.2 of Annex I to the CLP.

2.1.5.2.2. Supplemental hazard information

Some R-phrases under DSD were not covered by hazard classes in the GHS. They are included as supplemental hazard statements in Part 1 of Annex II to of CLP. The following EU hazard statements are important in connection with explosive properties:

Annex II: 1.1.1. EUH001 – 'Explosive when dry'

For explosive substances and mixtures as referred to in chapter 2.1 of part 2 of Annex I, placed on the market wetted with water or alcohols or diluted with other substances to suppress their explosives properties.

EUH001 must be assigned to explosives which are wetted, diluted, dissolved or suspended with a phlegmatizer in order to reduce or suppress their explosive properties (desensitized explosives in the sense of the foreseen new hazard class for desensitized explosives) and which do not meet the criteria of the hazard class of explosives.

Annex II: 1.1.6. EUH044 – 'Risk of explosion if heated under confinement'

For substances and mixtures not in themselves classified as explosive in accordance with section 2.1 of part 2 of Annex I, but which may nevertheless display explosive properties in practice if heated under sufficient confinement. In particular, substances which decompose explosively if heated in a steel drum do not show this effect if heated in less-strong containers.

Some substances and mixtures which may react explosively if heated under confinement are not covered adequately by the classification system. This may e.g. be the case for:

- substances or mixtures which are exempted from the class of explosives based on their packaging and according to results of the Test Series 6;
- substances or mixtures with a SADT of more than 75 °C for a 50 kg package which therefore cannot be classified as self-reactive.

EUH044 must be assigned to such substances or mixtures, in order to make the user aware about these properties.

2.1.5.3. Further communication requirements

According to Note 2 to Table 2.1.2, explosive properties of certain substances and mixtures which are exempted from classification as explosives must be communicated to the user via the SDS (when one is required).

Annex I: 2.1.3. Hazard communication

[...]

NOTE 2: Substances and mixtures, as supplied, with a positive result in Test Series 2 in Part I, Section 12, of the UN RTDG, Manual of Tests and Criteria, which are exempted from classification as explosives (based on a negative result in Test Series 6 in Part I, Section 16 of the UN RTDG, Manual of Tests and Criteria,) still have explosive properties. The user shall be informed of these intrinsic explosive properties because they have to be considered for handling – especially if the substance or mixture is removed from its packaging or is repackaged – and for storage. For this reason, the explosive properties of the substance or mixture shall be communicated in Section 2 (Hazards identification) and Section 9 (Physical and chemical properties) of the Safety Data Sheet and other sections of the Safety Data Sheet, as appropriate

2.1.6. Relation to transport classification

Division 1.1 – 1.6 within Class 1 of the UN RTDG Model Regulations covers explosive substances, mixtures and articles. Normally, the transport classification in accordance with the UN RTDG Model Regulations and the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) can be used one-to-one when deriving the CLP classification for explosives,

which are packaged in authorised transport packaging. See Annex VII of this guidance for additional information on transport classification in relation to CLP classification.

For the use of other packaging or for unpacked substances and mixtures the additional labelling provisions (see Section 2.1.5.2) have to be observed or re-testing is necessary.

2.1.7. Examples of classification for explosives

Examples are given below for the classification of substances. Equivalent information would be needed for mixtures.

2.1.7.1. Example of substances and mixtures fulfilling the classification criteria

a. RESULTS FROM APPLICATION OF THE ACCEPTANCE PROCEDURE

Step	Test	Conclusion	Rationale
0. General data: 0.1 Name of the substance / mixture: Hexanitrostilbene			
1. Is the substance / mixture a candidate for ammonium nitrate emulsion, suspension or gel, intermediate for blasting explosive (ANE)?		No	
2. Is the substance / mixture manufactured with the view to producing a practical explosive or pyrotechnic effect?		Yes	
3. Test Series 3 3.1 Thermal stability: 3.2 Impact sensitivity: 3.3 Friction sensitivity:	75 °C / 48 hour test (test 3(c)) BAM Fallhammer test (test 3(a)(ii)) BAM friction test (test 3(b)(i))	Result: '—', thermally stable Result: Limiting impact energy 5 J Result: Limiting load > 240 N	'—', not too dangerous in form tested '—', not too dangerous in form tested
4. Is the substance / mixture thermally stable?		Yes	
5. Is the substance / mixture too dangerous in the form in which it was tested?		No	

Step	Test	Conclusion	Rationale
6. Conclusion:		PROVISIONALLY ACCEPT INTO THIS CLASS	
10.1 Exit:		Apply the assignment procedure	

1 b. RESULTS FROM APPLICATION OF THE ASSIGNMENT PROCEDURE

Step	Test	Conclusion	Rationale
1. Is the substance a candidate for Division 1.5?		No Result: Package the substance	
2. Test Series 6			
2.1 Effect of initiation in the package:	Test 6(a) with detonator	Result: detonation, crater	
2.2 Effect of propagation:	Type 6(b) with detonator	Result: detonation of the whole stack of packages, crater	
2.4 Effect of fire engulfment:	Test 6(c) may be waived because of the result of 6(b) test.		
3. Is the result a mass explosion?		Yes	
4. Conclusion:		Assignment to Division 1.1	

2 **2.1.7.2. Example of substances and mixtures not fulfilling the**
3 **classification criteria**

4 This example is taken from the UN-MTC, Part I, Section 10.5.2, Figure 10.5.

5 a. RESULTS FROM APPLICATION OF THE ACCEPTANCE PROCEDURE

Step	Test	Conclusion	Rationale
0. General data:			
0.1 Name of the substance / mixture: 5-tert-butyl-2,4,6-trinitro-m-xylene (musk xylene)			
1. Is the substance / mixture a candidate for ammonium nitrate emulsion, suspension or gel,		No	

Step	Test	Conclusion	Rationale
intermediate for blasting explosive ANE?			
2. Is the substance / mixture manufactured with the view to producing a practical explosive or pyrotechnic effect?		No	
3. Test Series 1			
3.1 Propagation of Detonation:	UN gap test (test 1(a))	Result: '+', propagation of detonation	Fragmentation type 'F' '+', shows some explosive effects on heating under confinement
3.2 Effect of heating under confinement:	Koenen test (test 1(b))	Result: Limiting diameter 12.0 mm	
3.3 Effect of ignition under confinement:	Time/pressure test (test 1(c)(i))	Result: '—', no effect on ignition under confinement	
4. Is it an explosive substance / mixture?		Yes	
5. Test Series 2			
5.1 Sensitivity to shock:	UN gap test (test 2(a))	Result: '—', not sensitive to shock	Fragmentation type 'F' '+', violent effect on heating under confinement.
5.2 Effect of heating under confinement:	Koenen test (test 2(b))	Result: Limiting diameter 12.0 mm	
5.3 Effect of ignition under confinement:	Time/pressure test (test 2(c)(i))	Result: '—', no effect on ignition under confinement	
6. Is the substance / mixture too insensitive for acceptance into this class?		No	
Conclusion:		Substance to be considered for this class	
7. Test Series 3			
7.1 Thermal stability:	75 °C/48 hour test (test 3(c))	Result: '—', thermally stable	

Step	Test	Conclusion	Rationale
7.2 Impact sensitivity:	BAM Fallhammer test (test 3(a)(ii))	Result: Limiting impact energy 25 J", not too dangerous in form tested.	
7.3 Friction sensitivity:	BAM friction test (test 3(b)(i))	Result: Limiting load > 360 N	'—', not too dangerous in form tested
8. Is the substance / mixture thermally stable?		Yes	
9. Is the substance / mixture too dangerous in the form in which it was tested?		No	
10. Conclusion:		PROVISIONALLY ACCEPT INTO THIS CLASS	
10.1 Exit		Apply the assignment procedure The explosive properties shall be communicated in the safety data sheet in accordance with section 2.1.5.3 above.	

1 b. RESULTS FROM APPLICATION OF THE ASSIGNMENT PROCEDURE

Step	Test	Conclusion	Rationale
1. Is the substance a candidate for Division 1.5?		No Result: Package the substance	
2. Test Series 6			
2.1 Effect of initiation in the package:	Test 6(a) with detonator	Result: Only localised decomposition around detonator	No significant reaction
2.2 Effect of ignition in the package:	Test 6(a) with igniter	Result: Only localised decomposition around igniter	No significant reaction
2.3 Effect of propagation:	Type 6(b) test not required as no effect outside package between packages in 6(a) test		
2.4 Effect of fire engulfment:	Test 6	Result: Only slow burning with black smoke occurred.	No effects which would hinder fire fighting
3. Is the result a mass explosion?		No	
4. Is the major hazard that from dangerous projections?		No	
5. Is the major hazard radiant heat and/or violent burning but with no dangerous blast or projection hazard?		No	
6. Is there nevertheless a small hazard in the event of ignition or initiation?		No	
7. Is the substance manufactured with the view to producing a practical explosive or pyrotechnic effect?		No	
8. Conclusion:		NOT AN EXPLOSIVE	
8.1 Exit		Consider for another class (e.g. flammable solid)	

2.2. FLAMMABLE GASES (INCLUDING CHEMICALLY UNSTABLE GASES)

2.2.1. Introduction

The criteria for 'Flammable gases (including chemically unstable gases)' are found in Annex I, Section 2.2 of CLP and are identical to those in Chapter 2.2 of GHS.

2.2.2. Definitions and general considerations for the classification of flammable gases (including chemically unstable gases)

Annex I: 2.2.1. Definitions

2.2.1.1 Flammable gas means a gas or gas mixture having a flammable range with air at 20 °C and a standard pressure of 101.3 kPa.

2.2.1.2. A chemically unstable gas means a flammable gas that is able to explode even in the absence of air or oxygen.

The flammable range of a flammable gas is defined between the 'lower flammability limit' (LFL) in air and the 'upper flammability limit' (UFL) in air. In technical literature, the terms 'lower explosion limit' (LEL) and 'upper explosion limit' (UEL) are often used instead of the LFL and UFL, respectively.

The hazard class of flammable gases also covers chemically unstable gases as defined above.

2.2.3. Relation to other physical hazards

Annex I: 2.2.2. Classification criteria

[...]

Note: Aerosols shall not be classified as flammable gases; see section 2.3.

For flammable gases that are packaged in aerosol dispensers see 2.3 Aerosols. If classified as aerosols, they do not have to be classified as flammable gases in addition.

2.2.4. Classification of substances and mixtures as flammable gases (including chemically unstable gases)

2.2.4.1. Identification of hazard information

Many gases are classified as flammable gases in Annex VI of CLP and more gases are classified as flammable gases in the UN RTDG Model Regulations.

For gases that are not classified as flammable gases in Annex VI of CLP nor in the UN RTDG Model Regulations, there is ample scientific literature giving the flammability range for most gases (e.g. IEC 60079-20-1, *Explosive atmospheres – Part 20-1: Material characteristics for gas and vapour classification – Test methods and data as amended*).

In the case a gas or gas mixture needs to be tested for flammability, a recognised international standard must be used such as the EN 1839, *Determination of explosion limits of gases and vapours as amended* or ISO 10156, *Gases and gas mixtures – Determination of fire potential and oxidising ability for the selection of cylinder valves outlets as amended*.

Information on a number of chemically unstable gases can be found in the UN-MTC, Section 35. Tables 35.1 and 35.2 within UN-MTC, Section 35.3.2.1 contain information on a number of chemically unstable gases together with their classification and Category.

If information on other gases than the ones mentioned in the above tables is needed a test method for determination of chemical instability of gases and gas mixtures is described in UN-MTC, Section 35. However, it should be noted that this test method is not applicable to liquefied gas mixtures. In case the gaseous phase above a liquefied gas mixture may become chemically unstable after withdrawal, this should be communicated via the SDS.

2.2.4.2. Screening procedures and waiving of testing for gas mixtures

There are thousands of gas mixtures on the market and there are a limited number of test reports for the flammability of gas mixtures in the scientific literature. Tests to determine the flammability range are time consuming and expensive for gas mixtures which are often prepared on demand. In most of the cases, the formulator of the gas mixture will use a calculation method as described in ISO 10156 as amended (see Section 2.2.4.4) to determine if the mixture is flammable or not.

If the calculations in accordance with ISO 10156 as amended show that a gas mixture is not flammable it is also not classified as chemically unstable and therefore it is not necessary to carry out the tests for determining chemical instability for classification purposes.

Expert judgement should be applied to decide whether a flammable gas or gas mixture is a candidate for classification as chemically unstable in order to avoid unnecessary testing of gases where there is no doubt that they are stable. Functional groups indicating chemical instability in gases are triple bonds, adjacent or conjugated double-bonds, halogenated double-bonds and strained rings.

Gas mixtures containing only one chemically unstable gas are not considered as chemically unstable and therefore do not have to be tested for classification purposes if the concentration of the chemically unstable gas is below the higher of the following generic concentration limits:

- a. the lower explosion limit (LEL) of the chemically unstable gas; or
- b. 3 mole%.

Furthermore, for some gases there are also specific concentration limits available and these are indicated in the tables 35.1 and 35.2 within UN-MTC, Section 35.3.2.1.

2.2.4.3. Classification criteria

The criteria for the classification of flammable gases (including chemically unstable gases) are given in the following tables:

Annex I: 2.2.2. Table 2.2.1

Criteria for flammable gases

Category	Criteria
1	Gases, which at 20 °C and a standard pressure of 101.3 kPa: (a) are ignitable when in a mixture of 13 % or less by volume in air; or (b) have a flammable range with air of at least 12 percentage points regardless of the lower flammable limit.
2	Gases, other than those of Category 1, which, at 20 °C and a standard pressure of 101.3 kPa, have a flammable range while mixed in air.

Annex I: 2.2.2 Table 2.2.2

Criteria for chemically unstable gases

Category	Criteria
A	Flammable gases which are chemically unstable at 20 °C and a pressure of 101.3 kPa.
B	Flammable gases which are chemically unstable at a temperature greater than 20 °C and/or a pressure greater than 101.3 kPa.

2.2.4.4. Testing and evaluation of hazard information

ISO 10156 as amended describes a test method and a calculation method for the classification of flammable gases. The test method may be used in all cases, but must be used when the calculation method cannot be applied.

The calculation method applies to gas mixtures and can be applied when the T_{ci} for all flammable components and the K_k for all inert components are available. These are listed for a number of gases in ISO 10156 as amended. In the absence of T_{ci} value for a flammable gas, the value of the LFL can be used and ISO 10156 proposes the value of 1.5 where no K_k value is listed. The calculation method described in ISO 10156 as amended uses the criterion that a gas mixture is considered non-flammable in air if:

$$\sum_{i=1}^n \frac{A'_i}{T_{ci}} \leq 1$$

where:

$$A'_i = \frac{A_i}{\sum_{i=1}^n A_i + \sum_{k=1}^p K_k B_k}$$

and where:

A'_i is the equivalent content of the i :th flammable gas in the mixture, in %

T_{ci} is the maximum content of flammable gas i which, when mixed with nitrogen, is not flammable in air, in %

A_i is the molar fraction of the i :th flammable gas in the mixture, in %

B_k is the molar fraction of the k :th inert gas in the mixture, in %

K_k is the coefficient of equivalency of the inert gas k relative to nitrogen

n is the number of flammable gases in the mixture

p is the number of inert gases in the mixture

The principle of the calculation method is the following:

Where a gas mixture contains an inert diluent other than nitrogen, the volume of this diluent is adjusted to the equivalent volume of nitrogen using the equivalency coefficient for the inert gas K_k . From this the equivalent contents A'_i are then derived through [Equation 2.2.4.4.2](#), which should be viewed as the corresponding concentration of the flammable gases if nitrogen was the only inert gas present in the mixture. In [Equation 2.2.4.4.1](#) the equivalent contents are then compared to the constants T_{ci} , which have been experimentally found using nitrogen as the (only) inert gas.

It should be noted that ISO 10156 uses molar fractions in some of its equations. For most gases under normal (i.e. non-extreme) conditions, however, the volume fraction can be assumed to be equal to the molar fraction, which is the same as assuming ideal gas behaviour for all gases in the mixture. Furthermore, although normally a fraction is a number ranging from 0 to 1, in this case it is easier to express it as percentage, i.e. the fraction multiplied by 100.

The calculation method described in ISO 10156 as amended determines only if the mixture is flammable or not. It does not determine a flammability range and therefore the calculation method cannot determine if the mixture is flammable Category 1 or Category 2. Therefore, to be on the safe side, mixtures determined to be flammable according the calculation method are classified Flammable gas; Category 1. If, however, there is a need to distinguish between Category 1 and Category 2, the lower and the upper explosion limits have to be determined by using a suitable test method (e.g. EN 1839 or ISO 10156 as amended).

For mixtures containing both flammable and oxidising components, special calculation methods are described in ISO 10156 as amended.

Gases or compressed gas mixtures that are classified as flammable have to be considered for classification as chemically unstable in addition. If the screening procedures described in [Section 2.2.4.2](#) are not conclusive, the gas or gas mixture has to be tested. The test method is described in UN-MTC, Section 35. It uses the same equipment as the test method for oxidising gases according to ISO 10156 as amended and therefore could be applied by laboratories that also carry out the tests for oxidising gases.

2.2.4.5. Decision logic

Classification of flammable gases is laid down in the following flow-charts which are applicable according to CLP.

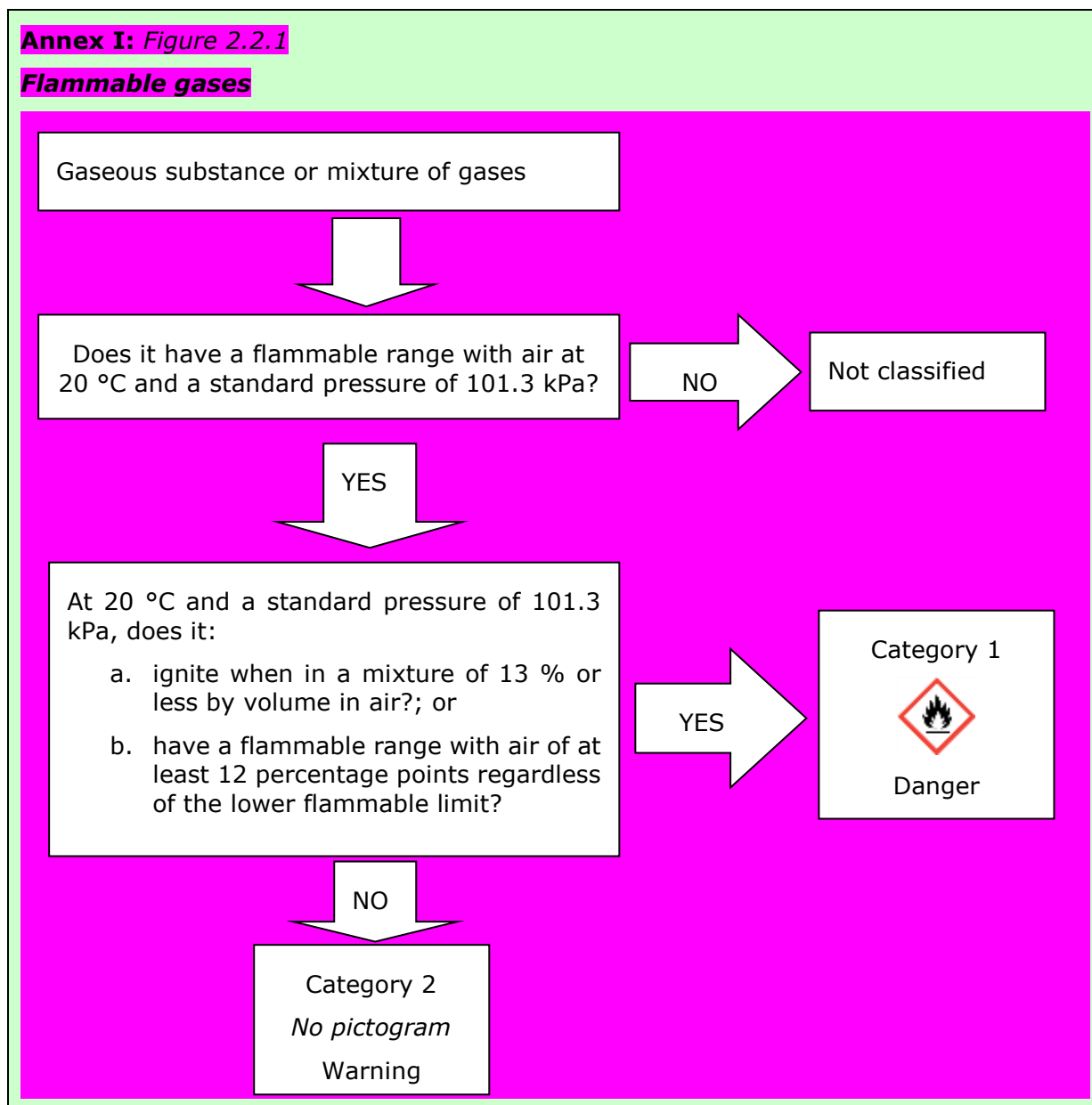


NOTE: The person responsible for the classification of flammable gases (including chemically unstable gases) should be experienced in this field and be familiar with the criteria for classification.

1 **2.2.4.5.1. Decision logic for flammable gases**

Annex I: Figure 2.2.1

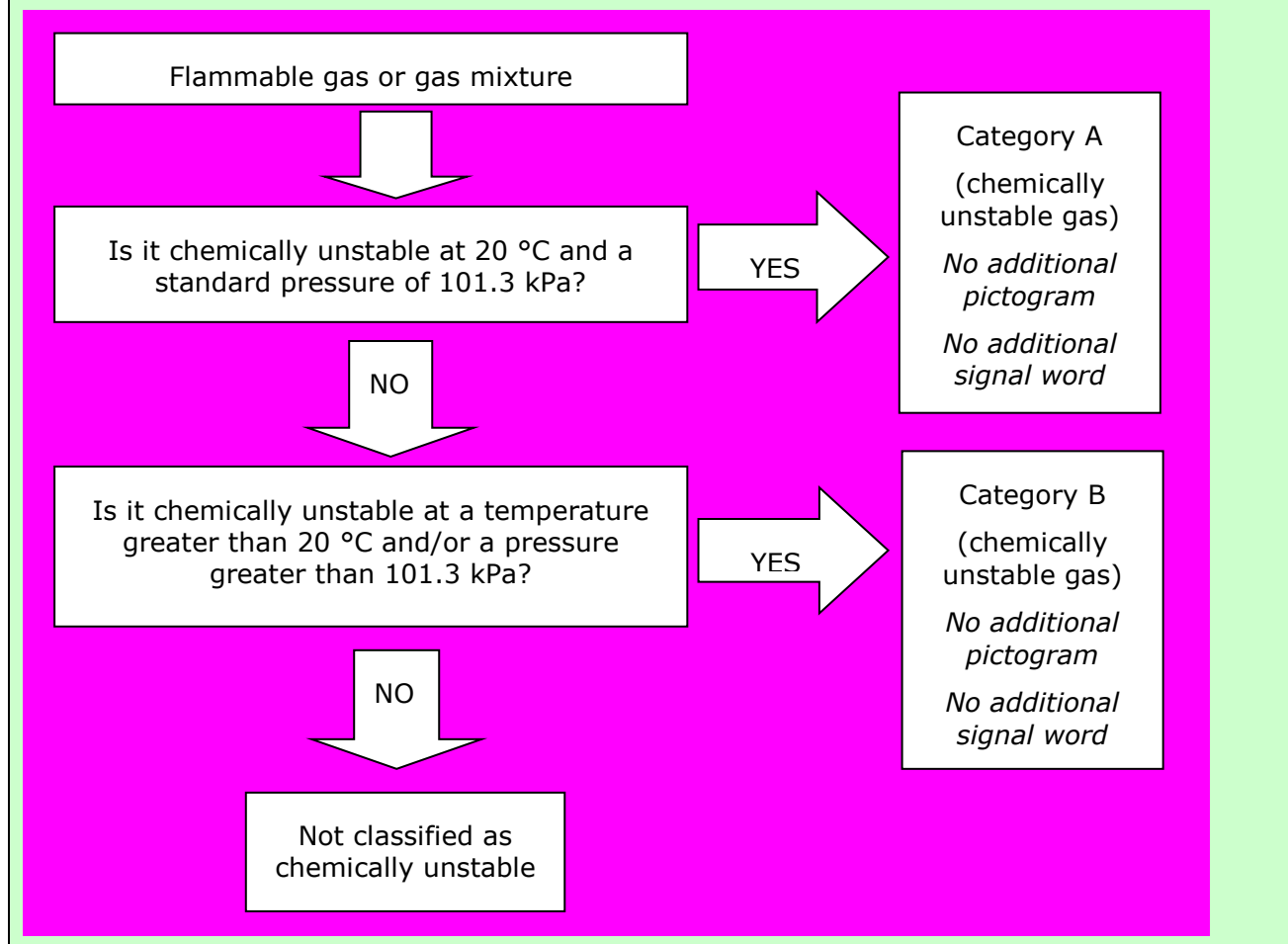
Flammable gases



1 2.2.4.5.2. Decision logic for chemically unstable gases

Annex I: Figure 2.2.2

Chemically unstable gases



2.2.5. Hazard communication for flammable gases (including chemically unstable gases)

2.2.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: 2.2.3. Table 2.2.3

Label elements for flammable gases (including chemically unstable gases)

Classification	Flammable gas		Chemically unstable gas	
	Category 1	Category 2	Category A	Category B
GHS Pictogram		No pictogram	No additional pictogram	No additional pictogram
Signal Word	Danger	Warning	No additional signal word	No additional signal word
Hazard Statement	H220: Extremely flammable gas	H221: Flammable gas	Additional hazard statement H230: May react explosively even in the absence of air	Additional hazard statement H231: May react explosively even in the absence of air at elevated pressure and/or temperature
Precautionary Statement Prevention	P210	P210	P202	P202
Precautionary Statement Response	P377 P381	P377 P381		
Precautionary Statement Storage	P403	P403		
Precautionary Statement Disposal				

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.2.6. Relation to transport classification

The criteria for flammable gases Category 1 correspond to the criteria that are in use for classifying flammable gases in the UN RTDG Model Regulations. Consequently all gases listed as flammable in the UN RTDG Model Regulations and in the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) must be classified as Flam.Gas 1; H220. See Annex VII for additional information on transport classification in relation to CLP classification.

2.2.7. Example of classification for flammable gases

Example mixture: 2 % (H₂) + 6 % (CH₄) + 27 % (Ar) + 65 % (He)

Calculation steps:

Step 1: Assign the gases and state their molar fractions, assuming the molar fractions are equal to the volume fractions (ideal gas behaviour for all gases).

H₂ is flammable gas 1,

yielding $A_1 = 2$ mole %

CH₄ is flammable gas 2,

yielding $A_2 = 6$ mole %

Ar is inert gas 1,

yielding $B_1 = 27$ mole %

He is inert gas 2,

yielding $B_2 = 65$ mole %

$n = 2$

since there are two flammable gases in the mixture

$p = 2$

since there are two inert gases in the mixture

Step 2: Look up the values of T_{ci} and K_k in ISO 10156 as amended.

$T_{c1} =$

5.5 mole %

$T_{c2} =$

8.7 mole %

$K_1 =$

0.55

$K_2 =$

0.9

Step 3: Calculate the equivalent gas contents A'_i for the flammable gases according to Equation 2.2.4.4.2

$$A'_1 = \frac{2}{(2+6) + (0.55 \times 27 + 0.9 \times 65)} = 2.46 \text{ mole \%}$$

$$A'_2 = \frac{6}{(2+6) + (0.55 \times 27 + 0.9 \times 65)} = 7.38 \text{ mole \%}$$

Step 4: Calculate the flammability of the gas mixture according to Equation 2.2.4.4.1

$$\sum_{i=1}^2 \frac{A'_i}{T_{ci}} = \frac{A'_1}{T_{c1}} + \frac{A'_2}{T_{c2}} = \frac{2.46}{5.5} + \frac{7.38}{8.7} = 1.29$$

Step 5: Compare the outcome to the criterion in [Equation 2.2.4.4.1](#)

Since $1.29 > 1$, this particular gas mixture is considered to be flammable.

2.3. AEROSOLS

2.3.1. Introduction

Identical criteria related to the flammability of aerosols are found in Annex I, Section 2.3 of CLP, Chapter 2.3 of GHS as well as in the Aerosol Dispensers Directive (ADD) 75/324/EEC.

2.3.2. Definitions and general considerations for the classification of aerosols

Annex I: 2.3.1. *Aerosols, this means aerosol dispensers, are any non-refillable receptacles made of metal, glass or plastics and containing a gas compressed, liquefied or dissolved under pressure, with or without a liquid, paste or powder, and fitted with a release device allowing the contents to be ejected as solid or liquid particles in suspension in a gas, as a foam, paste or powder or in a liquid state or in a gaseous state.*

2.3.3. Relation to other physical hazards

There is no direct relation to other physical hazards.

1. Annex I, 2.3.2.1.

[...]

Note 2:

Aerosols do not fall additionally within the scope of Sections [2.2](#) (flammable gases), [2.5](#) (gases under pressure), [2.6](#) (flammable liquids) and [2.7](#) (flammable solids). Depending on their contents, aerosols may however fall within the scope of other hazard classes, including their labelling elements.

2.3.4. Classification of aerosols

2.3.4.1. Classification criteria

Annex I: 2.3.2.1. *Aerosols shall be classified in one of the three categories of this hazard class, depending on their flammable properties and their heat of combustion. They shall be considered for classification in Category 1 or 2 if they contain more than 1% components (by mass) which are classified as flammable according to the following criteria set out in this Part:*

- Flammable gases (see section 2.2);*
 - Liquids with a flash point ≤ 93 °C, which includes Flammable Liquids according to section 2.6;*
 - Flammable solids (see section 2.7);*
- or their heat of combustion is at least 20kJ/g.*

Note 1:

Flammable components do not cover pyrophoric, self-heating or water-reactive substances and mixtures because such components are never used as aerosol contents.

[...]

2.3.2.2. *An aerosol shall be classified in one of the three categories for this Class on the basis of its components, of its chemical heat of combustion and, if applicable, of the results of the foam test (for foam aerosols) and of the ignition distance test and enclosed space test (for spray aerosols) in accordance with Figures 2.3.1(a) to 2.3.1(c) of this Annex and sub-sections 31.4, 31.5 and 31.6 of Part III of the UN RTDG, Manual of Tests and Criteria. Aerosols which do not meet the criteria for inclusion in Category 1 or Category 2 shall be classified in Category 3.*

Note:

Aerosols containing more than 1% flammable components or with a heat of combustion of at least 20 kJ/g, which are not submitted to the flammability classification procedures in this section shall be classified as aerosols, Category 1.

Under the ADD and also in UN-MTC, Section 31, flammability classification for aerosols refers to 'extremely flammable', 'flammable' and 'non-flammable'. This respectively corresponds to the terms 'Aerosol, Category 1', 'Aerosol, Category 2' and 'Aerosol, Category 3' which are used in CLP.

The following identical criteria can be found in both CLP and ADD:

The aerosol is classified as 'Aerosol, Category 3' if it contains 1 % or less flammable components³ **and** the chemical heat of combustion is less than 20 kJ/g.

The aerosol is classified as 'Aerosol, Category 1' if it contains 85 % or more flammable components **and** the chemical heat of combustion is 30 kJ/g or more.

All other aerosols should be submitted to the appropriate flammability classification procedures in order to select the appropriate Category 1, 2 or 3. However, if these are not submitted to the

³ Depending on their flash point value, also certain liquids not classified under CLP as Flam. Liq., Cat. 1, 2 or 3, will be considered as flammable components in an aerosol. The CLP hazard class of Flammable liquids covers liquids of flash point ≤ 60 °C while a liquid component in an aerosol is considered flammable when its flash point is ≤ 93 °C.

1 flammability classification procedures they must be automatically classified as 'Aerosol,
2 Category 1'.

3 The chemical heat of combustion is determined in accordance with CLP Annex I, 2.3.4.1 which
4 is identical to point 1.10 of the Annex to ADD.

5 **2.3.4.2. Testing and evaluation of hazard information**

6 Results from the ignition distance test, the enclosed space test and the foam flammability test
7 may be used for the classification related to the flammability of aerosols. These test methods
8 are described under point 6.3 of the Annex to ADD and are therefore available in all EU
9 languages. They are also described in the UN-MTC Section 31.

10 After the evaluation according to the appropriate criteria (see previous sections) the aerosol is
11 classified in one of the three categories.

12 **2.3.4.3. Decision logic**

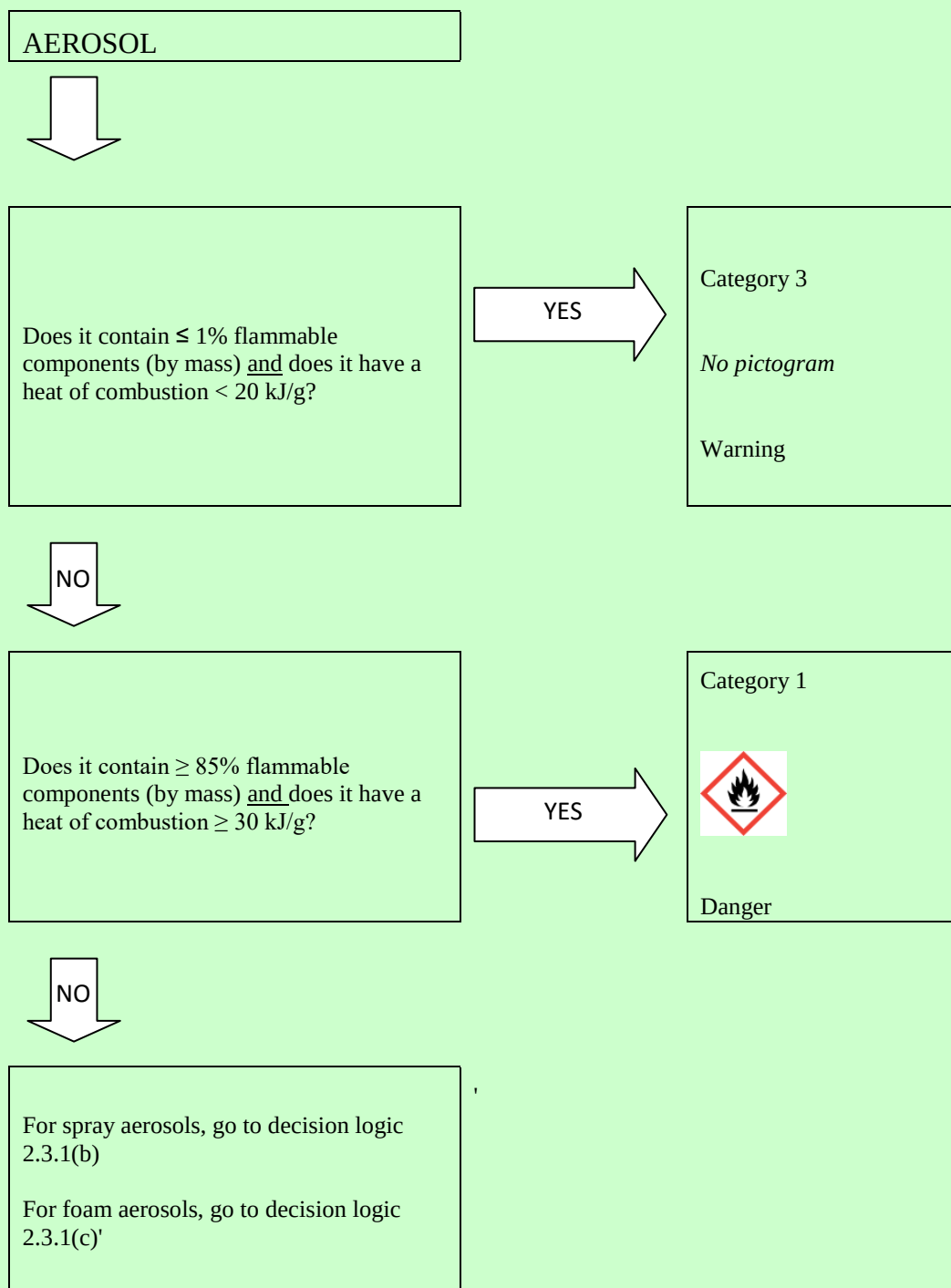
13 The classification procedure is also laid down in the following flow-charts which are applicable
14 according to CLP.



NOTE: The person responsible for the classification of aerosols should be experienced in this field and be familiar with the criteria for classification.

1 2.3.4.3.1. Decision logic for aerosols

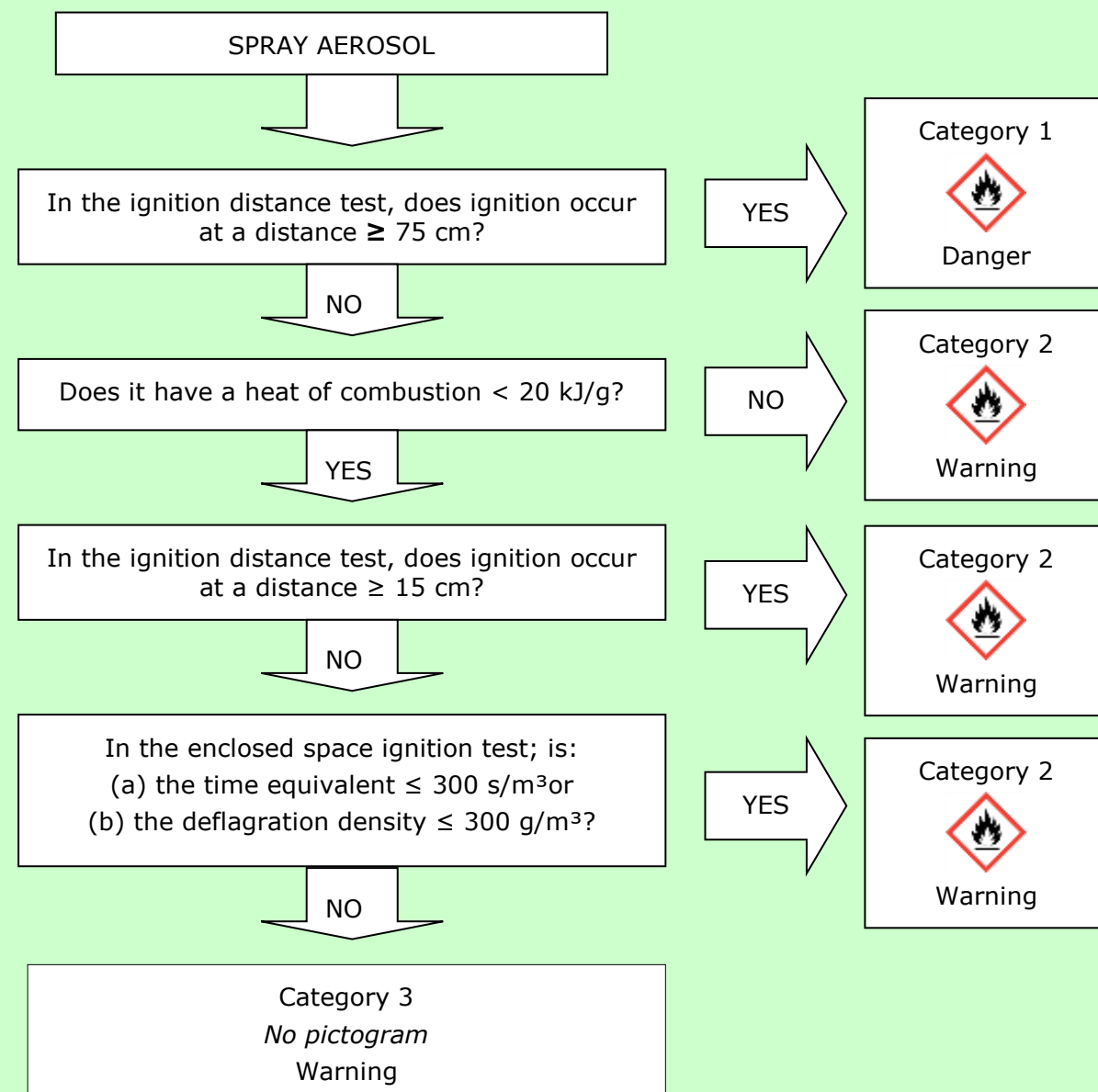
Annex I: Figure 2.3.1 (a)
for aerosols



1 2.3.4.3.2. Decision logic for spray aerosols

Annex I: Figure 2.3.1 (b)

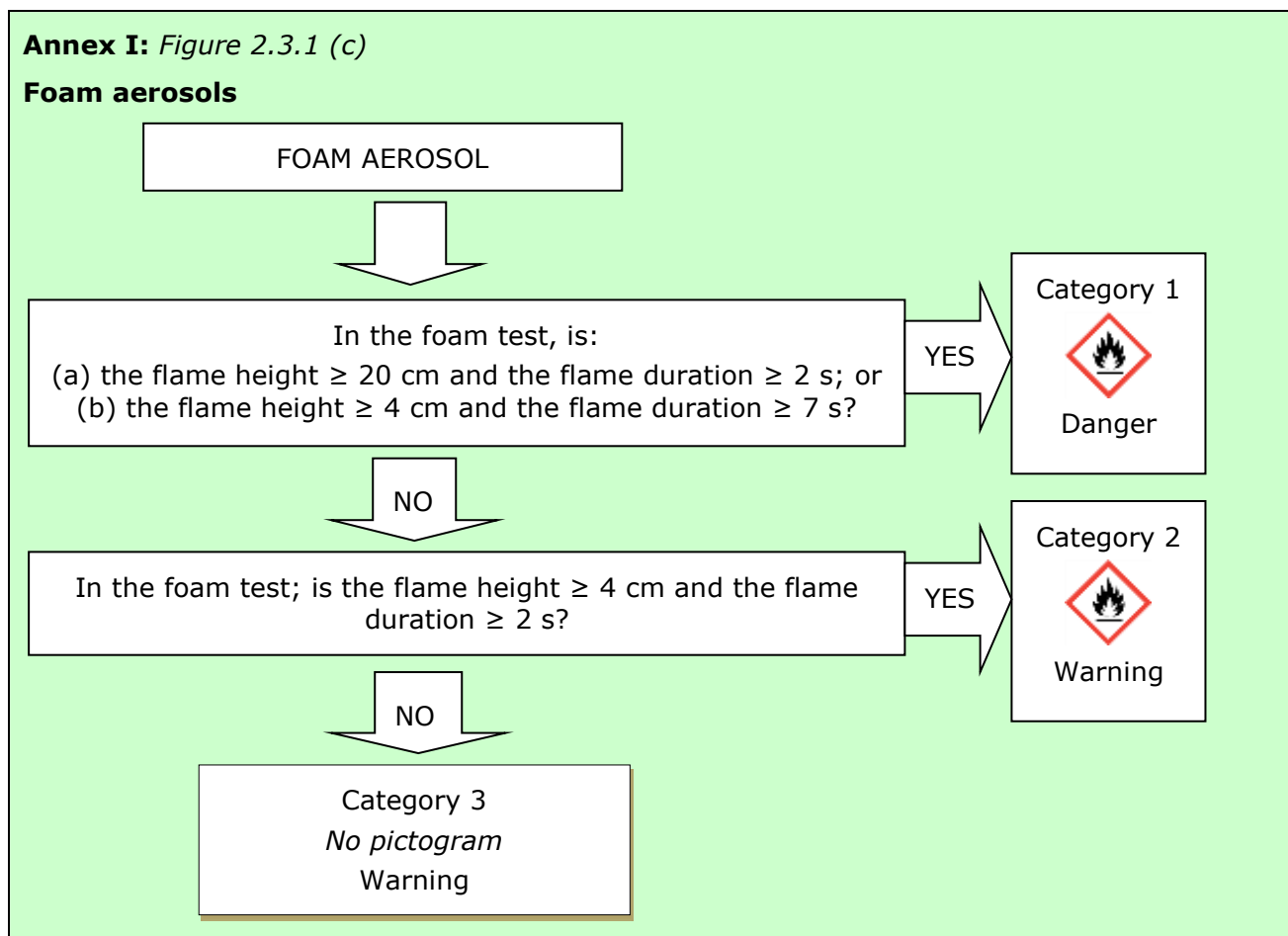
Spray aerosols



1 **2.3.4.3.3. Decision logic for foam aerosols**

Annex I: Figure 2.3.1 (c)

Foam aerosols



2.3.5. Hazard communication for aerosols

2.3.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: Table 2.3.1

Label elements for aerosols

Classification	Category 1	Category 2	Category 3
GHS Pictograms			No pictogram
Signal Word	Danger	Warning	Warning
Hazard Statement	H222: Extremely flammable aerosol H229: Pressurised container: May burst if heated.	H223: Flammable aerosol H229: Pressurised container: May burst if heated.	H229: Pressurised container: May burst if heated.
Precautionary Statement Prevention	P210 P211 P251	P210 P211 P251	P210 P251
Precautionary Statement Response			
Precautionary Statement Storage	P410 + P412	P410 + P412	P410 + P412
Precautionary Statement Disposal			

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.3.5.2. Additional labelling provisions

The ADD imposes additional labelling requirements on all aerosols, flammable or not.

For example:

Where an aerosol dispenser contains flammable components but is not classified as flammable (i.e. 'Aerosol, Category 3'), the quantity of flammable material contained in the aerosol dispenser must be stated clearly on the label, in the form of the following legible and indelible wording: 'X % by mass of the contents are flammable'.

2.3.6. Relation to transport classification

Aerosol dispensers (UN 1950) belong to Class 2 in the UN RTDG Model Regulations and in the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI). Flammability classification criteria are harmonised between CLP and in the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI).

Aerosols, Category 1 and 2 fall under Division 2.1 (sometimes referred to as Class 2.1 or Group F, FC, TF or TFC depending on their contents with hazardous properties). Aerosols, Category 3 fall under Division 2.2 (sometimes referred to as Class 2.2 or Group A, O, T, C, CO, TC or TOC depending on their contents with hazardous properties). See Annex VII for additional information on transport classification in relation to CLP classification.

2.3.7. Examples of classification for aerosols

For reasons of simplification the active materials chosen in the examples have been considered as non-combustible materials ($\Delta H_c = 0$ kJ/g). However this is not the case in practice.

2.3.7.1. Examples of aerosols fulfilling the classification criteria

Deodorant:

Composition:

Butane/propane:	70 % (flammable components, $\Delta H_c = 43.5$ kJ/g)
Ethanol:	25 % (flammable components, $\Delta H_c = 24.7$ kJ/g)
Others:	5 % (non-flammable components, $\Delta H_c = 0$ kJ/g)

This spray aerosol contains 95 % of flammable components, and its chemical heat of combustion equals 36.6 kJ/g ($= 0.70 \cdot 43.5 + 0.25 \cdot 24.7$).

This aerosol is classified as **Aerosol, Category 1**.

Air freshener (wet):

Composition:

Butane/propane:	30 % (flammable components, $\Delta H_c = 43.5$ kJ/g)
Others:	70 % (non-flammable components, $\Delta H_c = 0$ kJ/g)

This spray aerosol contains 30 % of flammable components and its chemical heat of combustion equals 13.1 kJ/g.

In the ignition distance test, the ignition occurs at less than 75 cm but more than 15 cm.

This aerosol is classified as **Aerosol, Category 2**.

Shaving foam:

Composition:

Butane/propane:	4 % (flammable components, $\Delta H_c = 43.5$ kJ/g)
Others:	96 % (non-flammable components, $\Delta H_c = 0$ kJ/g)

This foam aerosol contains 4 % of flammable components and its chemical heat of combustion equals 1.7 kJ/g.

In the foam test, the flame height is less than 4 cm and the flame duration less than 2 s.

This aerosol is classified as **Aerosol, Category 3**.

However, according to the requirements of ADD, the quantity of flammable components must be stated clearly on the label: '4% by mass of the contents are flammable'.

1 **2.3.7.2. Examples of aerosols not fulfilling the classification criteria**

2 By definition, all aerosol dispensers fall under one of the three categories for this hazard class.

3

2.4. OXIDISING GASES

2.4.1. Introduction

The requirements in Chapter 2.4 'Oxidising gases' of Annex I of CLP are identical to those in chapter 2.4 of the GHS.

2.4.2. Definitions and general considerations for the classification of oxidising gases

Annex I: 2.4.1. *Oxidising gas means any gas or gas mixture which may, generally by providing oxygen, cause or contribute to the combustion of other material more than air does.*

2.4.3. Relation to other physical hazards

Oxidising gases do not need to be classified in any other hazard class apart from 'Gases under pressure' where appropriate.

2.4.4. Classification of substances and mixtures as oxidising gases

2.4.4.1. Identification of hazard information

There are not many pure gases that are oxidising. Most oxidising gases are identified as such in the UN RTDG Model Regulations and in ISO 10156 *Gases and gas mixtures: Determination of fire potential and oxidizing ability for the selection of cylinder valve outlets* as amended.

2.4.4.2. Screening procedures and waiving of testing

There are thousands of gas mixtures containing oxidising gases on the market and there are very few test reports on oxidising potential of gas mixtures in the scientific literature. Tests according to ISO 10156 as amended in order to determine the oxidising potential are time consuming and expensive for gas mixtures which are often prepared on demand. In most of the cases, the formulator of the gas mixture will use a calculation method as described in ISO 10156 as amended.

2.4.4.3. Classification criteria

Annex I: 2.4.2. Table 2.4.1

Criteria for oxidising gases

Category	Criteria
1	Any gas which may, generally by providing oxygen, cause or contribute to the combustion of other material more than air does.
Note: <i>'Gases which cause or contribute to the combustion of other material more than air does' means pure gases or gas mixtures with an oxidising power greater than 23.5 % as determined by a method specified in ISO 10156 as amended.</i>	

Please note that ISO 10156-2:2005 has been integrated into the revised version ISO 10156:2010. ISO 10156:2010 supersedes EN 720-2:1996 and ISO 10156-2:2005.

2.4.4.4. Testing and evaluation of hazard information

ISO 10156 as amended describes a test method and a calculation method for the classification of oxidising gases. The test method may be used in all cases, but must be used when the calculation method cannot be applied.

The calculation method applies to gas mixtures and can be applied only when the C_i for all oxidising components and the K_k for all inert components are available. These are listed for a number of gases in ISO 10156 as amended. For gas mixtures the calculation method described in ISO 10156 as amended uses the criterion that a gas mixture should be considered as more oxidising than air if the 'Oxidising Power' (OP) of the gas mixture is higher than 0.235 (23.5 %).

The OP is calculated as follows:

Equation 2.4.4.4.1

$$OP = \frac{\sum_{i=1}^n x_i C_i}{\sum_{i=1}^n x_i + \sum_{k=1}^p K_k B_k}$$

Where:

x_i is the molar fraction of the i :th oxidising gas in the mixture, in %

C_i is the coefficient of oxygen equivalency of the i :th oxidising gas in the mixture

K_k is the coefficient of equivalency of the inert gas k relative to nitrogen

B_k is the molar fraction of the k :th inert gas in the mixture, in %

n is the number of oxidising gases in the mixture

p is the number of inert gases in the mixture

For mixtures containing both flammable and oxidising components, special calculation methods are described in ISO 10156 as amended.

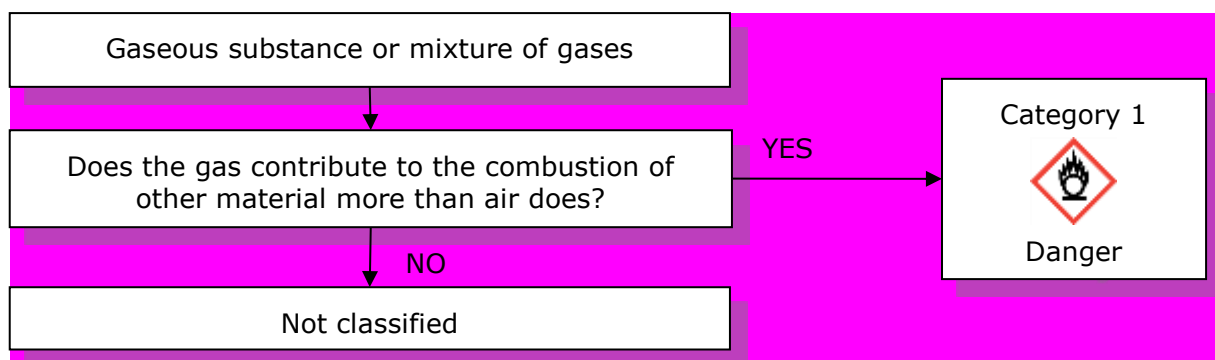
2.4.4.5. Decision logic

Classification of oxidising gases is done according to decision logic 2.4.4.1 as included in the GHS.



NOTE: The person responsible for the classification of oxidising gases should be experienced in this field and be familiar with the criteria for classification.

Figure 2.4.4–1 Decision logic for oxidising gases (Decision logic 2.4 of GHS)



2.4.5. Hazard communication for oxidising gases

2.4.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: Table 2.4.2

Label elements for oxidising gases

Classification	Category 1
GHS Pictogram	
Signal word	Danger
Hazard statement	H270: May cause or intensify fire; oxidiser
Precautionary Statement Prevention	P220 P244
Precautionary Statement Response	P370 + P376
Precautionary Statement Storage	P403
Precautionary Statement Disposal	

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.4.6. Relation to transport classification

Most oxidising gases are classified as such with subsidiary risk 5.1 in the UN RTDG Model Regulations. Consequently all gases listed as oxidising in the UN RTDG Model Regulations and in the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) must be classified as Ox. Gas 1. See Annex VII for additional information on transport classification in relation to CLP classification.

2.4.7. Example of classification for oxidising gases

2.4.7.1. Example of substances and mixtures not fulfilling the classification criteria

EXAMPLE OF A CLASSIFICATION USING THE CALCULATION METHOD OF ISO 10156 AS AMENDED

Example Mixture: 9 % (O₂) + 16 % (N₂O) + 75 % (N₂)

Calculation steps

Step 1: Ascertain the coefficient of oxygen equivalency (C_i) for the oxidising gases in the mixture and the nitrogen equivalency factors (K_k) for the non-flammable, non-oxidising gases.

$C_i (N_2O) =$	0.6 (nitrous oxide)
$C_i (O) =$	1 (oxygen)
$K_k (N_2) =$	1 (nitrogen)

Step 2: Calculate the Oxidising Power (OP) of the gas mixture according to [Equation 2.4.4.4.1](#)

$$OP = \frac{\sum_{i=1}^n x_i C_i}{\sum_{i=1}^n x_i + \sum_{k=1}^p K_k B_k} = \frac{0.09 \times 1 + 0.16 \times 0.6}{0.09 + 0.16 + 0.75 \times 1} = 0.186$$

0.186 < 0.235 (18.6 % < 23.5 %), therefore the mixture is not considered as an oxidising gas.

2.5. GASES UNDER PRESSURE

2.5.1. Introduction

The requirements in Chapter 2.5 'Gases under pressure' of Annex I of CLP are identical to those in Chapter 2.5 of GHS. The hazard class 'Gases under pressure' corresponds to Class 2 'Gases' in the UN RTDG Model Regulations.

2.5.2. Definitions and general considerations for the classification of gases under pressure

2.5.2.1. Definition of 'gas'

Annex I: 1.0. Gas means a substance which (i) at 50 °C has a vapour pressure greater than 300 kPa (absolute); or (ii) is completely gaseous at 20 °C at a standard pressure of 101.3 kPa;

This definition means that substances and mixtures are considered as gases when their boiling point or initial boiling point (BP) is not higher than 20 °C. Substances and mixtures with a boiling point or initial boiling point higher than 20 °C are liquids except those few that develop a vapour pressure higher than 300 kPa at 50 °C; these substances and mixtures are considered as gases because of the pressure hazard when packaged.

Hydrogen fluoride (HF) with a BP of 19.4 °C is a borderline line case that has always been classified as a liquid.

2.5.2.2. Definition of gases under pressure

Annex I: 2.5.1.1. Gases under pressure are gases or gas mixtures which are contained in a receptacle at a pressure of 200 kPa (gauge) or more at 20 °C, or which are liquefied or liquefied and refrigerated.

They comprise compressed gases, liquefied gases, dissolved gases and refrigerated liquefied gases.

This definition means in practice that compressed gases or dissolved gases that are packaged at a pressure less than 200 kPa are not classified for this hazard.

Dissolved gases packaged at a pressure less than 200 kPa (gauge) are liquids and should be classified as such if they have other hazardous properties, e.g. flammable liquids.

Also, liquids packaged under a layer of inert gas (e.g. nitrogen or helium) remain to be classified as liquids and not as gases under pressure.

2.5.3. Relation to other physical hazards

Gases under pressure may also need to be classified for the hazard classes flammable gases and oxidising gases where relevant.

2.5.4. Classification of substances and mixtures as gases under pressure

2.5.4.1. Identification of hazard information

Many gases are identified as such in the UN RTDG Model Regulations and many flammable gases and some oxidising gases are identified as gases in Annex VI of CLP. The UN RTDG Model Regulations identifies further if the gas can be packaged as a 'compressed gas', a 'liquefied gas',

a 'refrigerated liquefied gas' and a 'dissolved gas'. To determine whether a substance is a gas in case it is not listed in the UN RTDG Model Regulations and in case of doubt, the following physical characteristics are necessary:

- the boiling point;
- the vapour pressure at 50 °C.

See also *IR & CSA, Chapter R.7a: Endpoint specific guidance*, Section R.7.1.3 (Boiling point), R.7.1.5 (Vapour pressure).

For those substances that meet the definition of a gas (see Section 2.5.2), the critical temperature is also necessary. For the classification of gas mixtures based on the pseudo-critical temperature see Section 2.5.4.3.

The references according to Section 2.6.8 provide good quality data on boiling points, vapour pressure and the critical temperature of substances.

2.5.4.2. Classification criteria

Annex I: Table 2.5.1

Criteria for gases under pressure

Group	Criteria
Compressed gas	A gas which when packaged under pressure is entirely gaseous at - 50 °C; including all gases with a critical temperature $\leq$ - 50 °C.
Liquefied gas	A gas which, when packaged under pressure, is partially liquid at temperatures above - 50 °C. A distinction is made between: i) high pressure liquefied gas: a gas with a critical temperature between - 50 °C and + 65 °C; and ii) low pressure liquefied gas: a gas with a critical temperature above + 65 °C.
Refrigerated liquefied gas	A gas which when packaged is made partially liquid because of its low temperature.
Dissolved gas	A gas which when packaged under pressure is dissolved in a liquid phase solvent.
Note: Aerosols shall not be classified as gases under pressure. See section 2.3.	

2.5.4.3. Testing and evaluation of hazard information

The critical temperature of pure gases is well defined and can be found in technical literature, e.g. EN 13096 *Transportable gas cylinders — Conditions for filling gases into receptacles — Single component gases* as amended.

For gas mixtures, the classification is based on the 'pseudo-critical temperature' which can be estimated as the mole weighted average of the components' critical temperatures.

1 Pseudo-critical temperature = $\sum_{i=1}^n x_i \times T_{Crit_i}$

2 where x_i is the molar concentration of component i and T_{Crit_i} is the critical temperature (in °C
3 or in K) of the component i .

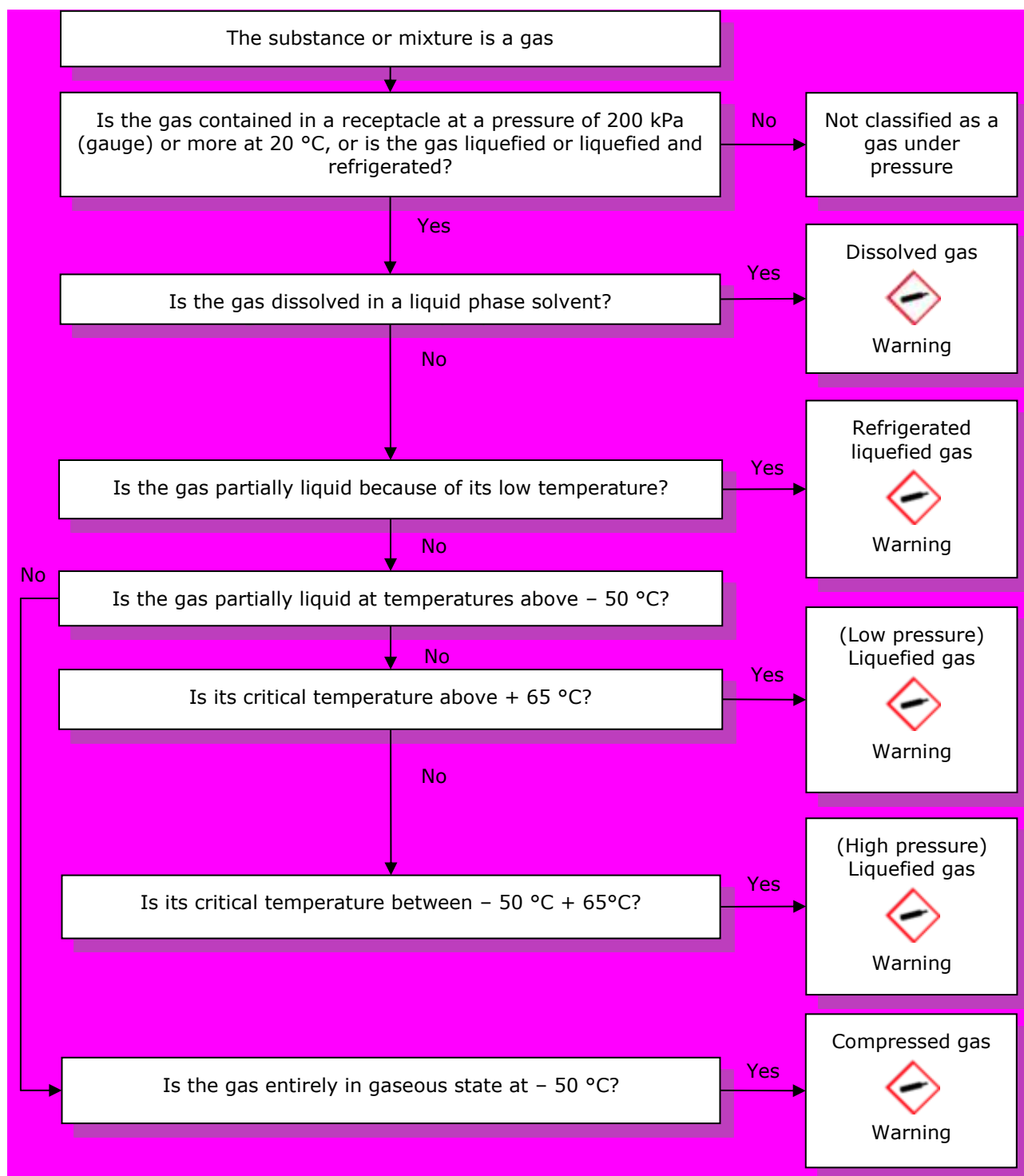
4 **2.5.4.4. Decision logic**

5 Classification of gases under pressure is done according to decision logic 2.5.4.1 as included in
6 the GHS.



NOTE: The person responsible for the classification of gases under pressure should be experienced in this field and be familiar with the criteria for classification.

1 **Figure 2.5.4—1 Decision logic for gases under pressure (Decision logic 2.5 of GHS)**



2.5.5. Hazard communication for gases under pressure

2.5.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: Table 2.5.2

Label elements for gases under pressure

Classification	Compressed gas	Liquefied gas	Refrigerated liquefied gas	Dissolved gas
GHS Pictogram				
Signal Word	Warning	Warning	Warning	Warning
Hazard Statement	H280: Contains gas under pressure; may explode if heated	H280: Contains gas under pressure; may explode if heated	H281: Contains refrigerated gas; may cause cryogenic burns or injury	H280: Contains gas under pressure; may explode if heated
Precautionary Statements Prevention			P282	
Precautionary Statements Response			P336 + P315	
Precautionary Statements Storage	P410 + P403	P410 + P403	P403	P410 + P403
Precautionary Statements Disposal				
Note: Pictogram GHS04 is not required for gases under pressure where pictogram GHS02 or pictogram GHS06 appears.				

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.5.6. Relation to transport classification

Gases are listed in UN RTDG Model Regulations and in the transport regulations (ADR, RID, ADN)⁴ with an indication of the physical state in their name for compressed gases (e.g. Argon, compressed), for refrigerated liquefied gas (e.g. Oxygen, refrigerated liquid) and for dissolved gas (e.g. Acetylene, dissolved). These indications of the physical state can be used to identify the group of gases under pressure according to CLP. The gas names without an indication of the physical state are 'liquefied gases' by default. See Annex VII for additional information on transport classification in relation to CLP classification.

⁴ The classification code according to the ADR, Sections 2.2.2.1.2 and 2.2.2.1.3 are: 1. Compressed gas; 2. Liquefied gas; 3. Refrigerated liquefied gas; 4. Dissolved gas. A asphyxiant; O oxidizing; F flammable; T toxic; TF toxic, flammable; TC toxic, corrosive; TO toxic, oxidizing; TFC toxic, flammable, corrosive; TOC toxic, oxidizing, corrosive.

2.5.7. Examples of classification for gases under pressure

2.5.7.1. Examples of substances and mixtures fulfilling the classification criteria

2.5.7.1.1. Example mixture: 9 % (O₂) + 16 % (N₂O) + 75 % (N₂)

Example mixture: 9 % (O₂) + 16 % (N₂O) + 75 % (N₂)

Calculation steps:

Step 1: Ascertain the critical temperatures in Kelvin for the gases in the mixture:

Oxygen (O₂):

$T_{Crit} = -118.4 \text{ °C} (= 154.75 \text{ K})$

Nitrous Oxide (N₂O):

$T_{Crit} = +36.4 \text{ °C} (= 309.55 \text{ K})$

Nitrogen (N₂):

$T_{Crit} = -147 \text{ °C} (= 126.15 \text{ K})$

Step 2: Calculate the pseudo-critical temperature:

$$0.09 \times 154.75 \text{ K} + 0.16 \times 309.55 \text{ K} + 0.75 \times 126.15 \text{ K} = 158.7 \text{ Kelvin} = -115.08 \text{ °C}$$

The pseudo-critical temperature is lower than -50 °C, therefore the mixture is a 'compressed gas'.

2.6. FLAMMABLE LIQUIDS

2.6.1. Introduction

The criteria for 'Flammable liquids' are found in Annex I, Section 2.6 of CLP and are **not** identical to those of GHS as the respective GHS Chapter 2.6 contains additional classification criteria - Category 4 for flammable liquids.

2.6.2. Definitions and general considerations for the classification of flammable liquids

Annex I: 2.6.1. *Flammable liquid means a liquid having a flash point of not more than 60 °C.*

The flash point is the lowest temperature of the liquid, corrected to a barometric pressure of 101.3 kPa, at which application of a test flame causes the vapour of the liquid to ignite momentarily and a flame to propagate across the surface of the liquid under the specified conditions of test. This means, the lower explosion limit is exceeded at the flash point.

2.6.3. Relation to other physical hazards

For flammable liquids that are packaged in aerosol dispensers, see Section 2.3 on Aerosols. If classified as flammable aerosols, they must not be classified as flammable liquids in addition (see Section 2.3).

2.6.4. Classification of substances and mixtures as flammable liquids

2.6.4.1. Identification of hazard information

For the decision if a substance or mixture is a liquid see Section 2.0.4.

For the classification of a substance or mixture as a flammable liquid, data on the flash point and on the boiling point (or the initial boiling point) are needed. For experimental determination of the flash point information on the viscosity of the liquid is needed, in order to select a suitable method. Furthermore, in order to make use of the derogation for classification in Category 3 according to Annex I Section 2.6.4.5 of CLP (see Section 2.6.4.3), information on sustained combustibility is necessary.

Experimentally determined data or data taken from reliable data sources are to be preferred over calculated ones. See also *IR & CSA, Chapter R.7a: Endpoint specific guidance*, Section R.7.1.3 (Boiling point), R.7.1.9 (Flash point).

The references in Section 2.6.8 provide good quality data on boiling points (all three references) and flash point (first reference) of substances.

Special care is required when viscous substances or mixtures are tested or when halogenated compounds are present (see Section 2.6.4.4.1).

2.6.4.2. Screening procedures and waiving of testing

2.6.4.2.1. Boiling point

Normally calculation methods based on increments give satisfying results for substances and mixtures. With respect to the criterion for distinguishing between Category 1 and 2 (boiling point of 35 °C) only that method with a mean absolute error lower than 5 °C could be recommended for screening.

2.6.4.2.2. Flash point

Calculation should work for pure liquids, neglecting impurities, if the vapour pressure curve and lower explosion limit are accurately known. For mixtures, calculation of the flash point is sometimes not reliable and at this time, it is not possible to predict what the accuracy of a calculated value is. Calculation can be used as a screening test for mixtures, and a flash point need not be determined experimentally if the calculated value using the method cited in CLP Annex I, 2.6.4.3 is 5 °C greater than the relevant classification criterion (23 °C and 60 °C, respectively). However, the restrictions outlined in the CLP Annex I, 2.6.4.2 must be taken account of.

Calculation based on structural similarity or properties is often only applicable to a narrowly defined set of substances. For mixtures they are not yet applicable.

Therefore for both flash point and boiling point experimental determination is recommended.

2.6.4.3. Classification criteria

A flammable liquid has to be classified in one of the 3 categories of this class.

Annex I: Table 2.6.1

Label elements for flammable liquids

Category	Criteria
1	Flash point < 23 °C and initial boiling point ≤ 35 °C
2	Flash point < 23 °C and initial boiling point > 35 °C
3	Flash point ≥ 23 °C and ≤ 60 °C ⁽¹⁾

(1) For the purpose of this Regulation gas oils, diesel and light heating oils having a flash point between > 55 °C and ≤ 75 °C may be regarded as Category 3.

Note:

Aerosols shall not be classified as flammable liquids; see section 2.3.

Annex I: 2.6.4.5. Liquids with a flash point of more than 35 °C and not more than 60 °C need not be classified in Category 3 if negative results have been obtained in the sustained combustibility test L.2, Part III, section 32 of the UN RTDG, Manual of Tests and Criteria.

Gas oils, diesel and light heating oils in the flash point range of 55 °C to 75 °C may be regarded as a whole. The reason is that these hydrocarbon mixtures have varying flash points in that range due to seasonal requirements (EN 590 Automotive fuels – Diesel- Requirements and Test Methods as amended). If they are regarded as a whole for CLP they have to be regarded as Category 3. This states however no preliminary decision with respect to downstream Regulations and legislation.

2.6.4.4. Testing and evaluation of hazard information

The assignment to the respective hazard category will determine the technical means to be taken to avoid dangerous events. In combination with other safety characteristics like explosion limits or auto ignition temperature this can lead to clear restrictions in the conditions of use. The relevant data are to be communicated via the CSR and SDS (see IR/CSA Part F: Chemical

Safety Report, Part G: Extending the SDS and Guidance on compilation of safety data sheets respectively).

2.6.4.4.1. Testing

Suitable methods are listed in CLP Annex I, Table 2.6.3.

In case of substances with a high decomposition potential, a method using small amounts of liquid (e.g. EN ISO 3679 *Determination of flash point - Rapid equilibrium closed cup method as amended*) is recommended to reduce the amount of substance under test.

The method to be used has to be chosen taking into account the properties of the liquid (viscosity, halogenated compounds present) and the scope of the standard.

For classification purposes it is recommended to use the mean of at least two test runs. One of these runs may be automated. In case of a deviation between manual and automated determination above the tolerance limits of the method, the lower value should be taken or the determination should be repeated with manual observation. If the experimentally determined flash point is found to be within ± 2 °C a threshold limit when using a non-equilibrium method, it is recommended to repeat the determination with an equilibrium method.

If no flash point is found up to 60 °C and (partly) halogenated compounds are present or if there is the possibility of loss of volatile flammable or non-flammable components (i.e. the liquid is a candidate for the assignment of EUH018, EUH209 or EUH209A) or if in doubt, the explosion limits should be determined in order to decide whether labelling with EUH018, EUH209 or EUH209A is appropriate. Determination of explosion limits should be carried out according to EN 1839 *Determination of explosion limits of gases and vapours as amended* or ISO 10156 *Gases and gas mixtures – Determination of fire potential and oxidising ability for the selection of cylinder valves outlets as amended* or EN 15794 *Determination of explosion points of flammable liquids as amended*.

Substances

For non-halogenated substances, the flash point is usually found 80 °C to 130 °C below the boiling point. Special care has to be taken when a sample contains impurities with a lower boiling point than the main compound. Even if their concentration is below 0.5 %, especially if their boiling point is substantially lower, they may have a strong effect on the test result. Impurities with a higher boiling point will normally have no effect on the flash point.

Within the respective scope, every standard is applicable.

Mixtures

The flash point may be lower than the lowest flash point of the components and non-volatile components may influence the flash point.

Equilibrium methods are advised if the boiling points of the components of the mixture cover a wide range of temperatures or their concentrations are very different. They are also advised in case of viscous mixtures (alternatively: test methods with low heating rates (1 °C per min) using a stirrer).

In case of viscous mixtures or if an inerting substance is present at low concentrations and this is a highly volatile compound, the ignitability of the mixture may depend on the temperature at which the tests are started. When an inerting substance is present temperature ranges may exist where the vapour phase is inerted and other temperature ranges where it is not.

Halogenated compounds

The difference between boiling point and flash point may be lower than with non-halogenated compounds.

It is highly recommended to run the tests under careful control with manual observation.

Test results may be very difficult to reproduce. In such cases, classification should be based on the lowest value found (flash or burning inside or outside the cup) or on the value obtained during the screening run if in the main trial performed in accordance with the standard, no flash could be found.

2.6.4.4.2. Evaluation of hazard information

Flash points determined by testing or from the mentioned internationally recognised qualified literature are to be preferred over those derived by calculation because of the error of most of the QSAR methods and their limited application range.

If in literature different flash points are found for the same substance the one found as evaluated or recommended has to be preferred.

If in literature different flash points are found for the same substance where none is found as evaluated/recommended the lower one has to be preferred because of safety reasons or an experimental determination should be carried out.

According to the criteria either Category 1, Category 2 or Category 3, including the relevant hazard statement and signal word, have to be assigned (see Section [2.6.5](#)). In case the criteria for EUH018, EUH209 or EUH209A are met, the liquid has to be labelled with the respective supplemental hazard statement as well. In the majority of cases EUH018 covers EUH209 and EUH209A.

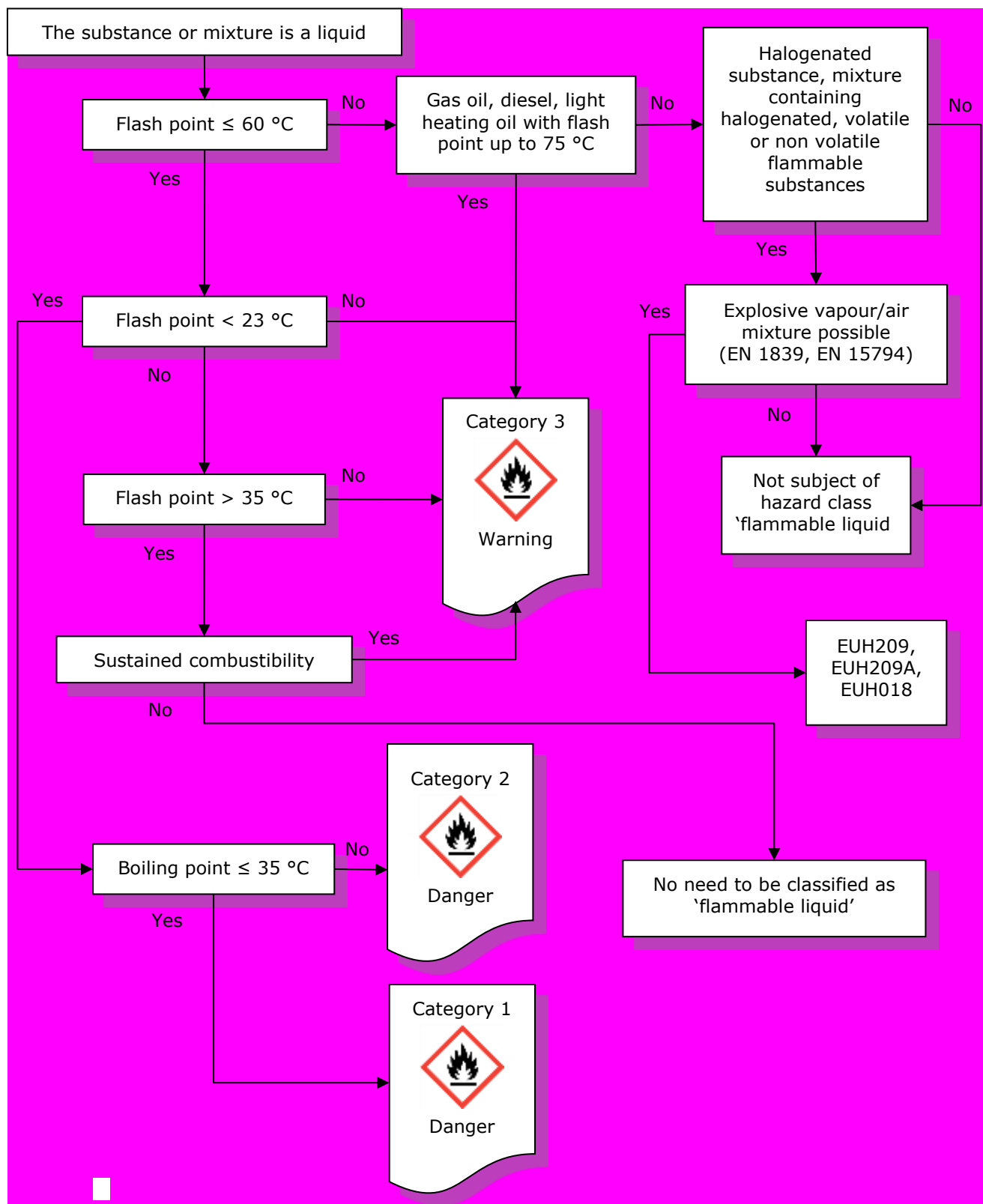
2.6.4.5. Decision logic

Compared to the decision logic 2.6 for flammable liquids contained in the GHS chapter 2.6.4.1, this decision logic below is amended to include derogations for gas oil, diesel, light heating, sustained combustibility and for phrases EUH018, EUH209 and EUH209A.



NOTE: The person responsible for the classification of flammable liquids should be experienced in this field and be familiar with the criteria for classification.

Figure 2.6.4—1 Amended GHS decision logic for flammable liquids to include derogations for gas oil, diesel, light heating, sustained combustibility and for phrases EUH018, EUH209 and EUH209A



2.6.5. Hazard communication for flammable liquids

2.6.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: 2.6.3. Table 2.6.2

Label elements for flammable liquids

Classification	Category 1	Category 2	Category 3
GHS Pictograms			
Signal Word	Danger	Danger	Warning
Hazard Statement	H224: Extremely flammable liquid and vapour	H225: Highly flammable liquid and vapour	H226: Flammable liquid and vapour
Precautionary Statement Prevention	P210 P233 P240 P241 P242 P243 P280	P210 P233 P240 P241 P242 P243 P280	P210 P233 P240 P241 P242 P243 P280
Precautionary Statement Response	P303 + P361 + P353 P370 + P378	P303 + P361 + P353 P370 + P378	P303 + P361 + P353 P370 + P378
Precautionary Statement Storage	P403 + P235	P403 + P235	P403 + P235
Precautionary Statement Disposal	P501	P501	P501

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.6.5.2. Additional labelling provisions for flammable liquids

Annex II: 1.1.4. EUH018 – 'In use, may form flammable/explosive vapour-air mixture'

For substances and mixtures not classified as flammable themselves, which may form flammable/explosive vapour-air mixtures. For substances this might be the case for halogenated hydrocarbons and for mixtures this might be the case due to a volatile flammable component or due to the loss of a volatile non-flammable component.

Substances or mixtures which do not show a flash point but do have an explosion range or may become flammable in use have to be labelled with EUH018.

Annex II: 2.9. Liquid mixtures containing halogenated hydrocarbons

For liquid mixtures which show no flashpoint or a flashpoint higher than 60 °C but not more than 93 °C and contain a halogenated hydrocarbon and more than 5 % highly flammable or flammable substances, the label on the packaging shall bear one of the following statements, depending on whether the substances referred to above are highly flammable or flammable:

EUH209 — 'Can become highly flammable in use' or

EUH209A — 'Can become flammable in use'



Note: EUH209 and EUH209A are limited to special types of mixtures whereas EUH018 covers a wider range of mixtures. In the majority of cases EUH018 covers EUH209 and EUH209A. Information about testing can be found in Section 2.6.4.4.1 paragraph 5.

2.6.6. Re-classification of substances and mixtures classified as flammable liquids according to DSD and DPD or already classified for transport

2.6.6.1. Relation to transport classification

Class 3 of the UN RTDG Model Regulations and the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) cover flammable liquids based on the same criteria as the CLP hazard class flammable liquid. In general there is a correspondence between transport packing groups and CLP hazard categories. However, in many cases specific exceptions apply. Further, the UN RTDG Model Regulations cover substances and mixtures transported above their flash point and desensitized explosives. In practice the information on flash point and boiling point needed for classification is available and it is recommended to classify based on the data rather than use direct translation. See Annex VII for additional information on transport classification in relation to CLP classification.

2.6.7. Examples of classification for flammable liquids

2.6.7.1. Examples of substances and mixtures fulfilling the classification criteria

2.6.7.1.1. Example 1

Mixture of: n-Butylacetate + p-Xylene + 1,3,5-Trimethylbenzene
(7.9 mol % + 60.3 mol % + 31.7 mol %)

Initial boiling point (calculated):	140 °C
-------------------------------------	--------

Flash point (calculated):	26 °C
---------------------------	-------

calculated flash point is within 5 °C to the limiting value of 23 °C
⇒ flash point has to be measured.

Dyn. Viscosity at 20 °C (DIN 53019):	8 mPas
--------------------------------------	--------

Flash point (EN ISO 3679):	30.0 °C
----------------------------	---------

⇒ According to boiling point and measured flash point result: **Flam.Liq. Category 3**

1 **2.6.7.1.2. Example 2**

Hydrocarbons and dichloromethane (70 vol % + 30 vol %)	
Initial Boiling point (calculated):	52 °C
Flash point:	no flash point according to a standard
⇒ Because the hydrocarbon part of the mixture has a flash point by itself (- 12 °C) the question 'Is an explosive vapour/air mixture possible' (EN 1839 as amended, EN 15794 as amended) or 'Can it become highly flammable / flammable during use?' has to be answered.	
Answer: Yes an explosion range exists; yes it can become highly flammable during use.	
⇒ According to the answer, the mixture has to be labelled with EUH018 or EUH209	
Note 1: In that case EUH018 covers EUH209	
Note 2: The EUH018 must only be assigned if the substance or mixture is classified as hazardous (Article 25 (1) of CLP)	
Cannot be classified as flammable liquid because the mixture has no flash point.	

2 **2.6.7.2. Examples of substances and mixtures not fulfilling the**
3 **classification criteria**

4 **2.6.7.2.1. Example 3**

Aqueous formulation of aliphatic polyurethane resin	
Boiling point (EC 440/2008, EU test method A.2):	92 °C
Dyn. Viscosity at 20 °C (DIN 53019 as amended):	1938 mPas
Sample is highly viscous, use low heating rate for flash point determination (1 °C /min).	
Flash point (EN ISO 13736 as amended):	42.5 °C
Sustained combustibility test (UN- MTC L.2) at 60.5 °C:	combustion not sustained
Sustained combustibility test (UN-MTC L.2) at 75 °C:	combustion not sustained
⇒ According to the flash point result: Category 3	
However, does not necessarily have to be classified as flammable liquid Category 3 because it did not sustain combustion.	

5 **2.6.8. References**

- 6 Brandes, E. and Möller, W.: *Safety Characteristic Data*, Volume 1, Flammable gases and liquids,
7 nw-Verlag, 2008
- 8 William M. Haynes *et al.* (2012) *CRC Handbook of Chemistry and Physics 93rd Edition*. CRC
9 Press, Taylor and Francis, Boca Raton, FL
- 10 O'Neil, Maryadele J. *et al.* © (2016, 2012) *The Merck Index - An Encyclopedia of Chemicals,*
11 *Drugs, and Biologicals* (14th Edition – Version 14.9). Merck Sharp & Dohme Corp., a subsidiary
12 of Merck & Co., Inc.

2.7. FLAMMABLE SOLIDS

2.7.1. Introduction

The criteria for 'Flammable solids' are found in Annex I, Section 2.7 of CLP and are identical to those in Chapter 2.7 of GHS.

2.7.2. Definitions and general considerations for the classification of flammable solids

Annex I: 2.7.1.1.

A flammable solid means a solid which is readily combustible, or may cause or contribute to fire through friction.

Readily combustible solids are powdered, granular, or pasty substances or mixtures which are dangerous if they can be easily ignited by brief contact with an ignition source, such as a burning match, and if the flame spreads rapidly.

Special consideration on particle size

Annex I: 2.7.2.3.

[...]

Note 1:

The test shall be performed on the substance or mixture in its physical form as presented. If for example, for the purposes of supply or transport, the same chemical is to be presented in a physical form different from that which was tested and which is considered likely to materially alter its performance in a classification test, the substance shall also be tested in the new form.

[...]

The finer the particle size of a solid substance or mixture, the greater the area exposed to air will be, and since flammability is a reaction with the oxygen in air, the particle size will greatly influence the ability to ignite. Hence it is very important that flammable properties for solids are investigated on the substance or mixture as it is actually presented (including how it can reasonably be expected to be used, see Article 8 (6) of CLP). This is indicated by the Note cited in CLP Annex I, 2.7.2.3. For further information please see Section 1.2 within this Guidance.

2.7.3. Relation to other physical hazards

Explosives, organic peroxides, self-reactive substances and mixtures as well as pyrophoric or oxidising solids should not be considered for classification as flammable solids since flammability is an intrinsic hazard in these classes.

However, flammable solids can present other physical hazards at the same time, i.e. they might be self-heating or corrosive or emit flammable gases in contact with water.

For flammable solids that are packaged in aerosol dispensers, see Section 2.3, Aerosols. If classified as flammable aerosols, they must not be classified as flammable solids in addition (see Section 2.7).

2.7.4. Classification of substances and mixtures as flammable solids

2.7.4.1. Identification of hazard information

For the classification of a substance or mixture as a flammable solid data on the following properties are needed:

- melting point;
- information on water reactivity;
- information on flash point for solids containing flammable liquids.

See also *IR & CSA, Chapter R.7a: Endpoint specific guidance*, Section R.7.1.2 (Melting/freezing point), R.7.1.9 (Flash point).

Many organic solid substances or mixtures fulfil the criteria to be classified as flammable solids. For inorganic solids, the classification as flammable is rather rare.

2.7.4.2. Screening procedures and waiving of testing

In general, a possible classification as a flammable solid should be considered for any solid organic substance or mixture containing such material. For inorganic material, testing may be waived in cases where the substance is commonly known to be not flammable (i.e. stable salts or metal oxides) or where a flammability hazard can be excluded by any other scientific reasoning. In many cases, a simple screening test (see Section 2.7.4.4) can be used to determine whether a solid should be classified as flammable. Solid substances and mixtures are classified as flammable according to their burning behaviour.

The test method as described in Part III, Sub-section 33.2.1.4.3.1 in the UN-MTC should be applied for screening purposes. Alternatively, the burning index (referred to as 'class number' in VDI 2263) as obtained from the Burning Behaviour test (VDI 2263, part 1) may be used. If a burning index of 3 or less is found, the substance or mixture should not be classified as a flammable solid and no further testing is required. However, if smouldering or a flame is observed, the full test must be carried out.

2.7.4.3. Classification criteria

The classification criteria are fully in accordance with the GHS system.

Annex I: 2.7.2.1. Powdered, granular or pasty substances or mixtures (except powders of metals or metal alloys – see 2.7.2.2) shall be classified as readily combustible solids when the time of burning of one or more of the test runs, performed in accordance with the test method described in Part III, sub-section 33.2.1, of the UN RTDG, Manual of Tests and Criteria, is less than 45 seconds or the rate of burning is more than 2,2 mm/s.

2.7.2.2. Powders of metals or metal alloys shall be classified as flammable solids when they can be ignited and the reaction spreads over the whole length of the sample in 10 minutes or less.

2.7.2.3. A flammable solid shall be classified in one of the two categories for this class using Method N.1 as described in 33.2.1 of the UN RTDG, Manual of Tests and Criteria in accordance with Table 2.7.1;

Table 2.7.1 Criteria for flammable solids	
Category	Criteria
1	Burning rate test Substances and mixtures other than metal powders: (a) wetted zone does not stop fire and (b) burning time < 45 seconds or burning rate > 2,2 mm/s Metal powders: burning time ≤ 5 minutes
2	Burning rate test Substances and mixtures other than metal powders: (a) wetted zone stops the fire for at least 4 minutes and (b) burning time < 45 seconds or burning rate > 2,2 mm/s Metal powders: burning time > 5 minutes and ≤ 10 minutes
[...] Note 2: Aerosols shall not be classified as flammable solids; see section 2.3.	

2.7.4.4. Testing and evaluation of hazard information

For safety reasons, it is advisable to test for explosive and self-reactive properties first and to rule out pyrophoric behaviour before performing this test. The classification test is described in Part III, Sub-section 33.2.1.4.3.2 of the UN-MTC. The sample should be tested in its commercially relevant form. Special care has to be taken that the sample forms an unbroken strip or powder train in the test mould. Large pieces that do not fit into the mould should be gently crushed. For pasty or sticking substances it may be helpful to line the mould with a thin plastic foil which is withdrawn after having formed the train. Classification is based upon the fastest burning rate / shortest burning time obtained in six test runs, unless a positive result is observed earlier. For substances and mixtures other than metal powders, the category is assigned depending on whether the wetted zone is able to stop the flame.

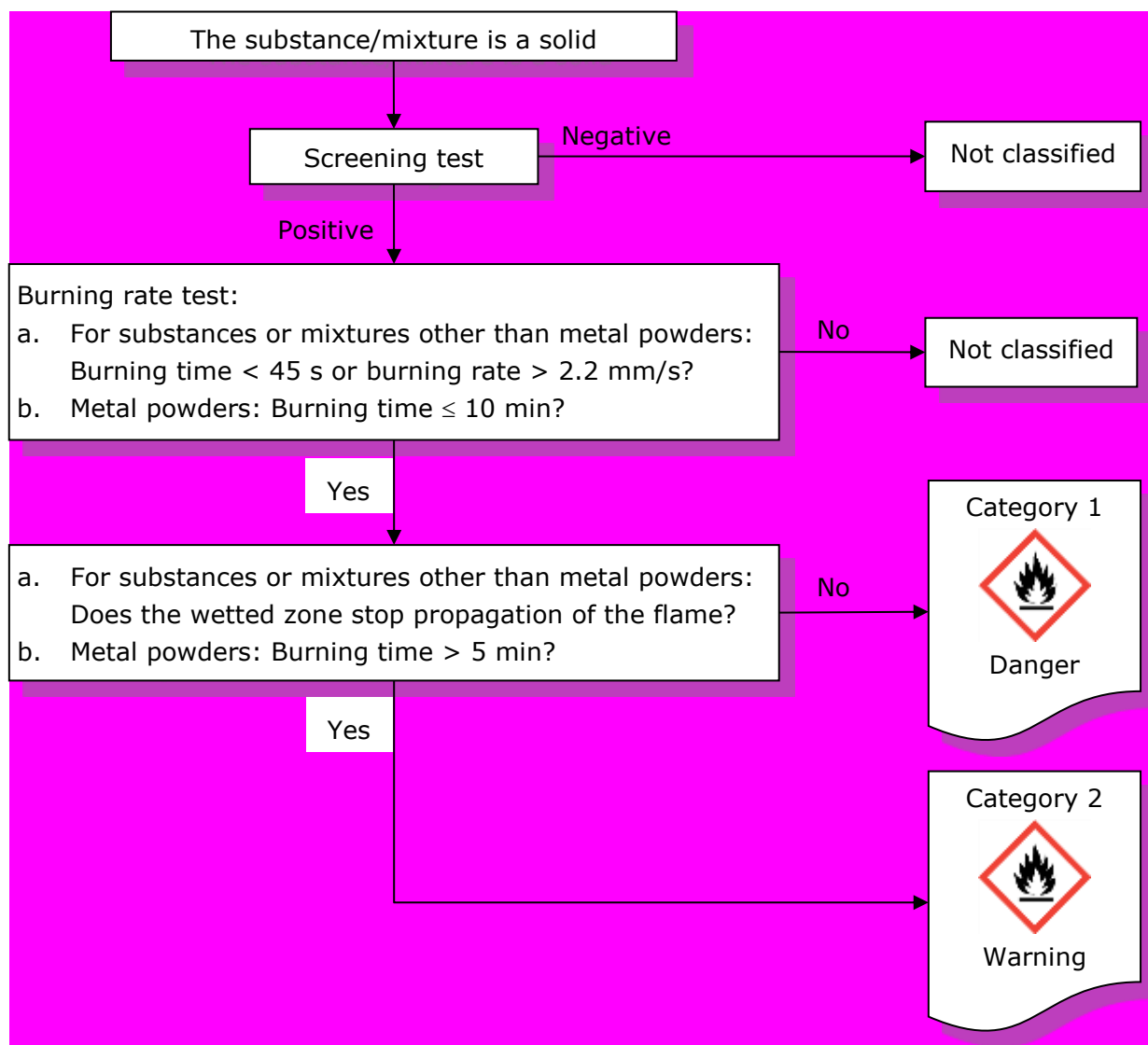
2.7.4.5. Decision logic

Classification of flammable solids is done according to decision logic 2.7.4 as included in the GHS.



NOTE: The person responsible for the classification of flammable solids should be experienced in this field and be familiar with the criteria for classification.

1 **Figure 2.7.4—1 Decision logic for flammable solids (Decision logic 2.7 of GHS)**



2
3

2.7.5. Hazard communication for flammable solids

2.7.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: 2.7.3. Table 2.7.2

Label elements for flammable solids

Classification	Category 1	Category 2
GHS Pictograms		
Signal Word	Danger	Warning
Hazard Statement	H228: Flammable Solid	H228: Flammable Solid
Precautionary Statement Prevention	P210 P240 P241 P280	P210 P240 P241 P280
Precautionary Statement Response	P370 + P378	P370 + P378
Precautionary Statement Storage		
Precautionary Statement Disposal		

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.7.6. Relation to transport classification

Division 4.1 within Class 4 of the UN RTDG Model Regulations covers flammable substances, solid desensitized explosives and self-reactive liquids or solids. If a transport classification according to the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) is available it should be kept in mind that transport classification is based on prioritisation of hazards (see UN RTDG Model Regulations, Section 2.0.3) and that flammable solids have a relatively low rank in the precedence of hazards. Therefore, the translation from transport classification to CLP should be only done if a transport classification for a flammable solid is explicitly available. The conclusion that a substance or mixture not classified as a flammable solid for transport should not be classified as a flammable solid according to CLP is, in general, not correct. See Annex VII for additional information on transport classification in relation to CLP classification.

2.7.7. Examples of classification for flammable solids

2.7.7.1. Example of substances and mixtures fulfilling the classification criteria

The following example shows a classification based on test data:

Test substance: 'Flammalene' (organic material, solid):	
Screening test (VDI 2263, part 1):	burning index: 5 (burning with an open flame or emission of sparks)
Conclusion: Substance is candidate for classification as a flammable solid, further testing required.	
UN Test N.1 (Test method for readily combustible solids):	Burning times for a distance of 100 mm (6 runs): 44 s; 40 s; 49 s; 45 s; 37 s; 41 s. Shortest burning time is less than 45 s; substance is a flammable solid. Wetted zone stops the fire, no reignition.
Conclusion: Classify as flammable solid, Category 2.	

2.7.7.2. Examples of substances and mixtures not fulfilling the classification criteria

Many inorganic salts and oxides are not flammable such as NaCl, NaBr, KI, FeO, MnO etc.

Urea or phthalic acid anhydride are examples of organic substances that would not be classified as flammable solids.

2.7.8. References

VDI guideline 2263, part 1, 1990, *Test methods for the Determination of the Safety Characteristics of Dusts*

2.8. SELF-REACTIVE SUBSTANCES AND MIXTURES

2.8.1. Introduction

The criteria for 'Self-reactive substances and mixtures' are found in Annex I, Section 2.8 of CLP and are identical to those in Chapter 2.8 of GHS.

In general, substances or mixtures classified as self-reactive substances and mixtures can decompose strongly exothermically when 50 kg are exposed to temperatures of 75 °C or lower depending on the Self-Accelerating Decomposition Temperature (SADT) of the substance or mixture.

Self-reactive substances and mixtures display a very wide range of properties. The most hazardous type is TYPE A of self-reactive substances and mixtures that are too dangerous to transport commercially though they can be stored safely with appropriate precautions. At the other end of the scale this classification includes substances and mixtures that only decompose slowly at temperatures well above the normal storage and transport temperatures (e.g. 75 °C).

The decomposition of self-reactive substances and mixtures can be initiated by heat, contact with catalytic impurities (e.g. acids, heavy-metal compounds, and bases), friction or impact. The rate of decomposition increases with temperature and varies with the substance or mixture. Decomposition, particularly if no ignition occurs, may result in the evolution of toxic gases or vapours. For certain self-reactive substances and mixtures, the temperature must be controlled during storage and handling. Some self-reactive substances and mixtures may decompose explosively, particularly if confined. This characteristic may be modified by the addition of diluents or by the use of appropriate packaging. Some self-reactive substances and mixtures burn vigorously. Self-reactive substances are, for example, some compounds of the types listed below:

- a. Aliphatic azo compounds ($-C-N=N-C-$);
- b. Organic azides ($-C-N_3$);
- c. Diazonium salts ($-CN_2^+Z^-$);
- d. N-nitroso compounds ($-N-N=O$); and
- e. Aromatic sulfohydrazides ($-SO_2-NH-NH_2$).

This list is not exhaustive and substances with other reactive groups, combination of groups and some mixtures of substances may have similar properties. Additional guidance on substances, which may have self-reactive properties, is given in Appendix 6, Section 5.1 of the UN-MTC.

Additional hazardous properties, resulting in subsidiary labelling, are indicated in the list of already classified self-reactive substances and mixtures included in the UN RTDG Model Regulations, Section 2.4.2.3.2.3.

Commercial self-reactive substances and mixtures are commonly formulated by dilution with solid and liquid substances with which they are compatible.

2.8.2. Definitions and general considerations for the classification of self-reactives

In CLP the following definition is given for self-reactive substances and mixtures:

Annex I: 2.8.1.1. Self-reactive substances or mixtures are thermally unstable liquid or solid substances or mixtures liable to undergo a strongly exothermic decomposition even without participation of oxygen (air). This definition excludes substances and mixtures classified according to this Part as explosives, organic peroxides or as oxidising.

2.8.1.2. A self-reactive substance or mixture is regarded as possessing explosive properties when in laboratory testing the formulation is liable to detonate, to deflagrate rapidly or to show a violent effect when heated under confinement.

General considerations

Annex I: 2.8.3. Hazard communication

Type G has no hazard communication elements assigned but shall be considered for properties belonging to other hazard classes.

2.8.3. Relation to other physical hazards

Neither the burning properties nor the sensitivity to impact and friction form part of the classification procedure for self-reactive substances and mixtures in CLP. These properties may be of importance in safe handling of self-reactive substances and mixtures (see additional tests in Section 2.8.4.3.2).

In addition, the following should be noted:

Explosive properties

The explosive properties do not have to be determined according to the CLP Annex I, Chapter 2.1, because explosive properties are incorporated in the decision logic for self-reactive substances and mixtures. Note that substances and mixtures may have explosive properties when handled under higher confinement.

2.8.4. Classification of substances and mixtures as self-reactive

2.8.4.1. Identification of hazard information

The classification of a self-reactive substance or mixture in one of the seven categories 'types A to G' is dependent on its detonation, deflagration and thermal explosion properties, its response to heating under confinement, its explosive power and the concentration and the type of diluent added to desensitize the substance or mixture. Specifications of acceptable diluents that can be used safely are given in the UN RTDG Model Regulations, Section 2.4.2.3.5.

The classification of a self-reactive substance or mixture as type A, B or C is also dependent on the type of packaging in which the substance or mixture is tested as it affects the degree of confinement to which the substance or mixture is subjected. This has to be considered when handling the substance or mixture; stronger packaging may result in more violent reactions when the substance or mixture decomposes. This is why it is important that storage and transport is done in packaging, allowed for the type of self-reactive substance and mixture, that conforms the requirements of the UN-packaging or IBC instruction (P520/IBC520) or tank instruction (T23).

The traditional aspects of explosive properties, such as detonation, deflagration and thermal explosion, are incorporated in the decision logic Figure 2.8.1 of CLP (see Section 2.8.4.4). Consequently, the determination of explosive properties as prescribed in the hazard class explosives needs not to be conducted for self-reactive substances and mixtures.

2.8.4.2. Classification criteria

According to CLP, substances and mixtures must be considered for classification in this hazard class as a self-reactive substance or mixture unless:

Annex I: 2.8.2.1. [...]

(a) they are explosives, according to the criteria given in 2.1;

(b) they are oxidising liquids or solids, according to the criteria given in 2.13 or 2.14, except that mixtures of oxidising substances, which contain 5 % or more of combustible organic substances shall be classified as self-reactive substances according to the procedure defined in 2.8.2.2;

(c) they are organic peroxides, according to the criteria given in 2.15;

(d) their heat of decomposition is less than 300 J/g; or

(e) their self-accelerating decomposition temperature (SADT) is greater than 75 °C for a 50 kg package (See UN RTDG, Manual of Test and Criteria, sub-sections 28.1, 28.2, 28.3 and Table 28.3.)

2.8.2.2. Mixtures of oxidising substances, meeting the criteria for classification as oxidising substances, which contain 5 % or more of combustible organic substances and which do not meet the criteria mentioned in (a), (c), (d) or (e) in 2.8.2.1, shall be subjected to the self-reactive substances classification procedure;

Such a mixture showing the properties of a self-reactive substance type B to F (see 2.8.2.3) shall be classified as a self-reactive substance.

[...]

1 In addition to the above, substances and mixtures must be considered for classification in this
2 hazard class unless:

Annex I: 2.8.4.2.

[...]

(a) There are no chemical groups present in the molecule associated with explosive or self-reactive properties; examples of such groups are given in Tables A6.1 and A6.2 in Appendix 6 of the UN RTDG, Manual of Tests and Criteria.

[...]

3 In the CLP decision logic (see Section 2.8.4.4), classification of self-reactive substances or
4 mixtures is based on performance based testing in both small scale tests and, where necessary,
5 some larger scale tests with the substance or mixture in its packaging. The concept of 'intrinsic
6 properties' is, therefore, not necessarily, applicable to this hazard class.

7 Self-reactive substances or mixtures are classified in one of the seven categories of 'types A to
8 G' according to the classification criteria given in Section 2.8.2.3 of Annex I, CLP. The
9 classification principles are given in the decision logic in Figure 2.8.1 of CLP (see Section
10 2.8.4.4) and the Test Series A to H, as described in the Part II of the UN-MTC, should be
11 performed.

Annex I: 2.8.2.3. Self-reactive substances and mixtures shall be classified in one of the seven categories of 'types A to G' for this class, according to the following principles:

- (a) any self-reactive substance or mixture which can detonate or deflagrate rapidly, as packaged, shall be defined as self-reactive substance TYPE A;
- (b) any self-reactive substance or mixture possessing explosive properties and which, as packaged, neither detonates nor deflagrates rapidly, but is liable to undergo a thermal explosion in that package shall be defined as self-reactive substance TYPE B;
- (c) any self-reactive substance or mixture possessing explosive properties when the substance or mixture as packaged cannot detonate or deflagrate rapidly or undergo a thermal explosion shall be defined as self-reactive substance TYPE C;
- (d) any self-reactive substance or mixture which in laboratory testing:
 - (i) detonates partially, does not deflagrate rapidly and shows no violent effect when heated under confinement; or
 - (ii) does not detonate at all, deflagrates slowly and shows no violent effect when heated under confinement; or
 - (iii) does not detonate or deflagrate at all and shows a medium effect when heated under confinement;
 shall be defined as self-reactive substance TYPE D;
- (e) any self-reactive substance or mixture which, in laboratory testing, neither detonates nor deflagrates at all and shows low or no effect when heated under confinement shall be defined as self-reactive substance TYPE E;
- (f) any self-reactive substance or mixture which, in laboratory testing, neither detonates in the cavitated state nor deflagrates at all and shows only a low or no effect when heated under confinement as well as low or no explosive power shall be defined as self-reactive substance TYPE F;
- (g) any self-reactive substance or mixture which, in laboratory testing, neither detonates in the cavitated state nor deflagrates at all and shows no effect when heated under confinement nor any explosive power, provided that it is thermally stable (SADT is 60 °C to 75 °C for a 50 kg package), and, for liquid mixtures, a diluent having a boiling point not less than 150 °C is used for desensitisation shall be defined as self-reactive substance TYPE G. If the mixture is not thermally stable or a diluent having a boiling point less than 150 °C is used for desensitisation, the mixture shall be defined as self-reactive substance TYPE F.

Where the test is conducted in the package form and the packaging is changed, a further test shall be conducted where it is considered that the change in packaging will affect the outcome of the test.

- 1 A list of currently classified self-reactive substances and mixtures is included in the UN RTDG
- 2 Model Regulations, Section 2.4.2.3.2.3.

3 **2.8.4.3. Testing and evaluation of hazard information**

4 **2.8.4.3.1. Thermal stability tests and temperature control**

- 5 In addition to the classification tests given in decision logic Figure 2.8.1 of CLP, the thermal
- 6 stability of the self-reactive substances and mixtures has to be assessed in order to determine
- 7 the SADT.

The SADT is defined as the lowest temperature at which self-accelerating decomposition of a substance or mixture may occur in the packaging as used in transport, handling and storage. The SADT is a measure of the combined effect of the ambient temperature, decomposition kinetics, package size and the heat transfer properties of the substance or mixture and its packaging.

There is no relation between the SADT of a self-reactive substance and mixture and its classification in one of the seven categories 'types A to G'. The SADT is used to derive safe handling, storage and transport temperatures (control temperature) and alarm temperature (emergency temperature).

Depending on its SADT a self-reactive substance and mixture needs temperature control and the rules as given in CLP Annex I, 2.8.2.4, consist of the following two elements:

1. Criteria for temperature control:

2. Self-reactive substances and mixtures need to be subjected to temperature control when the SADT is ≤ 55 °C.

3. Derivation of control and emergency temperatures:

Type of receptacle	SADT*	Control temperature	Emergency temperature
Single packagings and IBC's	20 °C or less	20 °C below SADT	10 °C below SADT
	over 20 °C to 35 °C	15 °C below SADT	10 °C below SADT
	over 35 °C	10 °C below SADT	5 °C below SADT
Tanks	< 50 °C	10 °C below SADT	5 °C below SADT

*i.e. the SADT of the substance/mixture as packaged for transport, handling and storage.

It should be emphasized that the SADT is dependent on the nature of the self-reactive substance or mixture itself, together with the volume and heat-loss characteristics of the packaging or vessel in which the substance or mixture is handled. The temperature at which self-accelerating decomposition occurs falls:

- as the size of the packaging or vessel increases; and
- with increasing efficiency of the insulation on the package or vessel.

The SADT is only valid for the substance or mixture as tested and when handled properly. Mixing the self-reactive substances and mixtures with other chemicals, or contact with incompatible materials (including incompatible packaging or vessel material) may reduce the thermal stability due to catalytic decomposition, and lower the SADT. This may increase the risk of decomposition and has to be avoided.

2.8.4.3.2. Additional considerations and testing

Explosive properties

The sensitivity of self-reactive substances and mixtures to impact (solids and liquids) and friction (solids only) may be of importance for the safe handling of the substances and mixtures, in the event that these substances and mixtures have pronounced explosive properties (e.g. rapid deflagration and/or violent heating under confinement). Test methods to determine these properties are described in Test Series 3 (a) (ii) and 3 (b) (i) of the UN-MTC. This information should be documented in the SDS.

Burning properties

Although there are currently no dedicated storage guidelines for self-reactive substances and mixtures (although in some countries under development), often the regulations for organic

peroxides are referred to. For storage classification the burning rate is commonly used, see Section 2.15 on organic peroxides.

Flash point

The flash point for liquid self-reactive substances or mixtures is only relevant in the temperature range where the product is thermally stable. Above the SADT of the self-reactive substance or mixture, flash point determination is not relevant because decomposition products are evolved.



NOTE: In case a flash point determination seems reasonable (expected flash point below the SADT) a test method using small amount of sample is recommended. In case the self-reactive substance or mixture is diluted or dissolved, the diluent may determine the flash point.

Auto-ignition temperature

The determination of the auto ignition temperature is not relevant for self-reactive substances and mixtures, because the vapours decompose during the execution of the test. Available test methods are for non-decomposing vapour phases. Auto ignition of self-reactive substance and mixtures vapours when they decompose, can never be excluded. This information should be documented in the SDS.

Self-ignition temperature

Also self-ignition temperature determination (test applicable for solids) is not relevant. The thermal stability of self-reactive substances and mixtures is quantitatively given by the SADT test.

Control and Emergency temperatures

The Control and Emergency temperatures are based on the SADT as determined by UN Test H.4. The Dewar vessel used in the UN Test H.4 is supposed to be representative for the substance or mixture handled in packages. For handling of the substance or mixture in larger quantities (IBCs/tanks/vessels etc.) and/or in better (thermally) insulated containers under more thermal insulated conditions, the SADT has to be determined for that quantity with the given degree of insulation. From that SADT the Control and Emergency temperatures can be derived (see also Section 0).

2.8.4.3.3. Additional classification considerations

Currently, the following properties are not incorporated in the classification of self-reactives under the CLP:

- mechanical sensitivity i.e. impact and friction sensitivity (for handling purposes);
- burning properties (for storage purposes);
- flash point for liquids; and
- burning rate for solids.

In addition to the GHS criteria CLP mentions that:

Annex I: 2.8.2.2

[...]

Where the test is conducted in the package form and the packaging is changed, a further test shall be conducted where it is considered that the change in packaging will affect the outcome of the test.

1 Please note that polymerising substances do not fulfil the criteria for classification as self-
2 reactives. However, there are on-going discussions at the UNSCEGHS on this subject.

3 **2.8.4.4. Decision logic**

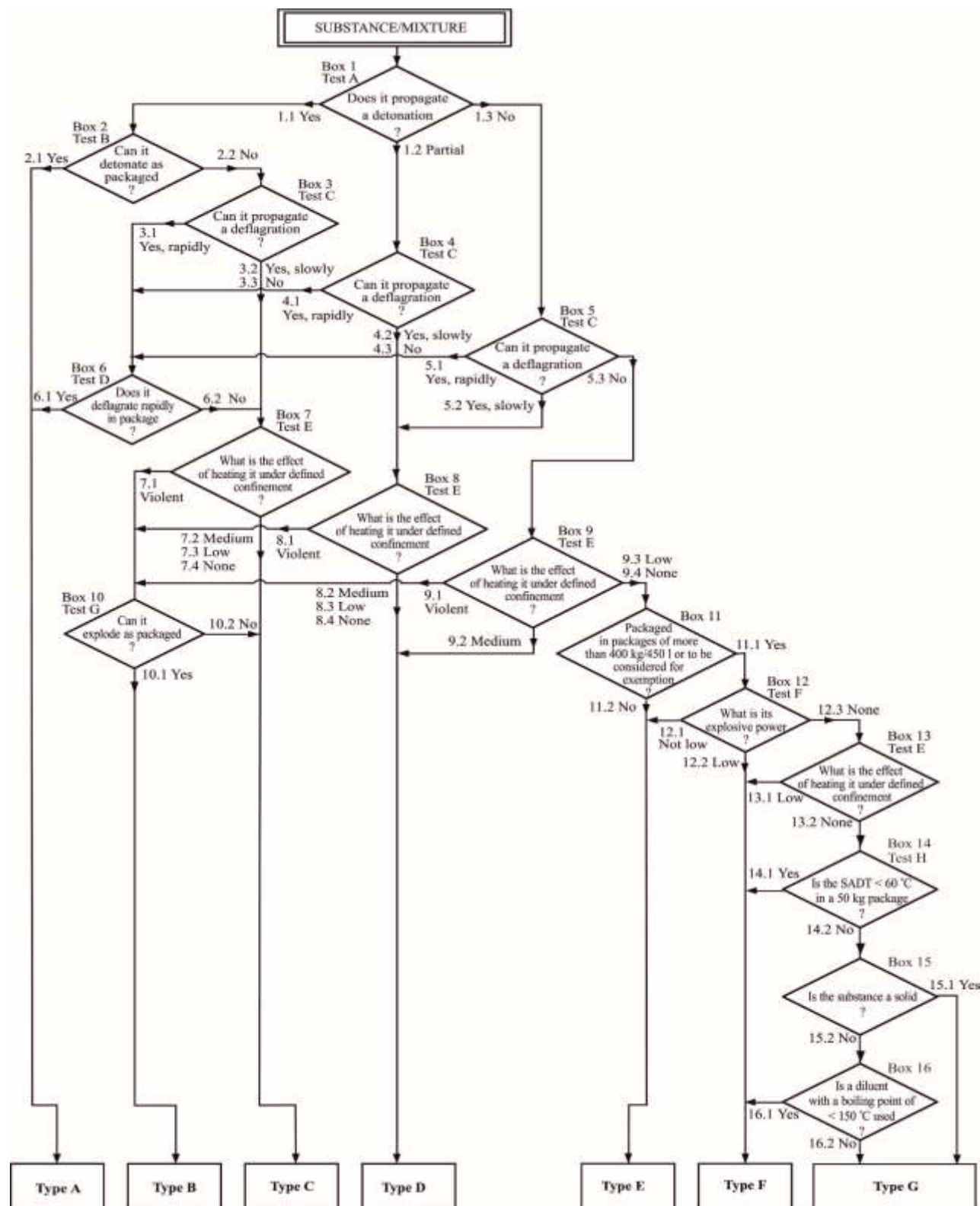
4 Classification of self-reactive substances and mixtures is done according to decision logic 2.8 as
5 included in the GHS.



NOTE: The person responsible for the classification of self-reactive substances and mixtures
should be experienced in this field and be familiar with the criteria for classification.

6

Figure 2.8.4–1 Decision logic 2.8 for self-reactive substances and mixtures



2.8.5. Hazard communication for self-reactives

2.8.5.1. Pictograms, signal words, hazard statements and precautionary statements

According to CLP the following label elements must be used for substances and mixtures meeting the criteria for this hazard class:

Annex I: Table 2.8.1

Label elements for self-reactive substances and mixtures

Classification	Type A	Type B	Type C & D	Type E & F	Type G ²
GHS pictograms					There are no label elements allocated to this hazard category
Signal Word	Danger	Danger	Danger	Warning	
Hazard Statement	H240: Heating may cause an explosion	H241: Heating may cause a fire or explosion	H242: Heating may cause a fire	H242: Heating may cause a fire	
Precautionary statement Prevention	P210 P234 P235 P240 P280	P210 P234 P235 P240 P280	P210 P234 P235 P240 P280	P210 P234 P235 P240 P280	
Precautionary statement Response	P370 + P372 + P380 + P373	P370 + P380 + P375 [+P378] ¹	P370 + P378	P370 + P378	
Precautionary statement Storage	P403 P411 P420	P403 P411 P420	P403 P411 P420	P403 P411 P420	
Precautionary statement Disposal	P501	P501	P501	P501	

¹ See the introduction to Annex IV for details on the use of square brackets.

² Type G has no hazard communication elements assigned but should be considered for properties belonging to other hazard classes.

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.8.6. Relation to transport classification according to DSD and DPD or already classified for transport

Division 4.1 within Class 4 of the UN RTDG Model Regulations covers flammable substances, solid desensitized explosives and self-reactive liquids or solids. A list of already classified self-reactive substances is included in UN RTDG Model Regulations, Section 2.4.2.3.2.3. This table includes self-reactive substances of various types from type B to type F. See Annex VII for additional information on transport classification in relation to CLP classification.

2.8.7. Examples of classification for self-reactives

2.8.7.1. Examples of substances and mixtures fulfilling the classification criteria

Substance to be classified: NP

Molecular formula: n.a.

According to CLP Annex I, Section 2.8.2.1, the substance has:

- an energy content of 1452 kJ/kg; and
- a SADT of 45 °C (in 50 kg package);

and consequently it has to be considered for classification in the hazard class self-reactive substances and mixtures.

Test results and classification according to CLP decision logic 2.8.1 for self-reactive substances and mixtures and the UN - MTC, Part II, is as follows:

CLASSIFICATION TEST RESULTS

1. Name of the self-reactive substance or mixture:	NP
2. General data	
2.1. Composition	NP, technically pure
2.2. Molecular formula	n.a.
2.3. Physical form	solid, fine powder
2.4. Colour	brown
2.5. Density (apparent)	460 kg/m ³
3. Detonation (test series A)	
Box 1 of the decision logic	Does the substance propagate a detonation?
3.1. Method	UN Test A.1: BAM 50/60 steel tube test
3.2. Sample conditions	technically pure substance
3.3. Observations	fragmented part of the tube: 12, 18cm

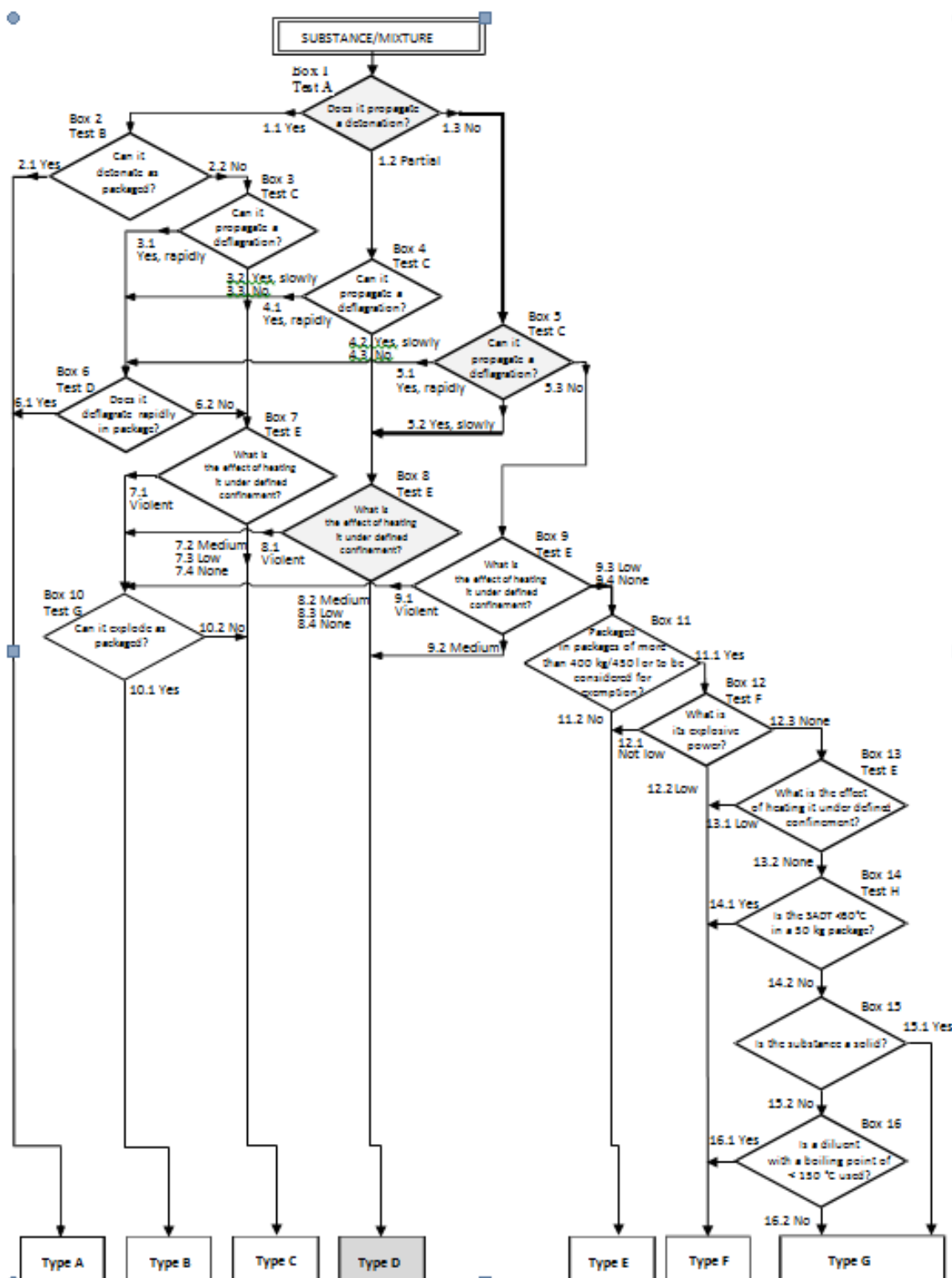
CLASSIFICATION TEST RESULTS	
3.4. Result	No
3.5. Exit	1.3
4. Deflagration (test series C)	
Box 5 of the decision logic	Does the substance propagate a deflagration?
4.1. Method 1	Time/pressure test (test C.1)
4.1.1. Sample conditions	ambient temperature
4.1.2. Observations	498, 966, 3395 ms
4.1.3. Result	Yes, slowly
4.2. Method 2	Deflagration test (test C.2)
4.2.1. Sample conditions	temperature: 20 °C
4.2.2. Observations	deflagration rate: 0.90, 0.87 mm/s
4.2.3. Result	Yes, slowly
4.3. Final result	Yes, slowly
4.4. Exit	5.2
5. Heating under confinement (test series E)	
Box 8 of the decision logic:	What is the effect of heating it under defined confinement?
5.1. Method 1	Koenen test (test E.1)
5.1.1. Sample conditions	
5.1.2. Observations	Limiting diameter: < 1.0 mm fragmentation type 'A'
5.1.3. Result	Low
5.2. Method 2	Dutch pressure vessel test (test E.2)
5.2.1. Sample conditions	
5.2.2. Observations	Limiting diameter: <1.0 mm (with 10 g), 1.0 mm (50 g)
5.2.3. Result	low
5.3. Final result	low
5.4. Exit	8.3
6. Thermal stability (outside of the decision logic)	
6.1. Method	Heat accumulation storage test (test H.4)
6.2. Sample conditions :	mass 232.5 g. Half life time of cooling of Dewar vessel with 400 ml water: 10.0 hrs.(representing substance in package)
6.3. Observations	self-accelerating decomposition at 45 °C no self-accelerating decomposition at 40 °C

CLASSIFICATION TEST RESULTS	
6.4. Result	SADT 45 °C (in 50 kg package)
7. General remarks	The decision logic is given in Figure 2.8.7—1
8. Final classification	
Hazard / hazard class:	Self-reactive substance, Type D, solid, temperature controlled
Label	Flame (GHS02)
Signal word	Danger
Hazard statement	H242: Heating may cause a fire
Temperature control	Needed based on SADT (45 °C, in package)
Control temperature*	35 °C (in package)
Emergency temperature*	40 °C (in package)

*See UN-MTC, table 28.2.

1
2

1 **Figure 2.8.7—1 Decision logic for self-reactive substance example: NP, technically pure**



2.9. PYROPHORIC LIQUIDS

2.9.1. Introduction

The criteria for 'Pyrophoric liquids' are found in Annex I, Section 2.9 of CLP and are identical to those in Chapter 2.9 of GHS.

Pyrophoricity, i.e. the ability to spontaneously ignite in air, is the result of a reaction of a substance or mixture with the oxygen in the air. The reaction is exothermic and has the particularity that it starts spontaneously, i.e. without the aid of a supplied spark, flame, heat or other energy source. Another way of saying this is that the auto-ignition temperature for a pyrophoric substance or mixture is lower than room (ambient) temperature.

Organo-metals and organo-metalloids may be suspected of being pyrophores, as well as their derivatives. Also organo-phosphines and their derivatives, hydrides and their derivatives and haloacetylene derivatives may show pyrophoricity (Urben, 2007).

There are also pyrophoric substances or mixtures that do not belong to the above mentioned groups of chemicals, i.e. the list above is not exhaustive. Since pyrophoric substances or mixtures ignite *spontaneously* in air, pyrophoricity is a very dangerous property. In case of doubt it should therefore be thoroughly investigated whether a given substance or mixture is pyrophoric. More information on pyrophoric substances can e.g. be found in *Bretherick's Handbook of Reactive Chemical Hazards* (Urben, 2007).

2.9.2. Definitions and general considerations for the classification pyrophoric liquids

The definition in CLP for pyrophoric liquids is as follows:

Annex I: 2.9.1. Definition

Pyrophoric liquid means a liquid substance or mixture which, even in small quantities, is liable to ignite within five minutes after coming into contact with air.

2.9.3. Relation to other physical hazards

Pyrophoric substances and mixtures will react spontaneously with air already in small amounts and more or less instantaneously (within minutes). This differentiates them from self-heating substances and mixtures, which also react spontaneously with air but only when in larger amounts and after an extended period of time (hours or days). While liquids in themselves generally do not exhibit self-heating properties due to the limited contact with air (which can occur only at the surface), liquids that are adsorbed onto solid particles should, in general, be considered for classification in the hazard class self-heating substances and mixtures, see Chapter 2.11 of this guidance.

Pyrophoricity may be expected for certain reactive metals and some of their compounds (e.g. hydrides and other organo-metal compounds). Many of these substances and mixtures will also react vigorously with water under the production of flammable gases. Such substances and mixtures may thus be classified in the hazard class substances and mixtures which in contact with water emit flammable gases in addition, see Chapter 2.12 of this guidance. It should be noted in this context that water-reactive substances and mixtures may also to some extent react with the humidity in air, although such a reaction is seldom vigorous. A substance or mixture that spontaneously ignites in air in accordance with the test procedures is to be considered pyrophoric, regardless of the reaction mechanism.

Liquids not classified as pyrophoric but that can burn may belong to the hazard class flammable liquids depending on their flash point and ability to sustain combustion, see Section 2.6 of this guidance.

2.9.4. Classification of substances and mixtures as pyrophoric liquids

2.9.4.1. Identification of hazard information

Since the tests to determine pyrophoricity are simple and require no special equipment, see Section 2.9.4.4 below, there is in general no reason to go to data sources instead of performing tests. Furthermore, the possibilities of waiving tests are ample both for known pyrophores and for substances and mixtures known not to be pyrophoric, see Section 2.9.4.2 below. If information anyway is taken from literature or other data sources, it is of utmost importance that the correct physical form is considered, see Section 2.0.4. Naturally, all data sources should be carefully evaluated with regard to reliability and scientific validity.

2.9.4.2. Screening procedures and waiving of testing

In case a liquid is known from practical handling to be pyrophoric no testing is necessary. Such liquids are classified as pyrophoric liquids without testing. This would also be the case if the liquid spontaneously ignites upon opening of the receptacle when trying to perform the tests for classification.

According to the additional classification considerations in CLP Annex I, 2.9.4, the classification procedure for pyrophoric liquids need not be applied when experience in manufacture or handling shows that the liquid does not ignite spontaneously on coming into contact with air at normal temperatures (i.e. the liquid is known to be stable at room temperature for prolonged periods of time (days)).

2.9.4.3. Classification criteria

Section 2.9.2.1 of Annex I of CLP specifies the classification criteria:

Annex I: Table 2.9.1

Criteria for pyrophoric liquids

Category	Criteria
1	The liquid ignites within 5 min when added to an inert carrier and exposed to air, or it ignites or chars a filter paper on contact with air within 5 min.

2.9.4.4. Testing and evaluation of hazard information

In Section 2.9.2.1 of Annex I of CLP reference to the test-methods are made:

Annex I: 2.9.2.1. A pyrophoric liquid shall be classified in a single category for this class using test N.3 in part III, sub-section 33.3.1.5 of the UN RTDG, Manual of Tests and Criteria according to Table 2.9.1:

The UN Test N.3 for pyrophoricity is quite simple and sufficiently described in Part III, Section 33 of the UN-MTC. No special equipment is needed. Essentially the substance or mixture is exposed to air to see if it ignites. For liquids which do not spontaneously ignite when poured, the surface in contact with air is increased using a filter paper. Ignition or charring of the filter paper is regarded as a positive response in the test, i.e. such a liquid is considered to be pyrophoric.

1 It is important that samples for testing of pyrophoric properties are carefully packed and sealed.
2 Furthermore, the material offered for testing should be freshly prepared, since the reactive
3 properties may diminish due to aging or agglomeration. Whenever experiments are to be done
4 one should be careful – a pyrophoric substance or mixture may well ignite already upon opening
5 the receptacle!

6 It should be noted that the mechanism of oxidation is, in general, very complex, and that the
7 humidity of air might influence the rate of reaction. Therefore a false negative may result when
8 performing the tests in an extremely dry environment, and this condition must be avoided when
9 performing the tests for classification for pyrophoricity. The filter paper test of UN Test N.3 for
10 pyrophoric liquids should be carried out at 25 ± 2 °C and a relative humidity of 50 ± 5 % (see
11 UN-MTC, Section 33.3.1.5).

12 **2.9.4.5. Decision logic**

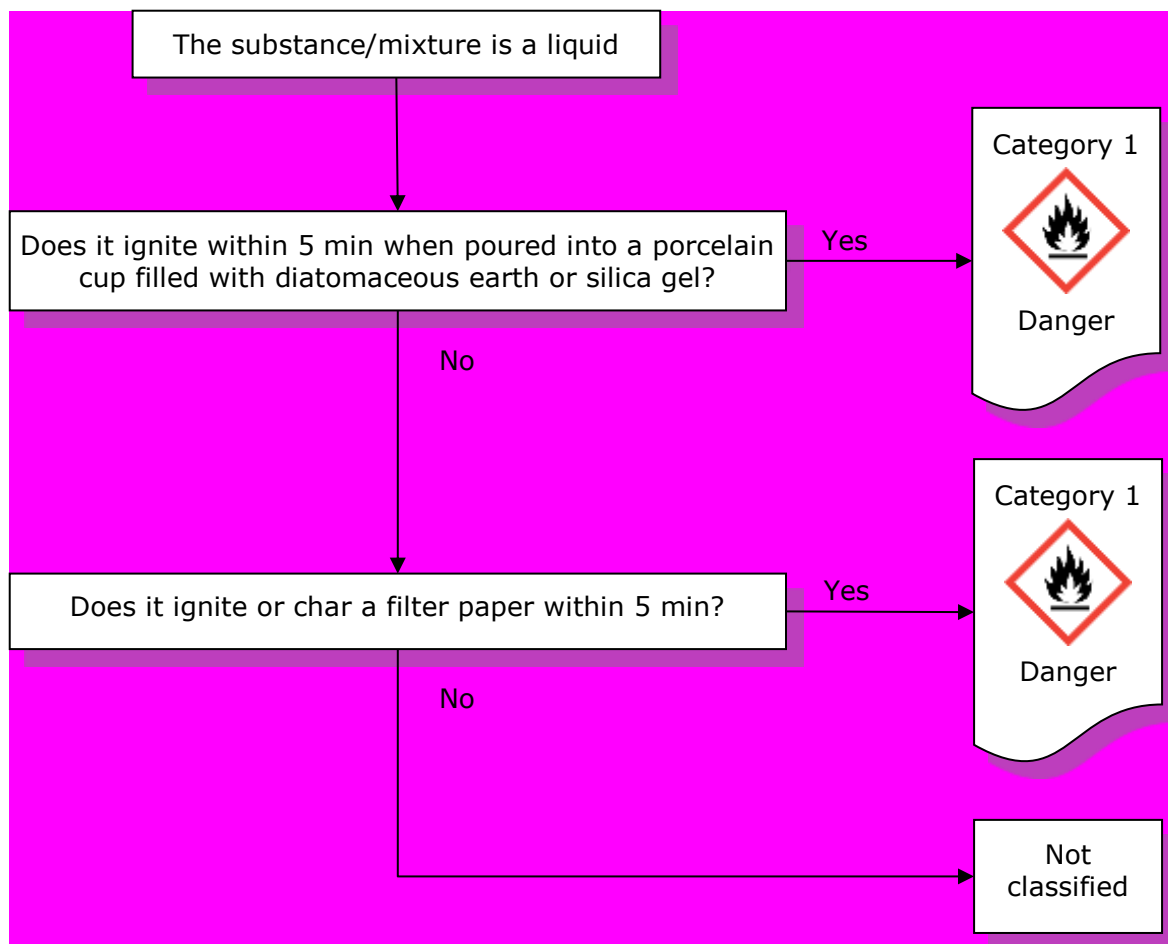
13 Classification of pyrophoric liquids is done according to decision logic 2.9.4.1 as included in the
14 GHS.



NOTE: The person responsible for the classification of pyrophoric liquids should be experienced in this field and be familiar with the criteria for classification.

2.9.4.5.1. Decision logic for pyrophoric liquids

Figure 2.9.4—1 Decision logic for pyrophoric liquids (Decision logic 2.9 of GHS)



2.9.5. Hazard communication for pyrophoric liquids

2.9.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: 2.9.3 Table 2.9.2 Label elements for pyrophoric liquids	
Classification	Category 1
GHS Pictogram	
Signal Word	Danger
Hazard Statement	H250: Catches fire spontaneously if exposed to air
Precautionary Statement Prevention	P210 P222 P231 + P232 P233 P280
Precautionary Statement Response	P302 + P334 P370 + P378
Precautionary Statement Storage	
Precautionary Statement Disposal	

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.9.6. Relation to transport classification

Division 4.2 within Class 4 of the UN RTDG Model Regulations covers pyrophoric solids, liquids and self-heating substances and mixtures. UN Test N.3 that is used for classification for pyrophoricity for liquids according to CLP is also used for classification in the subdivision pyrophoric substances and mixtures in Division 4.2: Substances liable to spontaneous combustion according to the UN RTDG Model Regulations. The criteria for Category 1 according to CLP (which is the only category for pyrophoric liquids) and for packing group I in Division 4.2 according to the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) are also exactly the same. Furthermore, all pyrophoric substances and mixtures are assigned to packing group I within Division 4.2, which is used exclusively for pyrophoric substances and mixtures.

Therefore, any liquid assigned to Division 4.2, packing group I according to the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) will be classified in Category 1 of the hazard class pyrophoric liquids according to CLP. See Annex VII for additional information on transport classification in relation to CLP classification.

2.9.7. Examples of classification for pyrophoric liquids

Please note that the substance and mixture names in this chapter are fictitious.

2.9.7.1. Examples of substances and mixtures fulfilling the classification criteria

2.9.7.1.1. Example 1

Name:	Pyrpherdine
Physical state:	Liquid
Pyrophoric properties:	Unknown, therefore the UN Test N.3 of the UN-MTC was applied. However, when opening the receptacle in order to perform the test, Pyrpherdine self-ignited.
Classification:	Pyrophoric liquid, Category 1

2.9.7.1.2. Example 2

Name:	Qulipyr
Physical state:	Liquid
Pyrophoric properties:	Unknown, therefore the UN Test N.3 of the UN-MTC was applied.
Test result:	When poured according to the test procedure, nothing happened. The procedure was repeated six times, each time giving a negative result (i.e. no ignition). Therefore Qulipyr was supplied to a filter paper in accordance with the test method. In the second trial the filter paper was charred within five minutes.
Classification:	Pyrophoric liquid, Category 1

2.9.7.2. Examples of substances and mixtures not fulfilling the classification criteria

2.9.7.2.1. Example 3

Name:	Notpyratal
Physical state:	Liquid
Pyrophoric properties:	Unknown, therefore UN Test N.3 of the UN-MTC was applied.
Test result:	When poured according to the test procedure nothing happened in either of six trials. Therefore Notpyratal was supplied to a filter paper in accordance with the test method, whereupon no ignition or charring occurred in either of three trials.
Classification:	Not a pyrophoric liquid

2.9.8. References

Urban, Peter G. (2007). *Bretherick's Handbook of Reactive Chemical Hazards*, Volumes 1-2 (7th Edition). Elsevier.

2.10. PYROPHORIC SOLIDS

2.10.1. Introduction

The criteria for 'Pyrophoric solids' are found in Annex I, Section 2.10 of CLP and are identical to those in Chapter 2.10 of GHS.

Pyrophoricity, i.e. the ability to spontaneously ignite in air, is the result of a reaction of a substance or mixture with the oxygen in the air. The reaction is exothermic and has the particularity that it starts spontaneously, i.e. without the aid of a supplied spark, flame, heat or other energy source. Another way of saying this is that the self-ignition temperature for a pyrophoric substance or mixture is lower than room (ambient) temperature.

Organo-metals and organo-metalloids may be suspected of being pyrophores, as well as their derivatives. Also organo-phosphines and their derivatives, hydrides and their derivatives, haloacetylene derivatives, and complex acetylides may show pyrophoricity (Urban, 2007). Furthermore, powders or fine particles of metals could be pyrophoric. However, although many solid metallic substances, like e.g. aluminium, would be suspected of being pyrophoric when considering their general reactivity, they form a protective oxide-coat upon reaction with air. This thin coat of metal oxide prevents the metal from reacting further, and hence such substances may not show pyrophoric behaviour in reality.

There are also pyrophoric solids that do not belong to the above mentioned groups of chemicals, i.e. the list above is not exhaustive. Since pyrophoric solids ignite *spontaneously* in air, pyrophoricity is a very dangerous property. In case of doubt it should therefore be thoroughly investigated whether a given solid is pyrophoric. More information on pyrophoric solids can e.g. be found in *Bretherick's Handbook of Reactive Chemical Hazards* (Urban, 2007).

2.10.2. Definitions and general considerations for the classification pyrophoric solids

The definition in CLP for pyrophoric solids is as follows:

Annex I: 2.10.1. Definition

Pyrophoric solid means a solid substance or mixture which, even in small quantities, is liable to ignite within five minutes after coming into contact with air.

Special consideration on particle size

Annex I: 2.10.2.1.

[...]

Note: The test shall be performed on the substance or mixture in its physical form as presented. If for example, for the purposes of supply or transport, the same chemical is to be presented in a physical form different from that which was tested and which is considered likely to materially alter its performance in a classification test, the substance shall also be tested in the new form.

The finer the particle size of a solid, the greater the area exposed to air will be, and since pyrophoricity is a reaction with the oxygen in air, the particle size will greatly influence the ability to spontaneously ignite. Hence it is very important that pyrophoric properties for solids are investigated on the substance or mixture as it is actually presented (including how it can reasonably be expected to be used, see Article 8 (6) of CLP). This is indicated by the Note cited in CLP Annex I, 2.10.2.1.

2.10.3. Relation to other physical hazards

Pyrophoric solids will react spontaneously with air already in small amounts and more or less instantaneously (within minutes). This differentiates them from self-heating substances and mixtures, which also react spontaneously with air but only when in larger amounts and after an extended period of time (hours or days). A solid which is not classified as a pyrophoric solid may thus belong to the hazard class self-heating substances and mixtures, and should be considered for classification in that hazard class, see Chapter 2.11 of this guidance.

Pyrophoricity may be expected for certain reactive metals and some of their compounds (e.g. hydrides and other organo-metal compounds). Many of these substances will also react vigorously with water under the production of flammable gases. Such substances may thus be classified in the hazard class substances and mixtures which in contact with water emit flammable gases in addition see Chapter 2.12 of this guidance. It should be noted in this context that water-reactive substances or mixtures may also to some extent react with the humidity in air, although such a reaction is seldom vigorous. A substance that spontaneously ignites in air in accordance with the test procedures is to be considered pyrophoric, regardless of the reaction mechanism.

Solids not classified as pyrophoric may still be able to burn rapidly if subjected to enough initiating energy, such as the flame from a gas burner, to start the reaction. Therefore they may be subject to classification in the hazard class flammable solids, see Chapter 2.7 of this guidance, i.e. they may be 'readily combustible solids'.

2.10.4. Classification of substances and mixtures as pyrophoric solids

2.10.4.1. Identification of hazard information

Since the tests to determine pyrophoricity are simple and require no special equipment, see Section 2.10.4.4 below, there is in general no reason to go to data sources instead of performing tests. Furthermore, the possibilities of waiving tests are ample both for known pyrophores and for substances and mixtures known not to be pyrophoric, see Section 2.10.4.2 below. If information is taken from literature or other data sources anyway, it is of utmost importance that the correct physical form is considered, see Section 2.0.4. Naturally, all data sources should be carefully evaluated with regard to reliability and scientific validity.

2.10.4.2. Screening procedures and waiving of testing

In case a solid is known from practical handling to be pyrophoric no testing is necessary. Such solids are classified as pyrophoric solids without testing. This would also be the case if the solid spontaneously ignites upon opening of the receptacle when trying to perform the tests for classification.

According to the additional classification considerations in CLP Annex I, 2.10.4, the classification procedure for pyrophoric solids need not be applied when experience in manufacture or handling shows that the substance or mixture does not ignite spontaneously on coming into contact with air at normal temperatures (i.e. the substance or mixture is known to be stable at room temperature for prolonged periods of time (days)).

2.10.4.3. Classification criteria

Section 2.10.2.1 of Annex I of CLP specifies the classification criteria:

Annex I: Table 2.10.1

Criteria for pyrophoric solids

Category	Criteria
1	The solid ignites within 5 minutes of coming into contact with air.

2.10.4.4. Testing and evaluation of hazard information

In Section 2.10.2.1 of Annex I of CLP reference to the test-methods are made:

Annex I: 2.10.2.1. A pyrophoric solid shall be classified in a single category for this class using test N.2 in part III, sub-section 33.3.1.4 of the UN RTDG, Manual of Tests and Criteria in accordance with Table 2.10.1:

UN Test N.2 for pyrophoricity is quite simple and sufficiently described in Part III, Section 33 of the UN-MTC. No special equipment is needed. Essentially the solid is exposed to air to see if it ignites.

It is important that samples for testing of pyrophoric properties are carefully packed and sealed. Furthermore, the material offered for testing should be freshly prepared, since the reactive properties may diminish due to aging or agglomeration. Whenever experiments are to be done one should be careful – a pyrophoric solid may well ignite already upon opening the receptacle!

It should be noted that the mechanism of oxidation is, in general, very complex, and that the humidity of air might influence the rate of reaction. It is known that certain metals will not react in dry air, whereas in the presence of moisture the reaction is almost instantaneous (often even trace amounts of moisture are sufficient). Therefore a false negative may result when performing the tests in an extremely dry environment, and this condition must be avoided when performing the tests for classification for pyrophoricity.

2.10.4.5. Decision logic

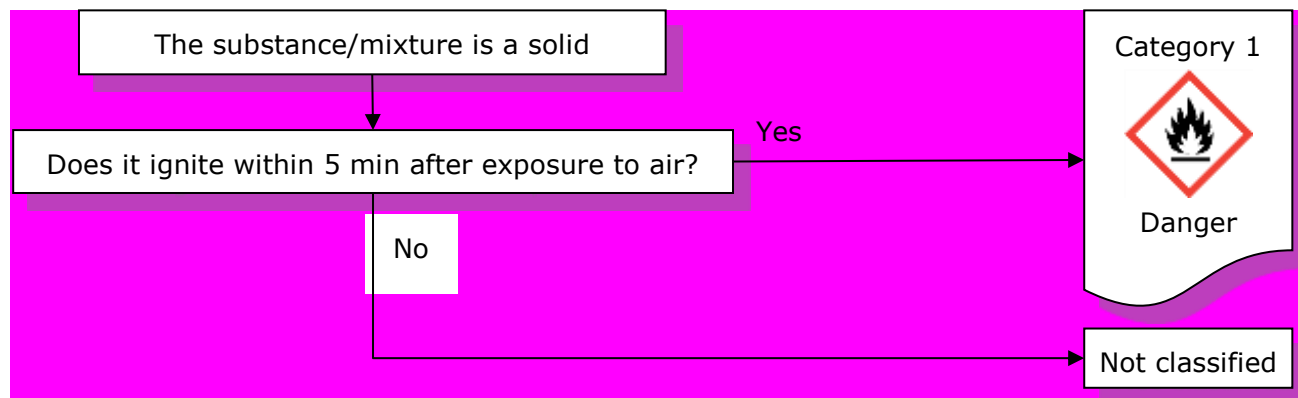
Classification of pyrophoric solids is done according to decision logic 2.10.4.1 as included in the GHS.



NOTE: The person responsible for the classification of pyrophoric solids should be experienced in this field and be familiar with the criteria for classification.

2.10.4.5.1. Decision logic for pyrophoric solids

Figure 2.10.4–1 Decision logic for pyrophoric solids (Decision logic 2.10 of GHS)



2.10.5. Hazard communication for pyrophoric solids

2.10.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: 2.10.3 Table 2.10.2 Label elements for pyrophoric solids	
Classification	Category 1
GHS Pictogram	
Signal Word	Danger
Hazard Statement	H250: Catches fire spontaneously if exposed to air
Precautionary Statement Prevention	P210 P222 P231 + P232 P233 P280
Precautionary Statement Response	P302 + P335 + P334 P370 + P378
Precautionary Statement Storage	
Precautionary Statement Disposal	

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.10.6. Relation to transport classification

Division 4.2 within Class 4 of the UN RTDG Model Regulations covers pyrophoric solids, liquids and self-heating substances and mixtures. The UN Tests N.2 that is used for classification for pyrophoricity for solids according to CLP is also used for classification in the subdivision pyrophoric substances and mixtures in Division 4.2: Substances liable to spontaneous combustion according to the UN RTDG Model Regulations. The criteria for Category 1 according to CLP (which is the only category for pyrophoric solids) and for packing group I in Division 4.2 according to the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) are also exactly the same. Furthermore, all pyrophoric substances and mixtures are assigned to packing group I within Division 4.2, which is used exclusively for pyrophoric substances and mixtures.

Therefore, any solid substance or mixture assigned to Division 4.2, packing group I according to the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) will be classified in Category 1 of the hazard class pyrophoric solids according to CLP. See Annex VII for additional information on transport classification in relation to CLP classification.

2.10.7. Examples of classification for pyrophoric solids

Please note that the substance and mixture names in this chapter are fictitious.

2.10.7.1. Examples of substances and mixtures fulfilling the classification criteria

2.10.7.1.1. Example 1

Name:	Pyroferil
Physical state:	Solid
Pyrophoric properties:	Pyroferil is known to self-ignite upon contact with air at ambient conditions.
Classification:	Pyrophoric solid, Category 1

2.10.7.1.2. Example 2

Name:	Zorapyrole
Physical state:	Solid
Pyrophoric properties:	Unknown, therefore the UN Test N.2 of the UN-MTC was applied.
Test result:	When poured from one meter height according to the test procedure, Zorapyrole self-ignited after two minutes already in the first trial.
Classification:	Pyrophoric solid, Category 1

2.10.7.2. Examples of substances and mixtures not fulfilling the classification criteria

2.10.7.2.1. Example 3

Name:	Nonopyr
Physical state:	Solid
Pyrophoric properties:	Nonopyr has been handled extensively in air and has never self-ignited. From the chemical structure no pyrophoricity is expected.
Classification:	Not a pyrophoric solid

1 **2.10.7.2.2. Example 4**

Name:	Pyronot
Physical state:	Solid
Pyrophoric properties:	Unknown, therefore UN Test N.2 of the UN-MTC was applied.
Test result:	When poured from one meter height according to the test procedure no ignition occurred within five minutes. The procedure was repeated six times and each time the result was negative.
Classification:	Not a pyrophoric solid

2 **2.10.8. References**

3 Urban, Peter G. (2007). *Bretherick's Handbook of Reactive Chemical Hazards*, Volumes 1-2 (7th
4 Edition). Elsevier.

2.11. SELF-HEATING SUBSTANCES AND MIXTURES

2.11.1. Introduction

The criteria for 'Self-heating substances and mixtures' are found in Annex I, Section 2.11 of CLP and are identical to those in Chapter 2.11 of GHS.

Self-heating is the result of an exothermic reaction of a substance or mixture with the oxygen in the air. Initially, the reaction rate may be very low. However, when the heat produced cannot be removed rapidly enough (i.e. heat accumulation), the substance or mixture will self-heat, with the possible consequence of self-ignition. The phenomenon can occur only where a large surface of substance or mixture is in contact with air or oxygen (for example, piles of powders, crystals, splinters, any other rough surface etc.). The initiation occurs usually at or near the centre of the substance or mixture pile with the available air in the interspace between the particles.

Since the surface area of a solid substance or mixture exposed to air increases with decreasing particle size, it follows that particle size and shape will greatly influence the propensity of a substance or mixture to self-heat. Therefore it is very important that self-heating properties for solids, and especially powders, are determined for the substance or mixture in the form it is supplied and expected to be used.

2.11.2. Definitions and general considerations for the classification of self-heating substances and mixtures

The definitions in CLP for self-heating substances and mixtures are as follows:

Annex I: 2.11.1.1. A self-heating substance or mixture is a liquid or solid substance or mixture, other than a pyrophoric liquid or solid, which, by reaction with air and without energy supply, is liable to self-heat; this substance or mixture differs from a pyrophoric liquid or solid in that it will ignite only when in large amounts (kilograms) and after long periods of time (hours or days).

2.11.1.2. Self-heating of a substance or a mixture is a process where the gradual reaction of that substance or mixture with oxygen (in the air) generates heat. If the rate of heat production exceeds the rate of heat loss, then the temperature of the substance or mixture will rise which, after an induction time, may lead to self-ignition and combustion.

2.11.3. Relation to other physical hazards

Pyrophoric solids and liquids should not be considered for classification as self-heating substances and mixtures.

2.11.4. Classification of self-heating substances and mixtures

2.11.4.1. Identification of hazard information

Self-heating is a very complex phenomenon which is influenced by many parameters (some of them being volume, temperature, particle shape and size, heat conductivity and bulk density). Therefore, self-heating behaviour cannot be predicted from any theoretical model. In some cases, properties might even differ between producers of seemingly very similar substances or mixtures. Differences in self-heating behaviour are especially to be anticipated where surface treatment occurs in the production process. Hence, all data sources should be carefully evaluated with regard to reliability and scientific validity.

It is of utmost importance that in compliance with Articles 5 and 6 of CLP authentic and representative material in the correct form and physical state be used for testing. In many

cases, a simple screening test (see Section 2.11.4.2) can be used to determine whether self-heating occurs or not.

2.11.4.2. Screening procedures and waiving of testing

Annex I: 2.11.4.2. The classification procedure for self-heating substances or mixtures need not be applied if the results of a screening test can be adequately correlated with the classification test and an appropriate safety margin is applied. Examples of screening tests are:

(a) The Greiner Oven test (VDI guideline 2263, part 1, 1990, Test methods for the Determination of the Safety Characteristics of Dusts) with an onset temperature 80 K above the reference temperature for a volume of 1 l;

(b) The Bulk Powder Screening Test (Gibson, N. Harper, D.J. Rogers, R. Evaluation of the fire and explosion risks in drying powders, Plant Operations Progress, 4 (3), 181-189, 1985) with an onset temperature 60 K above the reference temperature for a volume of 1 l.

EU test method A.16 as described in Regulation (EC) No 440/2008 checks for self-heating properties. However, the method used is generally inappropriate for a sound assessment, and the findings do not lead to a classification. Therefore, special care must be taken if results from EU test method A.16 are interpreted towards a CLP classification for self-heating substances and mixtures.

In general, the phenomenon of self-heating applies only to solids. The surface of liquids is not large enough for reaction with air and the test method is not applicable to liquids. Therefore liquids are not classified as self-heating. However, if liquids are adsorbed on a large surface (e.g. on powder particles), a self-heating hazard should be considered.

Substances or mixtures with a low melting point (< 160 °C) should not be considered for classification in this class since the melting process is endothermic and the substance-air surface is drastically reduced. However, this criterion is only applicable if the substance or mixture is **completely molten** up to this temperature.

2.11.4.3. Classification criteria

A self-heating substance or mixture must be classified in one of the two categories for this class if, in a test performed in accordance with UN Test N.4 in Part III, Sub-section 33.3.1.6 of the UN-MTC, the result meets the criteria according to following table:

Annex I: Table 2.11.1

Criteria for self-heating substances and mixtures

Category	Criteria
1	A positive result is obtained in a test using a 25 mm sample cube at 140 °C
2	(a) a positive result is obtained in a test using a 100 mm sample cube at 140 °C and a negative result is obtained in a test using a 25 mm cube sample at 140 °C and the substance or mixture is to be packed in packages with a volume of more than 3 m³; or (b) a positive result is obtained in a test using a 100 mm sample cube at 140 °C and a negative result is obtained in a test using a 25 mm cube sample at 140 °C, a positive result is obtained in a test using a 100 mm cube sample at 120 °C and the substance or mixture is to be packed in packages with a volume of more than 450 litres; or

	(c) a positive result is obtained in a test using a 100 mm sample cube at 140 °C and a negative result is obtained in a test using a 25 mm cube sample at 140 °C and a positive result is obtained in a test using a 100 mm cube sample at 100 °C.
Note The test shall be performed on the substance or mixture in its physical form as presented. If, for example, for the purposes of supply or transport, the same chemical is to be presented in a physical form different from that which was tested and which is considered likely to materially alter its performance in a classification test, the substance shall also be tested in the new form. 2.11.2.3. Substances and mixtures with a temperature of spontaneous combustion higher than 50 °C for a volume of 27 m³ shall not be classified as a self-heating substance or mixture. 2.11.2.4. Substances and mixtures with a spontaneous ignition temperature higher than 50 °C for a volume of 450 litres shall not be assigned to Category 1 of this class.	

2.11.4.4. Testing and evaluation of hazard information

A self-heating substance or mixture must be classified in one of the two categories for this class using UN Test N.4 in Part III, Sub-section 33.3.1.6 of the UN-MTC.

2.11.4.4.1. General remarks

If self-heating behaviour cannot be ruled out by a screening test, further testing becomes necessary. UN Test N.4 as described in the latest version of the UN-MTC should be used.

Explosive substances and mixtures should not be tested according to this method. For safety reasons, it is advisable to test for explosive and self-reactive properties and to rule out pyrophoric behaviour before performing this test. The oven should be equipped with an appropriate pressure-release device in case an energetic decomposition is triggered by a temperature rise. For samples containing flammable solvents explosion protection measures have to be taken.

The tests may be performed in any order. It is suggested to start with the 25 mm sample cube at 140 °C. If a positive result is obtained, the substance or mixture must be classified as a self-heating substance or mixture, Category 1, and no further testing is necessary.

The test procedure need not be applied if the substance or mixture is completely molten at 160 °C.

2.11.4.4.2. Sample preparation

The sample (powder or granular) in its commercial form should be used and should not be milled or ground. It should be filled to the brim of the sample container and the container tapped several times. If the sample settles, more is added. If the sample is heaped it should be levelled to the brim. The sample container is placed in the oven as described in the UN-MTC.

2.11.4.4.3. Criteria and evaluation

A positive result is obtained if spontaneous ignition occurs or if the temperature of the sample exceeds the oven temperature by 60 K. The testing time is 24 hours. The time count starts when the temperature in the centre of the sample has reached a value of 2 K below the oven temperature. This is especially important when the sample contains solvents which evaporate under the test conditions or when larger test volumes are used for extrapolation purposes (see below).

Before starting UN Test N.4, the decomposition behaviour of the sample should be known. In general, it is sufficient to perform a screening with Differential Scanning Calorimetry. Special

care with respect to the interpretation of the test data is necessary when exothermic decomposition may occur at the test temperatures. In such cases, a test under an inert atmosphere (i.e. nitrogen) should be run to determine the temperature rise due to decomposition. Careful flushing with the chosen inert gas is essential in such cases since otherwise much air may be retained between the crystals of the sample in the container.

2.11.4.5. Decision logic

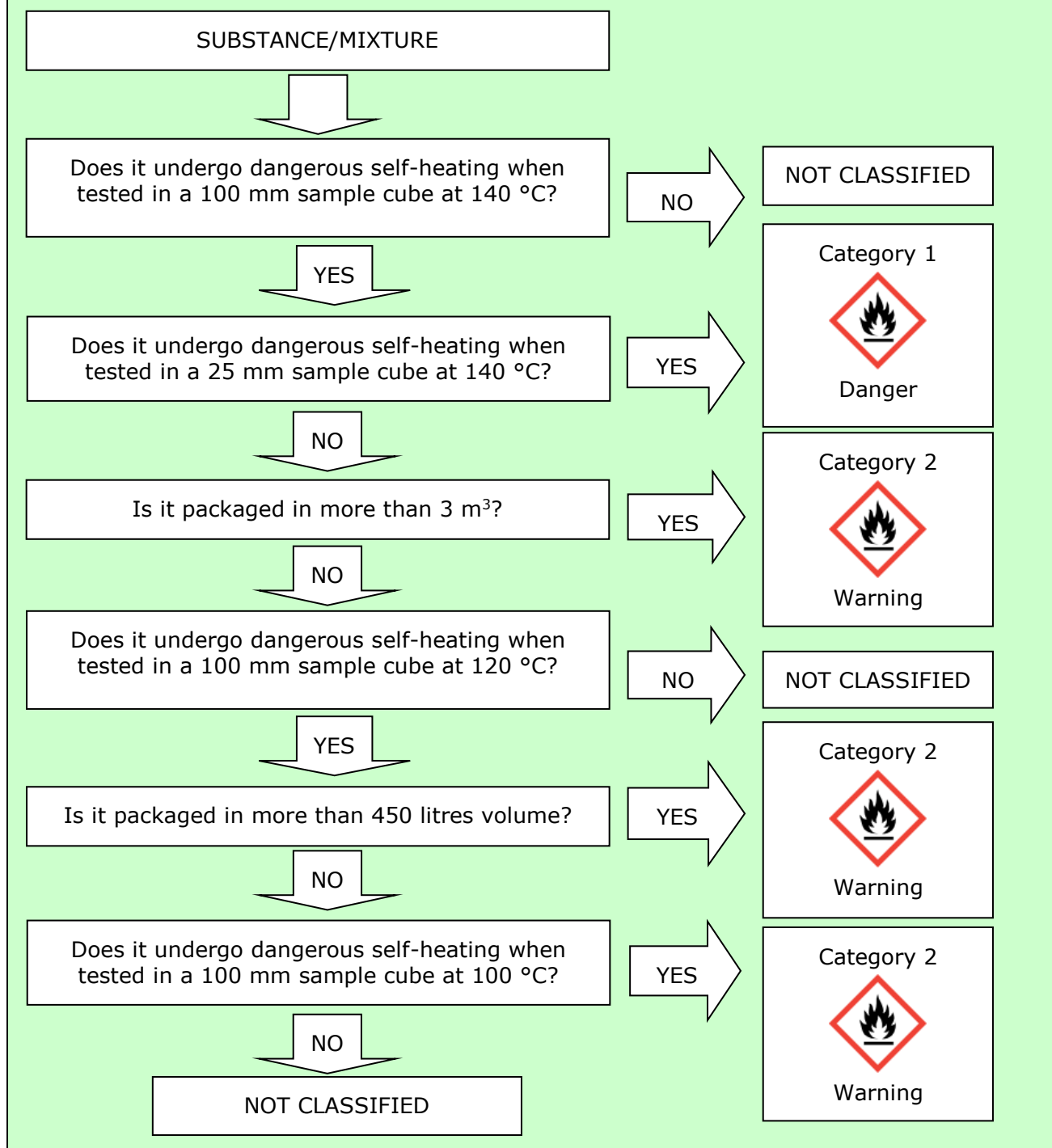
The following decision logic for self-heating substances and mixtures is applicable according to CLP.



NOTE: The person responsible for the classification of self-heating substances and mixtures should be experienced in this field and be familiar with the criteria for classification.

Annex I: Figure 2.11.1.

Self-heating substances and mixtures



2.11.4.6. Exemption

The following exemptions apply (see Section 2.11.4.3):

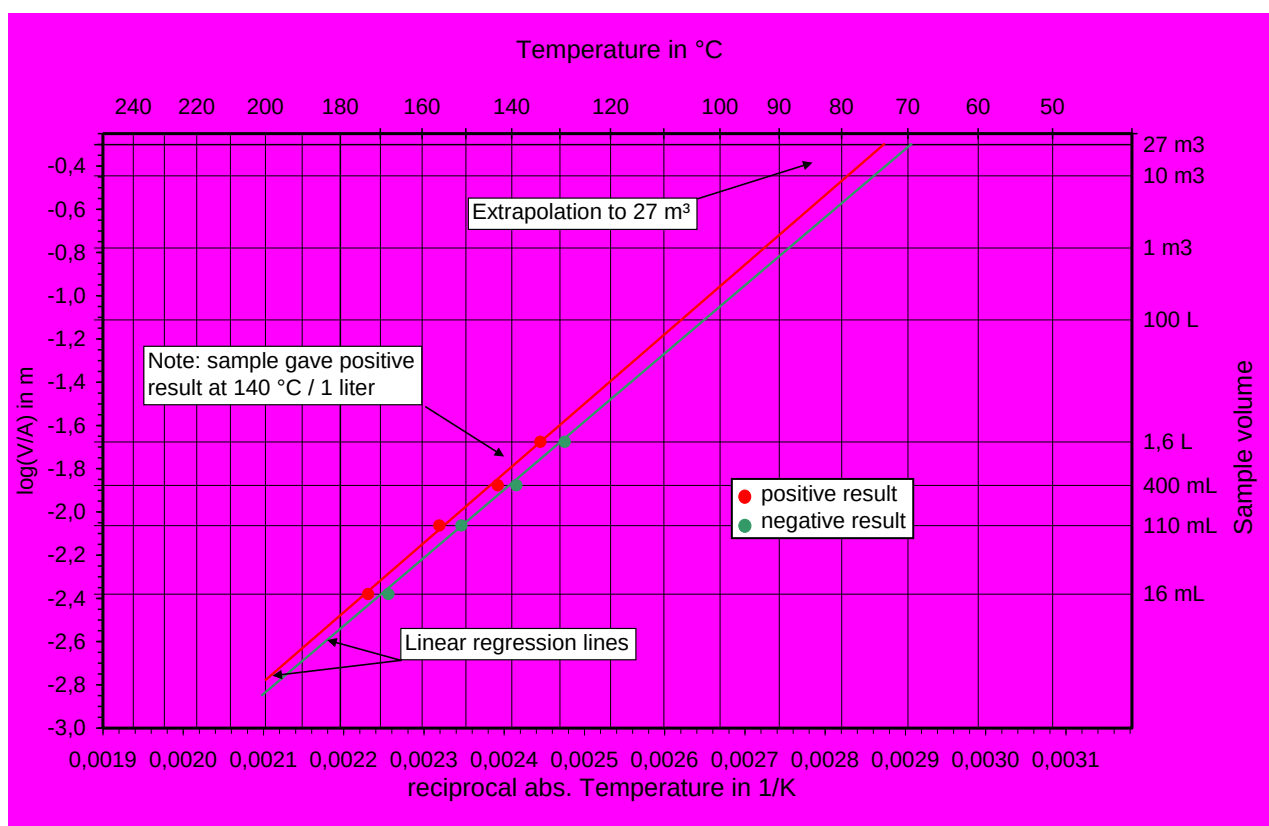
- Substances and mixtures with a temperature of spontaneous combustion higher than 50 °C for a volume of 27 m³ must not be classified as a self-heating substance or mixture.

- Substances and mixtures with a spontaneous ignition temperature higher than 50 °C for a volume of 450 litres must not be assigned to Category 1 of this class.

However, the UN-MTC does not provide any guidance on how these values should be determined. The UN test regime is based on the assumption of a cubic sample shape. For the extrapolation to larger volumes, an improved model has to be used. According to Grewer (Grewer, 1994), plotting the logarithm of the volume to surface ratio ($\log(V/A)$) versus the reciprocal temperature gives good results without knowledge of the Frank-Kamenetzskii (Frank-Kamenetzskii, 1969) shape factor.

The critical temperature for a volume of 450 l or 27 m³ can be found by extrapolation of the critical temperature in a $\log(V/A)$ vs. $1/T$ plot (see Figure 2.11.4–1):

Figure 2.11.4–1 Extrapolation towards large volumes



The test setup is essentially the same as in UN Test N.4 of the UN-MTC but now the sample size and possibly the shape are systematically varied. The criteria of section 2.11.4.3 apply as well.

The critical temperature must be determined over a range of at least four different volumes and with a volume not smaller than 16 ml. If possible, larger volumes should be also tested. The borderline temperature should be determined as precisely as possible. For small volumes (< 1 litre), the temperature rise due to self-heating may be considerably less than 60 K; in this case a noticeable temperature rise is interpreted as a positive result.

A conservative approach is required for the evaluation. The uncertainty of measurement must be taken into account. The extrapolation must be based on a linear regression of the negative and positive borderline data sets in the $\log(V/A)$ vs. $1/T$ diagram. The maximum permissible difference between a positive and a negative result should be 5 K. An exemption may be claimed if the more conservative endpoint for the particular volume is well beyond 50 °C (i.e. 55 °C or higher).

2.11.5. Hazard communication for self-heating substances and mixtures

2.11.5.1. Pictograms, signal words, hazard statements and precautionary statements

Annex I: Table 2.11.2

Label elements for self-heating substances and mixtures

Classification	Category 1	Category 2
GHS Pictograms		
Signal Word	Danger	Warning
Hazard Statement	H251: Self-heating; may catch fire	H252: Self-heating in large quantities; may catch fire
Precautionary Statement Prevention	P235 P280	P235 P280
Precautionary Statement Response		
Precautionary Statement Storage	P407 P413 P420	P407 P413 P420
Precautionary Statement Disposal		

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.11.6. Relation to transport classification

Division 4.2 – substances and mixtures liable to spontaneous combustion – within Class 4 of the UN RTDG Model Regulations comprises the following entries:

- pyrophoric substances and mixtures ;
- self-heating substances and mixtures.

Whereas pyrophoric substances and mixtures in the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) are assigned to packing group I, self-heating substances and mixtures are assigned to packing groups II and III. In cases where a substance or mixture is classified in Division 4.2, packing group II or III, the translation into the CLP system is straightforward.

It should be kept in mind that transport classification is based on prioritisation of hazards (see UN RTDG Model Regulations, Section 2.0.3) and that self-heating substances and mixtures have a relatively low rank in the precedence of hazards. Therefore, the translation from the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) to CLP should be only done if a transport classification as self-heating is explicitly available. The conclusion that a substance or mixture not classified as self-heating for transport should not be classified as a self-heating

substance or mixture according to CLP is, in general, not correct. See Annex VII for additional information on transport classification in relation to CLP classification.

2.11.7. Examples of classification for self-heating substances and mixtures

2.11.7.1. Examples of substances and mixtures fulfilling the classification criteria

- many organometallic compounds, especially substances or mixtures containing transition metals;
- many organic substances or mixtures; the tendency to self-heat increases with decreasing particle size;
- many metals, especially catalysts.

2.11.7.2. Examples of substances and mixtures not fulfilling the classification criteria

In general, liquids show no self-heating behaviour unless adsorbed on a large surface.

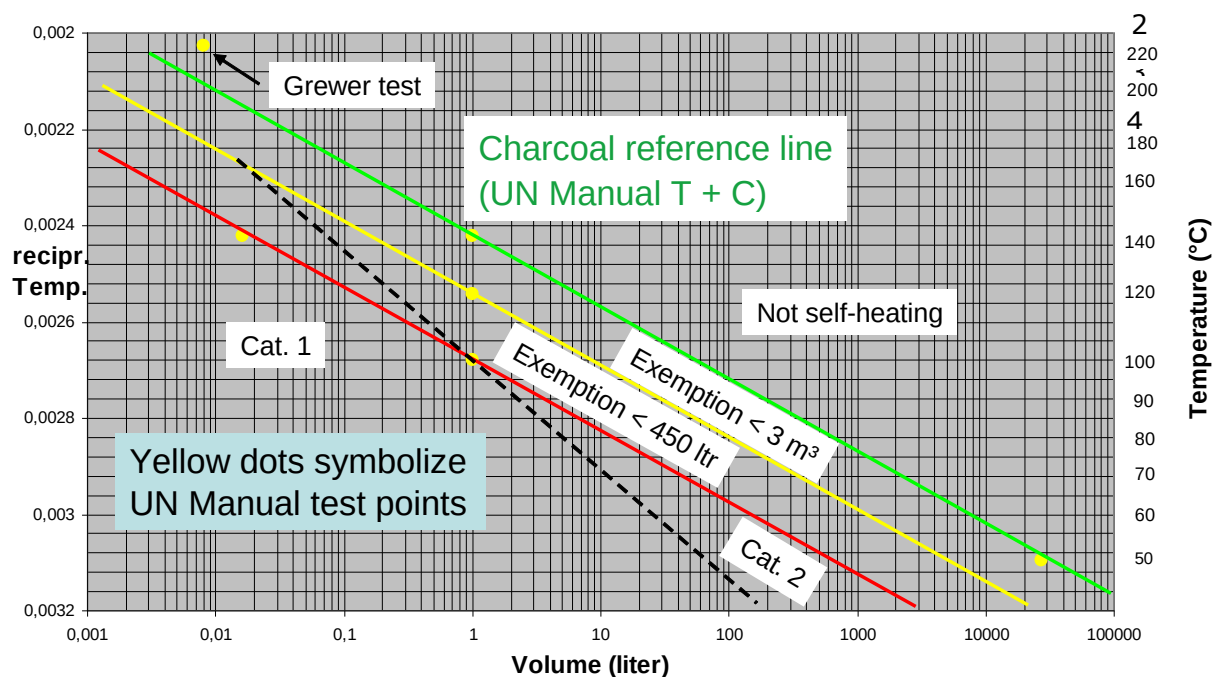
Scientific background

A basic model for the thermal explosion of solids was first developed by Frank-Kamenetskii (Frank-Kamenetskii, 1969). It is based on the assumption that only the heat loss by thermal conduction is relevant for the phenomenon. In this case, the critical criterion for a thermal runaway reaction can be described as a linear relationship between the reciprocal absolute temperature and the logarithm of volume.

The classification scheme of the UN for self-heating substances and mixtures is based on charcoal as a reference system. The critical temperature for a 1 litre cube of charcoal is 140 °C and for a cube of 27 m³ 50 °C. When a parallel line is drawn in the 1/T vs. logarithm of volume diagram from the reference points 1 litre / 120 °C and 1 litre / 100 °C, the corresponding volumes for a critical temperature of 50 °C are found to be 3 m³ and 450 l, respectively (see [Figure 2.11.7—1](#)). The black dotted line in [Figure 2.11.7—1](#) separates Category 1 from Category 2. For examples of results following the Test N.2 see Section 33.3.1.4.5 of UN-MTC.

However, the slope of the line in the 1/T vs. volume diagram depends on the individual activation energy of the substance or mixture, and therefore it may vary within certain limits. It must be born in mind that this test regime has been developed to facilitate classification and that it may not suffice to solve safety issues in storage.

1 **Figure 2.11.7—1 Volume dependency of the critical temperature for charcoal**



5 **2.11.8. References**

6 Grewer, T. (1994). *Thermal hazards of chemical reactions*, Elsevier.

7 Frank-Kamenetzskii, D.A. (1969). *Diffusion and heat transfer in chemical kinetics*, 2nd edition,
8 Plenum Press, New York, London.

9

2.12. SUBSTANCES AND MIXTURES WHICH, IN CONTACT WITH WATER, EMIT FLAMMABLE GASES

2.12.1. Introduction

The criteria for 'Substances and mixtures which, in contact with water, emit flammable gases' are found in Annex I, Section 2.12 of CLP and are identical to those in Chapter 2.12 of GHS.

Depending on the chemical structure and/or the physical state (e.g. particle size) substances or mixtures may be able to react with water (even damp / air humidity) under normal ambient temperature conditions. Sometimes this reaction can be violent and/or with significant generation of heat. Especially if gases are evolved this reaction may become very dangerous during use. In addition, it is important to know whether a substance or mixture emits flammable gases after contact with water because special precautions are necessary especially with regard to explosion protection.

Examples are demonstrated in the following table.

Table 2.12.1—1 Examples of hazards, depending on the property of the emitted gas, when substances and mixtures are in contact with water

Type of emitted gas	Example of the hazard	CLP Reference
Gas (in general)	- Heating up of the substance - Splashing of the substance and thus e.g. contact with skin etc. or additional risk during fire fighting - Pressure rise and bursting of e.g. the packaging, tank	Annex II, 1.1.3: Supplemental hazard information: EUH014*
Flammable gas	- Ignition - Flash of fire	Annex I, 2.12: H260/H261
Toxic gas	Damage to health: intoxication (acute)	Annex II, 1.2.1: Supplemental hazard information: EUH029

* For supplemental hazard information: see Section 2.12.4.2

2.12.2. Definitions and general considerations for the classification of substances and mixtures which, in contact with water, emit flammable gases

The following definition is given in CLP for substances and mixtures which, in contact with water, emit flammable gases (CLP Annex I, 2.12).

Annex I: 2.12.1. Substances or mixtures which, in contact with water, emit flammable gases means solid or liquid substances or mixtures which, by interaction with water, are liable to become spontaneously flammable or to give off flammable gases in dangerous quantities.

2.12.3. Relation to other physical hazards

If the chemical identity of the emitted gas is unknown, the gas must be tested for flammability (unless it ignites spontaneously). Other than under DSD/DPD, pyrophoric liquids and pyrophoric solids have to be considered for classification in this hazard class as well and data about pyrophoric properties are needed prior to testing for this hazard class.

2.12.4. Classification of substances and mixtures which, in contact with water, emit flammable gases

2.12.4.1. Identification of hazard information

For the classification of substances and mixtures which, in contact with water, emit flammable gases the following data are needed, if applicable:

- chemical structure;
- water solubility;
- chemical identity and flammability of the emitted gas;
- pyrophoric properties of the tested substance or mixture;
- particle size in case of solids;
- friability in case of solids;
- hazard properties in general;
- information concerning the experience in production or handling.

See also *IR & CSA, Chapter R.7a: Endpoint specific guidance*, Section R.7.1.7 (Water solubility), R.7.1.14 (Granulometry).

Information about the chemical structure is used to check whether the substance or mixture contains metals and/or metalloids.

The water solubility is used to decide whether the substance or mixture is soluble in water to form a stable mixture. This may also be decided based on information concerning experience in handling or use, e.g. the substance or mixture is manufactured with water or washed with water (see Section 2.12.4.4.1).

The chemical identity of the emitted gas is used to decide whether the evolved gas is flammable or not. If the chemical identity of the emitted gas is unknown, the gas must be tested for flammability (see Section 0).

In case of pyrophoric substances and mixtures the UN Test N.5 of the UN-MTC, Part III, Section 33.4.1.3.1 must be executed under nitrogen atmosphere. Therefore, data about pyrophoric properties are needed prior to testing.

The melting point, boiling point and information about viscosity are necessary to identify the physical state of the substance or mixture. See also *IR & CSA, Chapter R.7a: Endpoint specific guidance*, Section R.7.1.2 (Melting point/freezing point), R.7.1.3 (Boiling point), R.7.1.18 (Viscosity).

Even though the UN Test N.5 can be applied to both, solids and liquids, these data are necessary to decide whether information concerning the friability (for solids) in accordance with the test method is necessary.

The particle size and the friability of a solid substance or mixture are crucial parameters for the classification of substances and mixtures which, in contact with water, emit flammable gases. These parameters have a significant effect on the test result. Thus specific requirements

regarding the particle size and the friability are prescribed in the UN Test N.5. For further details regarding the test procedure see Section 2.12.4.4.1.

The references in Section 2.12.8 provide good quality data on physical hazards.

2.12.4.2. Screening procedures and waiving of testing

For the majority of substances and mixtures, flammability as a result of contact with water is not a typical property and testing can be waived based on a consideration of the structure and experiences in handling and use.

Annex I: 2.12.4.1. The classification procedure for this class need not be applied if:

- a) the chemical structure of the substance or mixture does not contain metals or metalloids; or
- b) experience in handling and use shows that the substance or mixture does not react with water, e.g. the substance is manufactured with water or washed with water; or
- c) the substance or mixture is known to be soluble in water to form a stable mixture.

2.12.4.3. Classification criteria

Annex I: Table 2.12.1

Criteria for substances or mixtures which in contact with water emit flammable gas

Category	Criteria
1	Any substance or mixture which reacts vigorously with water at ambient temperatures and demonstrates generally a tendency for the gas produced to ignite spontaneously, or which reacts readily with water at ambient temperatures such that the rate of evolution of flammable gas is equal to or greater than 10 litres per kilogram of substance over any one minute.
2	Any substance or mixture which reacts readily with water at ambient temperatures such that the maximum rate of evolution of flammable gas is equal to or greater than 20 litres per kilogram of substance per hour, and which does not meet the criteria for Category 1.
3	Any substance or mixture which reacts slowly with water at ambient temperatures such that the maximum rate of evolution of flammable gas is equal to or greater than 1 litre per kilogram of substance per hour, and which does not meet the criteria for Categories 1 and 2.

Note:

The test shall be performed on the substance or mixture in its physical form as presented. If for example, for the purposes of supply or transport, the same chemical is to be presented in a physical form different from that which was tested and which is considered likely to materially alter its performance in a classification test, the substance must also be tested in the new form.

2.12.2.2. A substance or mixture shall be classified as a substance or mixture which in contact with water emits flammable gases if spontaneous ignition takes place in any step of the test procedure.

2.12.4.4. Testing and evaluation of hazard information

2.12.4.4.1. Testing procedure

Care must be taken during testing as the emitted gas might be toxic or corrosive.

The testing procedure for substances and mixtures which in contact with water emit flammable gases is sensitive to a number of influencing factors and therefore must be carried out by experienced personnel. Some of these factors are described in the following:

2. Apparatus / measuring technique

In UN Test N.5 no special laboratory apparatus / measuring technique to determine the rate of gas evolution is required and no reference material is prescribed. As demonstrated in the past by a round robin test (Kunath, K. *et al.* 2011), the gas evolution rate measured by different apparatuses may vary widely. Therefore in order to avoid measuring and classification errors adequate quality control measures are necessary to validate the results and should be noted in the test report.

3. Particle size and/or friability

The particle size of a solid has a significant effect on the test result. Therefore, if for solids the percentage of powder with a particle size of less than 500 µm constitutes more than 1 % of the total mass, or if the substance or mixture is friable, then the complete sample must be ground to a powder before testing to account for a possible reduction in particle size during handling and transport.

In certain cases, grinding may not be applicable and/or the sample cannot be ground completely to a particle size of less than 500 µm (e.g. metal granules).

Information on these pre-treatments and the respective procedures, the particle size and the friability has to be provided in the test report.

4. Atmospheric parameters

Variations of the atmospheric parameters (mainly air pressure and temperature) during the test have a considerable influence on the test result. Therefore the substance or mixture must be tested at 20 °C, i.e. make sure that the test apparatus is acclimatised to 20 °C.

On the other hand it is difficult to regulate and stabilise the air pressure during the testing. To characterise this influencing factor and to avoid false positive results, an additional 'blank test' is highly recommended. The results of the blank test should be noted in the test report.

5. Test with demineralised (distilled) water

The UN Test N.5 is performed with demineralised (distilled) water. In practice, contact with water can be to water in the liquid state (fresh water, sea water) or humid air, respectively. Note that the reactivity and thus the gas evolution rate observed in practice may differ from the gas evolution rate value measured using demineralised water. This should be taken into account when handling substances and mixtures which in contact with water emit flammable gases.

6. Stirring procedures during the test

Stirring of the sample or water mixture during the test may have a considerable effect on the test result (e.g. significant increase or decrease of the gas evolution rate). Therefore, the sample or water mixture should not be stirred continuously during the test, e.g. by an automatic magnetic stirrer, even if the test sample has hydrophobic properties and moistening of the sample becomes impossible (see Kunath K. *et al.*, 2011).

7. Spontaneous ignition

Spontaneous ignition of the evolved gas without contact with an additional ignition source, i.e. without the flame of the gas burner results in classification as Category 1. This does not

necessarily mean that the evolved gas is pyrophoric but often the heat of reaction is sufficient to ignite the evolved gas (e.g. the hydrogen evolved when sodium reacts with water).

2.12.4.4.2. Evaluation of hazard information

In order to accurately interpret the test results the evaluating person must have sufficient experience in the application of the test methods and in the disturbing / influencing factors as described above.

The evaluation of data comprises two steps:

- evaluation of all available data; and
- identification of the study or studies giving rise to the highest concern (key studies).

The criteria for assignment to Category 2 or 3 are gas evolution rates of 20 and 1 litre per kilogram of substance or mixture per hour, respectively, but for Category 1 the relevant criterion is 10 litres per kilogram of substance or mixture over any one minute period (if the gas does not ignite spontaneously). This has to be considered while testing and for correct evaluation of the test results.

The assignment to the respective hazard class/category will further determine the technical means to be taken to avoid dangerous events which, in combination with other safety characteristics such as i) explosion limits, ii) flash points (applicable only for liquids) or iii) self-ignition temperature, can lead to clear restrictions in the conditions of use.

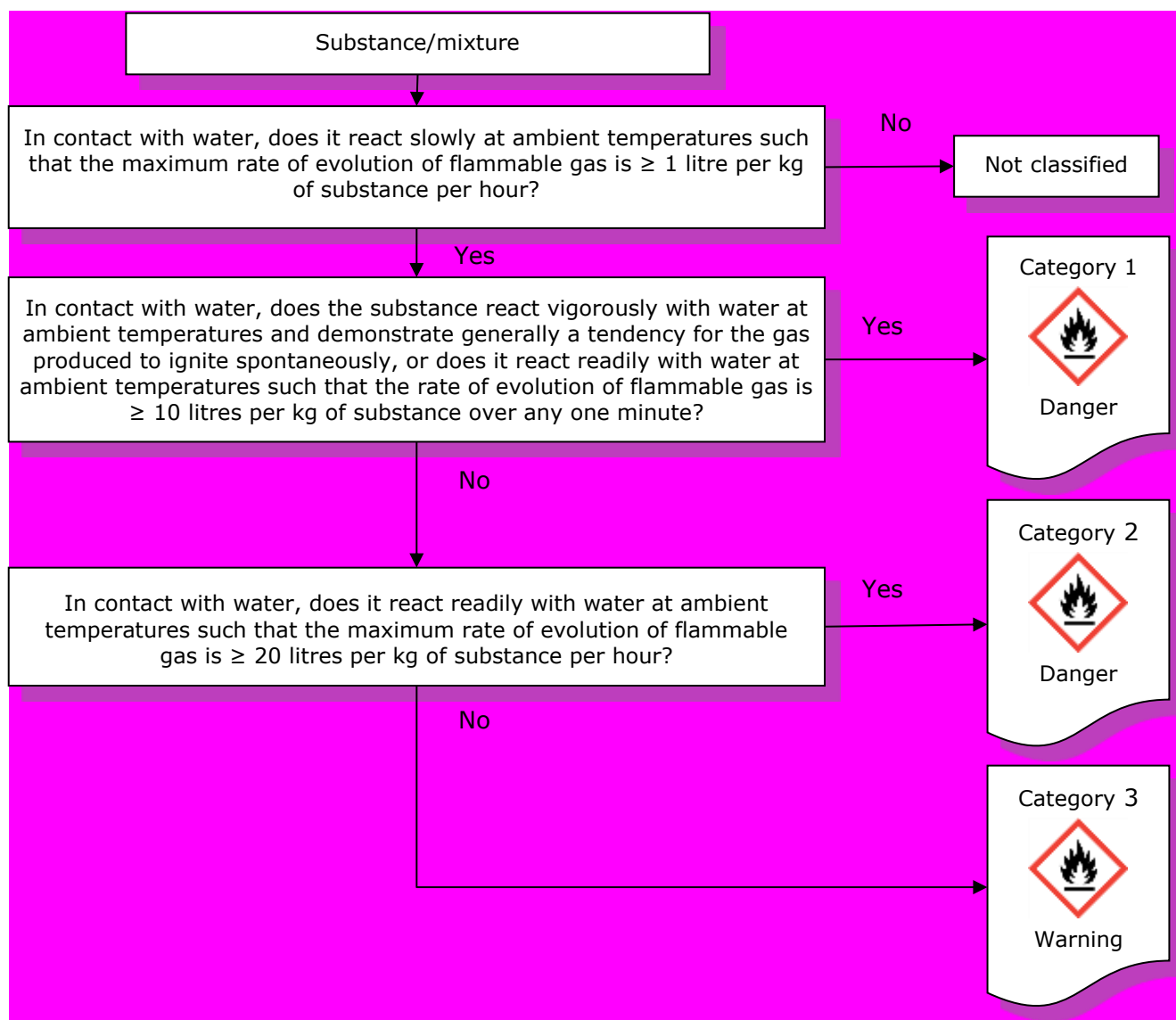
2.12.4.5. Decision logic

Classification of substances and mixtures which, in contact with water, emit flammable gases is done according to decision logic 2.12.4.1 as included in the GHS.



NOTE: The person responsible for the classification of substances and mixtures which, in contact with water, emit flammable gases should be experienced in this field and be familiar with the criteria for classification.

1 **Figure 2.12.4—1 Decision logic for substances and mixtures which, in contact with water, emit**
2 **flammable gases (Decision logic 2.12 of GHS)**



2.12.5. Hazard communication for substances and mixtures which, in contact with water, emit flammable gases

2.12.5.1. Pictograms, signal words, hazard statements and precautionary statements for substances and mixtures

Annex I: Table 2.12.2

Label elements for substances or mixtures which in contact with water emit flammable gases

Classification	Category 1	Category 2	Category 3
GHS Pictograms			
Signal Word	Danger	Danger	Warning
Hazard Statement	H260: In contact with water releases flammable gases which may ignite spontaneously	H261: In contact with water releases flammable gases	H261: In contact with water releases flammable gases
Precautionary Statement Prevention	P223 P231 + P232 P280	P223 P231 + P232 P280	P231 + P232 P280
Precautionary Statement Response	P302 + P335 + P334 P370 + P378	P302 + P335 + P334 P370 + P378	P370 + P378
Precautionary Statement Storage	P402 + P404	P402 + P404	P402 + P404
Precautionary Statement Disposal	P501	P501	P501

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.12.5.2. Additional labelling provisions

Annex II of CLP provides the following additional labelling provisions for water-reactive substances and mixtures. These statements must be assigned in accordance with CLP, Article 25 (1), to substances and mixtures classified for physical, health or environmental hazards. There are no criteria or test methods provided for these EUH statements.

Annex II: 1.1.3. EUH014 – 'Reacts violently with water'

For substances and mixtures which react violently with water, such as acetyl chloride, alkali metals, titanium tetrachloride.

Annex II: 1.2.1. EUH029 - 'Contact with water liberates toxic gas'

For substances and mixtures which in contact with water or damp air, evolve gases classified for acute toxicity in category 1, 2 or 3 in potentially dangerous amounts, such as aluminium phosphide, phosphorus pentasulphide.

2.12.6. Relation to transport classification

Division 4.3 within Class 4 of the UN RTDG Model Regulations covers substances and mixtures which in contact with water emit flammable gasses. Substances and mixtures which are classified and/or labelled in Division 4.3 in the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) are classified as substances and mixtures which, in contact with water, emit flammable gases under CLP. See Annex VII for additional information on transport classification in relation to CLP classification.

2.12.7. Examples of classification for substances and mixtures which, in contact with water, emit flammable gases

2.12.7.1. Example of a substance fulfilling the classification criteria

Many different types of chemicals may belong to the hazard class of substances and mixtures which, in contact with water, emit flammable gases, for example, alkali metals, alkyl aluminium derivatives, alkyl metals, metal hydrides, metal phosphides, certain metal powders. A comprehensive list can be found in *Bretherick's Handbook of Reactive Chemical Hazards* (Urban, 2007).

2.12.7.1.1. Example 1

Pyrophoric substance fulfilling the criteria for CLP classification	
Substance:	Magnesium alkyls (Index No. 012-003-00-4)
Chemical structure:	R ₂ Mg
Flammable gas:	Hydrogen
Gas evolution rate:	not applicable
Spontaneous ignition:	not possible due to the nitrogen atmosphere during the UN Test N.5
DSD classification:	F; R14-17
Transport classification:	-
Reference:	Former Annex I to DSD and Annex VI to CLP
⇒ CLP Classification:	Water-react. 1; H260 Pyr. Sol. 1; H250

Supplemental Hazard Information:

EUH014

2.12.7.2. Example of a substance not fulfilling the classification criteria

2.12.7.2.1. Example 2

Manganese ethylene bis (dithiocarbamate) complex with zinc salt 88 % (Mancozeb)

Gas evolution rate:	0 litres per kilogram of substance per hour.
Spontaneous ignition:	not applicable
Transport classification:	not Class 4.3
Reference:	UN Test N.5, UN-MTC Table 33.4.1.4.5
⇒ CLP Classification:	Not classified as substance which, in contact with water, emit flammable gases

2.12.8. References

William M. Haynes et al. (2012) *CRC Handbook of Chemistry and Physics 93rd Edition*. CRC Press, Taylor and Francis, Boca Raton, FL

GESTIS-database on hazardous substances:
<http://www.dguv.de/bgia/en/gestis/stoffdb/index.jsp>

O'Neil, Maryadele J. et al. (2016, 2012) *The Merck Index - An Encyclopaedia of Chemicals, Drugs, and Biologicals* (14th Edition – Version 14.9). Merck Sharp & Dohme Corp., a subsidiary of Merck & Co., Inc.

Urban, Peter G. (2007). *Bretherick's Handbook of Reactive Chemical Hazards*, Volumes 1-2 (7th Edition). Elsevier.

Kunath, K., Lüth, P., Uhlig, S. (2011). *Interlaboratory test on the method UN Test N.5 / EC A.12 "Substances which, in contact with water, emit flammable gases" 2007*. Short report. BAM Bundesanstalt für Materialforschung und -prüfung. Berlin. ISBN 978-3-9814634-1-5.
http://www.bam.de/de/service/publikationen/publikationen_medien/short_report_rv_un_n_5.pdf

2.13. OXIDISING LIQUIDS

2.13.1. Introduction

The criteria for 'Oxidising liquids' are found in Annex I, Section 2.13 of CLP and are identical to those in Chapter 2.13 of GHS.

The hazard class oxidising liquids comprises liquid substances and mixtures whose hazard is characterised by the fact that, in contact with other materials, they are able to cause or contribute to the combustion of those materials. The other materials do not necessarily have to belong to a certain hazard class in order to be able to be affected by the presence of oxidising substances or mixtures. This is for example the case when a solid material (e.g. wood) is soaked with an oxidising liquid.

Certain combinations of combustible materials and oxidising substances or mixtures may even result in spontaneous combustion, thermal instability or form an explosive mixture, this means that they may have explosive properties or may be regarded as self-reactive substances or mixtures.

Although widely known as oxidising materials, their hazard and behaviour might be better understood by considering them to be fire enhancing substances or mixtures.

The hazards communication of oxidising liquids intends to communicate the property that it may cause fire or explosion or that it may intensify fire.

Apart from the combustion hazard, the production of toxic and/or irritating fumes may cause an additional hazard. For example, when nitrates are involved in a fire, nitrous fumes may be formed.

The testing procedure and criteria for oxidising substances or mixtures do not work properly for ammonium nitrate compounds or solutions, ammonium nitrate based fertilizers and ammonium nitrate emulsions, suspensions or gels. Therefore for classification and labelling of substances or mixtures containing ammonium nitrate, known experience should be used and expert judgement should be sought. For the classification procedures for ammonium nitrate emulsions, suspensions or gels – intermediate for blasting explosives, see Chapter 2.1 of this guidance.

Annex I: 2.13.4.3

In the event of divergence between test results and known experience in the handling and use of substances or mixtures which shows them to be oxidising, judgments based on known experience shall take precedence over test results.

2.13.2. Definitions and general considerations for the classification of oxidising liquids

The CLP text comprises the following definition for oxidising liquids.

Annex I: 2.13.1. Definition

Oxidising liquid means a liquid substance or mixture which, while in itself not necessarily combustible, may, generally by yielding oxygen, cause, or contribute to, the combustion of other material.

2.13.3. Relation to other physical hazards

Oxidising liquids that are mixed with combustible materials or reducing agents may have explosive properties and should be considered for classification in the hazard class Explosives (including the applicable screening procedures), see Chapter 2.1 of this guidance.

In rare cases, mixtures with oxidising liquids may exhibit self-reactive behaviour, see Chapter 2.8 of this guidance. Expert judgement should be sought in case of doubt.

The classification procedure and criteria for oxidising substances or mixtures is not applicable for organic peroxides. Under DSD organic peroxides were considered to be oxidising substances or mixtures because of the presence of the –O–O– bond. The majority of the organic peroxides do not possess oxidising properties; their main hazards are reactivity and flammability. Under CLP organic peroxides are comprised in a separate hazard class (CLP Annex I, 2.15) and they must not be considered according to the procedures described for oxidising liquids. Organic peroxides were classified as oxidising (O; R7) according to the DSD, which was not appropriate since the vast majority of them do not exhibit oxidising properties.

Inorganic oxidising liquids are not flammable and therefore do not have to be subjected to the classification procedures for the hazard classes flammable liquids or pyrophoric liquids. Also other liquids that are classified as oxidising liquids are normally not flammable, although a few exemptions may exist. Expert judgement should be sought in case of doubt.

2.13.4. Classification of substances and mixtures as oxidising liquids

2.13.4.1. Identification of hazard information

Oxidising liquids may cause, or contribute to, the combustion of other material. Although the definition states that they generally do this by yielding oxygen, halogens can behave in a similar way. Therefore, any substance or mixture containing oxygen and/or halogen atoms should in principle be considered for inclusion into the hazard class oxidising liquids. This does not necessarily mean that every substance or mixture containing oxygen and/or halogen atoms should be subjected to the full testing procedure.

2.13.4.1.1. Screening procedures and waiving of testing

Liquids that are classified as explosives should not be subjected to the testing procedures for oxidising liquids.

Organic peroxides should be considered for classification within the hazard class organic peroxides, see Chapter 2.15 of this guidance.

Experience in the handling and use of substances or mixtures which shows them to be oxidising is an important additional factor in considering classification as oxidising liquids. In the event of divergence between test results and known experience, judgement based on known experience should take precedence over test results.

Before submitting a substance or a mixture to the full test procedure, an evaluation of its chemical structure may be very useful as it may prevent unnecessary testing. The person applying this procedure should have sufficient experience in testing and in theoretical evaluation of hazardous substances and mixtures. The following text provides a guideline for the theoretical evaluation of potential oxidising properties on basis of its composition and chemical structure. In case of doubt, the full test must be performed.

For organic substances or mixtures the classification procedure for this hazard class need not to be applied if:

- a. the substance or mixture does not contain oxygen, fluorine or chlorine; or
- b. the substance or mixture contains oxygen, fluorine or chlorine and these elements are chemically bonded only to carbon or hydrogen.

For inorganic substances or mixtures, the classification procedure for this hazard class need not be applied if they do not contain oxygen or halogen.

On basis of this theoretical evaluation only a distinction can be made between 'potentially oxidising' (i.e. further testing required) and 'non-oxidising' (i.e. no further testing for this hazard class required). It is not possible to assign a hazard category on basis of a theoretical evaluation.

Any substance or mixture that complies with the above waiving criteria can be safely regarded to have no oxidising properties and, hence, needs not to be tested and needs not to be regarded as an oxidising liquid. However, such a substance or mixture may still possess other hazardous properties that require classification into another hazard class.

In case a mixture of an oxidising substance and a non-hazardous inert substance is offered for classification, the following should be taken into account:

- An inert material by definition does not contribute to the oxidising capability of the oxidising substance. Hence, the mixture can never be classified into a more severe hazard category.
- If an oxidising substance is mixed with an inert material, the oxidising capability of the mixture does not linearly decrease with decreasing content of oxidising substance. The relationship is more or less logarithmic and depends on the characteristics of the oxidising substance. For instance, a mixture containing 50 % of a strong oxidiser and 50 % of an inert material may retain 90 % of the oxidising capability of the original oxidising component. Non-testing classification of mixtures based solely on test data for the original oxidising substance should therefore be done with extreme care and only, if sufficient experience in testing exists.
- The determination of the oxidising properties of an aqueous solution of solid oxidising substances and the classification as an oxidising mixture is not necessary provided that the total concentration of all solid oxidisers in the aqueous solution is less than or equal to 20 % (w/w).

2.13.4.2. Classification criteria

The testing procedures for oxidising liquids are based on the capability of an oxidising liquid to enhance the combustion of a combustible material. Therefore, substances and mixtures that are submitted for classification testing are mixed with a combustible material. In principle, dried fibrous cellulose is used as a combustible material. The mixture of the potentially oxidising liquid and cellulose is then ignited and its behaviour is observed and compared to the behaviour of reference materials.

For liquids the mixture with cellulose is ignited under confinement in an autoclave and the pressure rise rate that is caused by the ignition and the subsequent reaction is recorded. The pressure rise rate is compared to that of three reference material mixtures. The higher the pressure rise rate, the stronger the oxidising capability of the liquid tested.

Annex I: 2.13.2.1.

An oxidising liquid shall be classified in one of the three categories for this class using test O.2 in Part III, sub-section 34.4.2 of the UN RTDG, Manual of Tests and Criteria in accordance with Table 2.13.1:

Table 2.13.1

Criteria for oxidising liquids

Category	Criteria
----------	----------

1	Any substance or mixture which, in the 1:1 mixture, by mass, of substance (or mixture) and cellulose tested, spontaneously ignites; or the mean pressure rise time of a 1:1 mixture, by mass, of substance (or mixture) and cellulose is less than that of a 1:1 mixture, by mass, of 50 % perchloric acid and cellulose.
2	Any substance or mixture which, in the 1:1 mixture, by mass, of substance (or mixture) and cellulose tested, exhibits a mean pressure rise time less than or equal to the mean pressure rise time of a 1:1 mixture, by mass, of 40 % aqueous sodium chlorate solution and cellulose; and the criteria for Category 1 are not met.
3	Any substance or mixture which, in the 1:1 mixture, by mass, of substance (or mixture) and cellulose tested, exhibits a mean pressure rise time less than or equal to the mean pressure rise time of a 1:1 mixture, by mass, of 65 % aqueous nitric acid and cellulose; and the criteria for Category 1 and 2 are not met.

1 For additional information regarding the use of non-testing data see Section 2.13.4.3 below and
2 Urban, 2007 (see Section 2.13.7).

3 **2.13.4.3. Testing and evaluation of hazard information**

4 The test methods for oxidising liquids are designed to give a final decision regarding their
5 classification. Apart from testing, also experience in the handling and use of substances or
6 mixtures which shows them to be oxidising is an important additional factor in considering
7 classification in this hazard class. In the event of divergence between test results and known
8 experience, judgement based on known experience should take precedence over test results.
9 However, a substance or mixture must not be classified into a less severe Category based on
10 experience only.

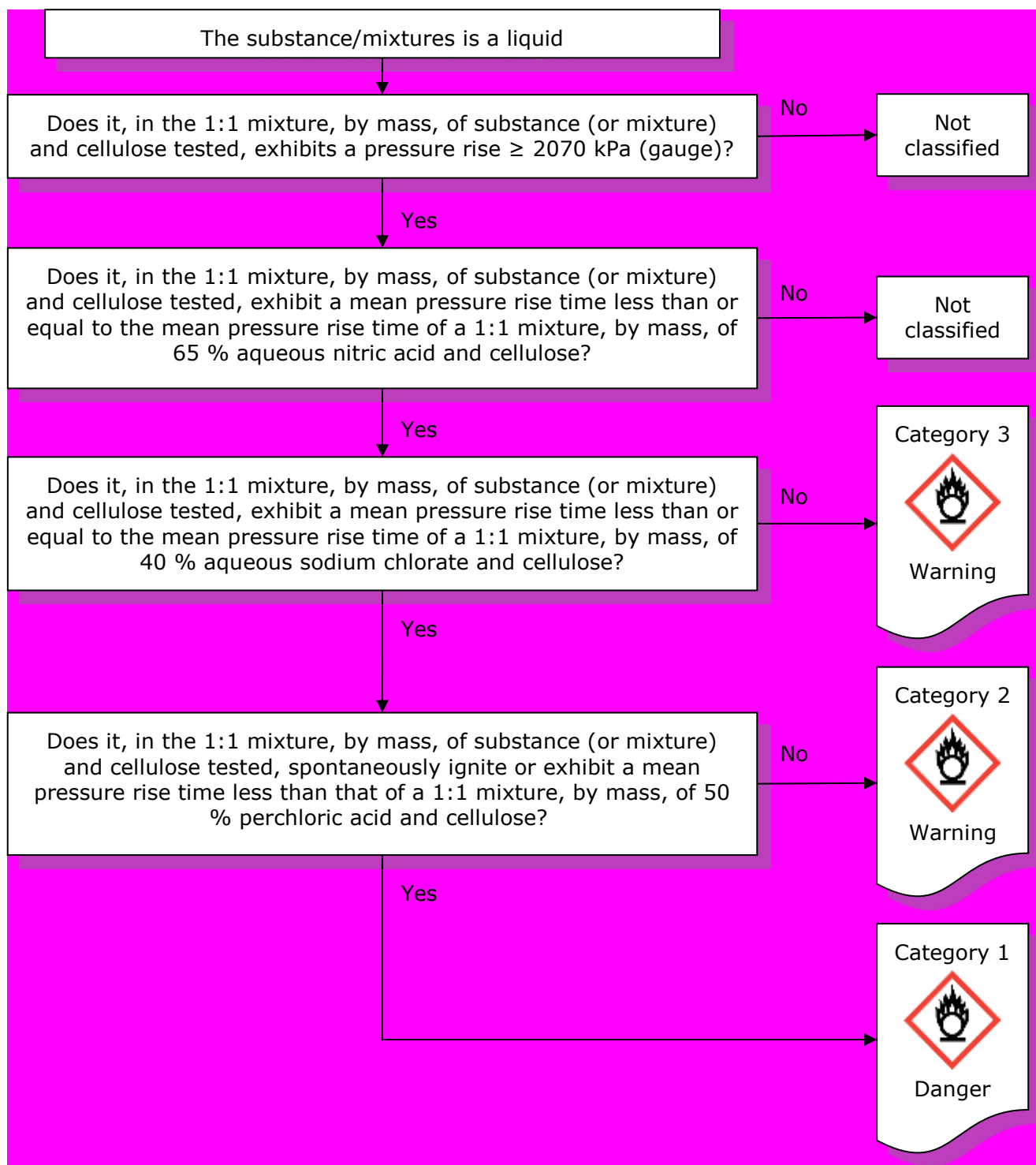
11 **2.13.4.4. Decision logic**

12 Classification of oxidising liquids is done according to decision logic 2.13 as included in the GHS.



NOTE: The person responsible for the classification of oxidising liquids should be experienced in this field and be familiar with the criteria for classification.

1 **Figure 2.13.4—1 Decision logic for oxidising liquids (Decision logic 2.13 of GHS)**



2.13.4.5. Hazard communication for oxidising liquids

2.13.4.5.1. Pictograms, signal words, hazard statements and precautionary statements

The pictograms and hazard statements are designed to indicate that oxidising substances and mixtures may cause or contribute to fire or explosion and therefore in principle should be separated from combustible materials.

Annex I : Table 2.13.2

Label elements for oxidising liquids

	Category 1	Category 2	Category 3
GHS Pictograms			
Signal Word	Danger	Danger	Warning
Hazard Statement	H271: May cause fire or explosion; strong oxidiser	H272: May intensify fire; oxidiser	H272: May intensify fire; oxidiser
Precautionary Statement Prevention	P210 P220 P280 P283	P210 P220 P280	P210 P220 P280
Precautionary Statement Response	P306 + P360 P371 + P380 + P375 P370 + P378	P370 + P378	P370 + P378
Precautionary Statement Storage	P420		
Precautionary Statement Disposal	P501	P501	P501

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.13.5. Relation to transport classification

Division 5.1 within Class 5 of the UN RTDG Model Regulations covers oxidising liquids and oxidising solids, using the same tests and criteria as the CLP. Therefore, a liquid substance or mixture classified as Division 5.1 (sometimes referred to as Class 5.1) according to any of the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) is normally also classified as an oxidising liquid according to the CLP. Packing Groups I, II and III of the transport regulations correspond directly to Categories 1, 2 and 3 of the CLP, respectively. See Annex VII for additional information on transport classification in relation to CLP classification.

2.13.6. Examples of classification for oxidising liquids

2.13.6.1. Examples of substances and mixtures fulfilling the classification criteria

The list of substances and mixtures fulfilling the criteria for classification is only presented for information purposes. This list is not exhaustive. For examples of results see Section 34.4.2.5 of UN-MTC.

- Ferric nitrate, saturated aqueous solution
- Lithium perchlorate, saturated aqueous solution
- Magnesium perchlorate, saturated aqueous solution
- Perchloric acid, 55 %
- Sodium nitrate, 45 % aqueous solution

2.13.6.2. Examples of substances and mixtures not fulfilling the classification criteria

- Nickel nitrate, saturated aqueous solution
- Potassium nitrate, 30 % aqueous solution
- Silver nitrate, saturated aqueous solution

2.13.7. Reference

Urban, Peter G. (2007). *Bretherick's Handbook of Reactive Chemical Hazards, Volumes 1-2* (7th Edition). Elsevier.

2.14. OXIDISING SOLIDS

2.14.1. Introduction

The criteria for 'Oxidising solids' are found in Annex I, Section 2.14 of CLP and are identical to those in Chapter 2.14 of GHS.

The hazard class oxidising solids comprises substances and mixtures whose hazard is characterised by the fact that, in contact with other materials, they are able to cause or contribute to the combustion of those materials. The other materials do not necessarily have to belong to a certain hazard class in order to be affected by the presence of an oxidising solid. This is for example the case when a liquid fuel (e.g. gas oil) mixes with an oxidising solid. Certain combinations of combustible materials and oxidising substances or mixtures may even result in spontaneous combustion, thermal instability or form an explosive mixture, this means that they may have explosive properties or may be regarded as self-reactive substances or mixtures.

Although widely known as 'oxidising materials', their hazard and behaviour might be better understood by considering them to be 'fire enhancing substances'.

The hazards communication of oxidising solids intends to communicate the property that it may cause fire or explosion or that it may intensify fire.

Apart from the combustion hazard, the production of toxic and/or irritating fumes may cause an additional hazard. For example, when nitrates are involved in a fire, nitrous fumes may be formed.

The testing procedure and criteria for oxidising substances or mixtures do not work properly for ammonium nitrate, ammonium nitrate compounds, ammonium nitrate based fertilizers and ammonium nitrate gels. Therefore, for classification and labelling of substances and mixtures containing ammonium nitrate, known experience should be used and expert judgement should be sought. For the classification procedures for ammonium nitrate gels – intermediate for blasting explosives, see Section [2.1](#) of this guidance.

Annex I: 2.14.4.3

In the event of divergence between test results and known experience in the handling and use of substances or mixtures which shows them to be oxidising, judgments based on known experience shall take precedence over test results.

2.14.2. Definitions and general considerations for the classification of oxidising solids

The CLP text comprises the following definition for oxidising solids.

Annex I: 2.14.1. Definition

Oxidising solid means a solid substance or mixture which, while in itself is not necessarily combustible, may, generally by yielding oxygen, cause, or contribute to, the combustion of other material.

Special consideration on particle size

The oxidising properties of a solid depend on its particle size. Smaller particles enable a more intimate contact between the solid oxidiser and a combustible solid. The smaller the particle size, the higher the oxidising capability of the solid. As a consequence, it may happen that large

particles of a certain solid are considered to be non-hazardous, while small particles of the same solid need to be classified into the hazard class of oxidising solids.

Hence it is very important that oxidising properties for solids are investigated on the substance or mixture as it is actually presented (including how it can reasonably be expected to be used, see Article 8 (6) of CLP). This is indicated by the Note 2 cited in CLP Annex I, 2.14.2.1.

Annex I: 2.14.2.1.

[...]

Note 2: The test shall be performed on the substance or mixture in its physical form as presented. If for example, for the purposes of supply or transport, the same chemical is to be presented in a physical form different from that which was tested and which is considered likely to materially alter its performance in a classification test, the substance shall also be tested in the new form.

2.14.3. Relation to other physical hazards

Oxidising solids that are mixed with combustible materials or reducing agents may have explosive properties and should be considered for classification in the hazard class Explosives (including the applicable screening procedures), see Chapter 2.1 of this guidance.

In rare cases, mixtures with oxidising solids may exhibit self-reactive behaviour, see Chapter 2.8 of this guidance. Expert judgement should be sought in case of doubt.

The classification procedure and criteria for oxidising substances and mixtures is not applicable for organic peroxides. Under DSD organic peroxides were considered to be oxidising substances because of the presence of the –O–O– bond. The majority of the organic peroxides do not possess oxidising properties; their main hazards are reactivity and flammability. Under CLP organic peroxides are comprised in a separate hazard class (CLP Annex I, 2.15) and they must not be considered according to the procedures described for oxidising solids. Organic peroxides were classified as oxidising (O; R7) according to the DSD, which was not appropriate since the vast majority of them do not exhibit oxidising properties.

Inorganic oxidising solids are not flammable and therefore do not need to be subject to the classification procedures for the hazard classes flammable solids or pyrophoric solids. Also other solids that are classified as oxidising solids are normally not flammable, although a few exemptions may exist. Expert judgement should be sought in case of doubt.

2.14.4. Classification of substances and mixtures as oxidising solids

2.14.4.1. Identification of hazard information

Oxidising solids may cause, or contribute to, the combustion of other material. Although the definition states that they generally do this by yielding oxygen, halogens can behave in a similar way. Therefore, any substance or mixture containing oxygen and/or halogen atoms should in principle be considered for inclusion into the hazard categories oxidising solids. This does not necessarily mean that every substance or mixture containing oxygen and/or halogen atoms should be subjected to the full testing procedure.

2.14.4.1.1. Screening procedures and waiving of testing

Solids that are classified as explosives should not be subjected to the testing procedures for oxidising solids.

Organic peroxides should be considered for classification within the hazard class organic peroxides, see Chapter 2.15 of this guidance.

Experience in the handling and use of substances or mixtures which shows them to be oxidising is an important additional factor in considering classification as oxidising solids. In the event of divergence between test results and known experience, judgement based on known experience should take precedence over test results.

Before submitting a substance or a mixture to the full test procedure, an evaluation of its chemical structure may be very useful as it may prevent unnecessary testing. The person applying this procedure should have sufficient experience in testing and in theoretical evaluation of hazardous substances and mixtures. The following text provides a guideline for the theoretical evaluation of potential oxidising properties on basis of its composition and chemical structure. In case of doubt, the full test must be performed.

For organic substances or mixtures the classification procedure for this hazard class need not to be applied if:

- a. the substance or mixture does not contain oxygen, fluorine or chlorine; or
- b. the substance or mixture contains oxygen, fluorine or chlorine and these elements are chemically bonded only to carbon or hydrogen.

For inorganic substances or mixtures, the classification procedure for this hazard class need not be applied if they do not contain oxygen or halogen.

On basis of this theoretical evaluation only a distinction can be made between 'potentially oxidising' (i.e. further testing required) and 'non-oxidising' (i.e. no further testing for this hazard class required). It is not possible to assign a hazard category on basis of a theoretical evaluation.

Any substance or mixture that complies with the above waiving criteria can be safely regarded to have no oxidising properties and, hence, needs not to be tested and needs not to be regarded as an oxidising solid. However, such a substance or mixture may still possess other hazardous properties that require classification into another hazard class.

In case a mixture of an oxidising substance and a non-hazardous inert substance is offered for classification, the following should be taken into account:

- An inert material by definition does not contribute to the oxidising capability of the oxidising substance. Hence, the mixture can never be classified into a more severe hazard category.
- If an oxidising substance is mixed with an inert material, the oxidising capability of the mixture does not linearly decrease with decreasing content of oxidising substance. The relationship is more or less logarithmic and depends on the characteristics of the oxidising substance. For instance, a mixture containing 50 % of a strong oxidiser and 50 % of an inert material may retain 90 % of the oxidising capability of the original oxidising component. Non-testing classification of mixtures based solely on test data for the original oxidising substance should therefore be done with extreme care and only, if sufficient experience in testing exists.

2.14.4.2. Classification criteria

The testing procedures for oxidising solids are based on the capability of an oxidising solid to enhance the combustion of a combustible material. Therefore, solids that are submitted to classification testing are mixed with a combustible material. In principle, dried fibrous cellulose is used as a combustible material. The mixture of the potentially oxidising solid and cellulose is then ignited and its behaviour is observed and compared to the behaviour of reference material mixtures.

- 1 For solids the mixture with cellulose is ignited at atmospheric conditions and the time necessary
2 for the combustion reaction to consume the mixture is recorded. The faster the combustion
3 rate, the stronger the oxidising capability of the solid tested.

Annex I: 2.14.2.1. *An oxidising solid shall be classified in one of the three categories for this class using test O.1 in Part III, sub section 34.4.1 or test O.3 in Part III, sub-section 34.4 3 of the UN RTDG, Manual of Tests and Criteria, in accordance with Table 2.14.1:*

Table 2.14.1

Criteria for oxidising solids

Category	Criteria using test O.1	Criteria using test O.3
1	<i>Any substance or mixture which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning time less than the mean burning time of a 3:2 mixture, (by mass), of potassium bromate and cellulose.</i>	<i>Any substance or mixture which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate greater than the mean burning rate of a 3:1 mixture (by mass) of calcium peroxide and cellulose.</i>
2	<i>Any substance or mixture which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning time equal to or less than the mean burning time of a 2:3 mixture (by mass) of potassium bromate and the criteria for Category 1 are not met.</i>	<i>Any substance or mixture which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate equal to or greater than the mean burning rate of a 1:1 mixture (by mass) of calcium peroxide and cellulose and the criteria for Category 1 are not met.</i>
3	<i>Any substance or mixture which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning time equal to or less than the mean burning time of a 3:7 mixture (by mass) of potassium bromate and cellulose and the criteria for Categories 1 and 2 are not met.</i>	<i>Any substance or mixture which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate equal to or greater than the mean burning rate of a 1:2 mixture (by mass) of calcium peroxide and cellulose and the criteria for Categories 1 and 2 are not met.</i>

Note 1

Some oxidising solids also present explosion hazards under certain conditions (when stored in large quantities). Some types of ammonium nitrate may give rise to an explosion hazard under extreme conditions and the 'Resistance to detonation test' (IMSBC Code (International Maritime Solid Bulk Cargoes Code, IMO), Appendix 2, Section 5) can be used to assess this hazard. Appropriate information shall be made in the SDS.

- 4 Note 1 may also apply to other oxidising ammonium salts. Experience indicates that the
5 conditions required for ammonium nitrate to present an explosion hazard involve a combination
6 of factors, such as storage in large volumes (multiple tonnes) and either contamination (e.g.
7 with metals, acids, organics) or excessive heat (e.g. under conditions of fire). The resistance to
8 detonation (RTD) test is extensively described in Regulation (EC) No 2003/2003 for ammonium
9 nitrate.
- 10 For additional information regarding the use of non-testing data see Section [2.14.4.3](#) below and
11 Urban, 2007 (see Section [2.14.7](#)).

2.14.4.3. Testing and evaluation of hazard information

The test methods⁵ for oxidising solids are designed to give a final decision regarding their classification. To be reminded that experience in the handling and use of substances or mixtures, besides testing, is an important additional factor in considering classification in this hazard class.

2.14.4.4. Decision logic

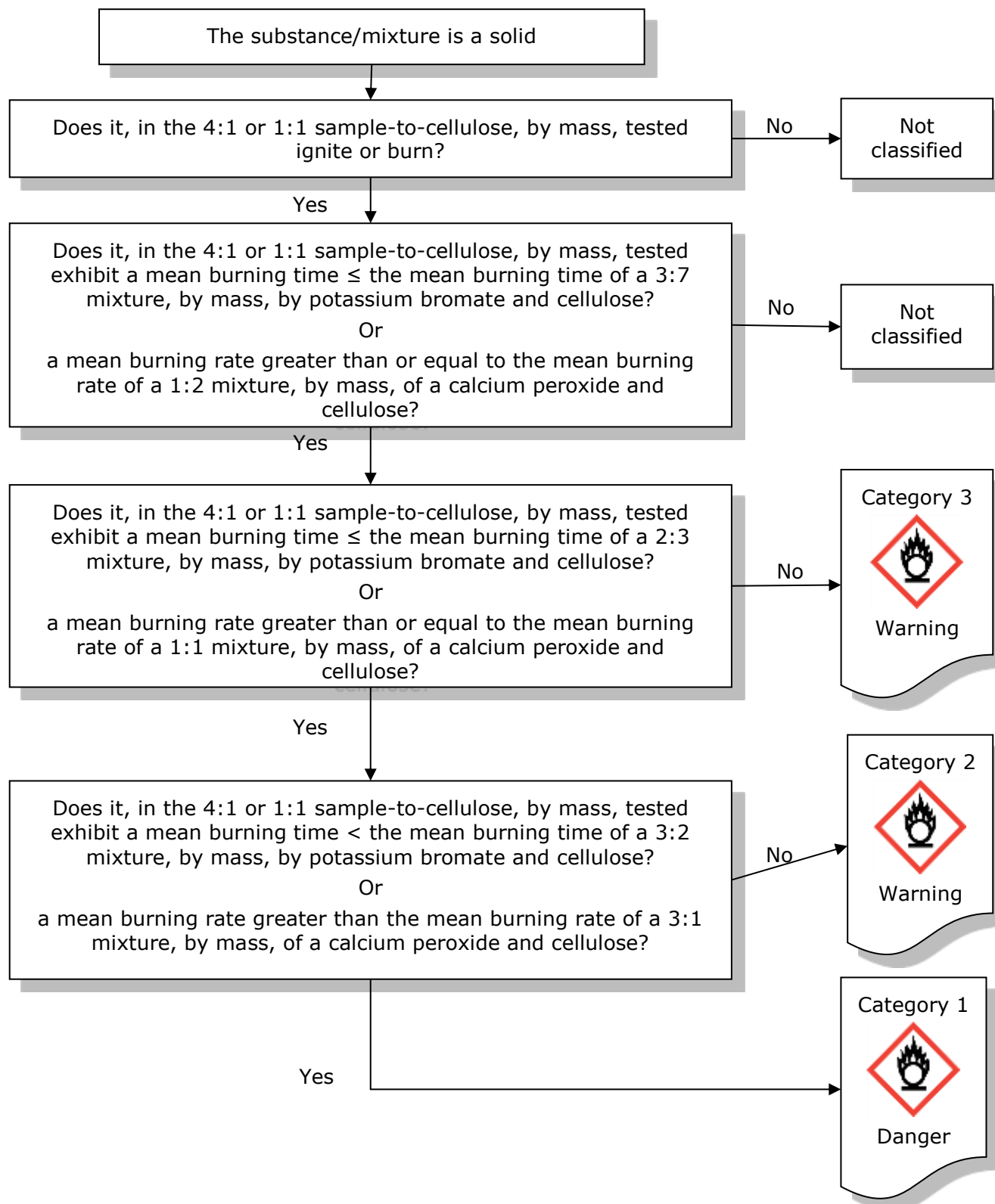
Classification of oxidising solids is done according to decision logic 2.14 as included in the GHS.



NOTE: The person responsible for the classification of oxidising solids should be experienced in this field and be familiar with the criteria for classification.

⁵ As from December 2012 an alternative test method for oxidising solids, Test O.3, has been included in the UN MTC (see document ST/SG/AC.10/40/Add.2). Test O.3 is an improved version of Test O.1 using a different reference substance and gravimetric measurements of the burning rate. Reference to Test O.3 has been included in the 5th revised edition of the GHS.

1 **Figure 2.14.4—1 Decision logic for oxidising solids (Decision logic 2.14 of GHS)**



2.14.4.5. Hazard communication for oxidising solids

2.14.4.5.1. Pictograms, signal words, hazard statements and precautionary statements

The pictograms and hazard statements are designed to indicate that oxidising substances and mixtures may cause or contribute to fire or explosion and therefore in principle should be separated from combustible materials.

Annex I: Table 2.14.2			
Label elements for oxidising solids			
	<i>Category 1</i>	<i>Category 2</i>	<i>Category 3</i>
<i>GHS Pictograms</i>			
<i>Signal Word</i>	<i>Danger</i>	<i>Danger</i>	<i>Warning</i>
<i>Hazard Statement</i>	<i>H271: May cause fire or explosion; strong oxidiser</i>	<i>H272: May intensify fire; oxidiser</i>	<i>H272: May intensify fire; oxidiser</i>
<i>Precautionary Statement Prevention</i>	<i>P210 P220 P280 P283</i>	<i>P210 P220 P280</i>	<i>P210 P220 P280</i>
<i>Precautionary Statement Response</i>	<i>P306 + P360 P371 + P380 + P375 P370 + P378</i>	<i>P370 + P378</i>	<i>P370 + P378</i>
<i>Precautionary Statement Storage</i>	<i>P420</i>		
<i>Precautionary Statement Disposal</i>	<i>P501</i>	<i>P501</i>	<i>P501</i>

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.14.5. Relation to transport classification

Division 5.1 within Class 5 of the UN RTDG Model Regulations covers oxidising liquids and oxidising solids, using the same tests and criteria as the CLP. Therefore, a solid substance or mixture classified as Division 5.1 (sometimes referred to as Class 5.1) according to any of the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) is normally also classified as an oxidising solid according to the CLP. Packing Groups I, II and III of the transport regulations correspond directly to Categories 1, 2 and 3 of the CLP, respectively. See Annex VII for additional information on transport classification in relation to CLP classification.

2.14.6. Examples of classification for oxidising solids

2.14.6.1. Examples of substances and mixtures fulfilling the classification criteria

The list of substances and mixtures fulfilling the criteria for classification is only presented for information purposes. This list is not exhaustive. For examples of results see section 34.4.1.5 of UN-MTC.

- Calcium nitrate, anhydrous
- Chromium trioxide
- Potassium nitrite
- Potassium perchlorate
- Potassium permanganate
- Sodium chlorate
- Sodium nitrite
- Sodium nitrate
- Strontium nitrate, anhydrous

2.14.6.2. Examples of substances and mixtures not fulfilling the classification criteria

- Calcium nitrate, tetrahydrate
- Cobalt nitrate, hexahydrate

2.14.7. Reference

Urban, Peter G. (2007). *Bretherick's Handbook of Reactive Chemical Hazards, Volumes 1-2* (7th Edition). Elsevier.

2.15. ORGANIC PEROXIDES

2.15.1. Introduction

The criteria for 'Organic peroxides' are found in Annex I, Section 2.15 of CLP and are identical to those in Chapter 2.15 of GHS.

The hazard class organic peroxides is unique in the respect that it is the only category to which chemicals are assigned on the basis of their chemical structure. Organic peroxides cannot be seen as an 'intrinsic property'; it is a family of chemical substances and mixtures which may have various properties. However, the type of peroxide is determined by testing.

2.15.2. Definitions and general considerations for the classification of organic peroxides

In CLP, the following definition is given for organic peroxides.

Annex I: 2.15.1. Definition

Organic peroxides means liquid or solid organic substances which contain the bivalent -O-O- structure and may be considered derivatives of hydrogen peroxide, where one or both of the hydrogen atoms have been replaced by organic radicals. The term organic peroxide includes organic peroxide mixtures (formulations) containing at least one organic peroxide. Organic peroxides are thermally unstable substances or mixtures, which can undergo exothermic self-accelerating decomposition. In addition, they can have one or more of the following properties:

(i) be liable to explosive decomposition;

(ii) burn rapidly;

(iii) be sensitive to impact or friction;

(iv) react dangerously with other substances.

2.15.1.2. *An organic peroxide is regarded as possessing explosive properties when in laboratory testing the mixture (formulation) is liable to detonate, to deflagrate rapidly or to show a violent effect when heated under confinement.*

2.15.3. Relation to other physical hazards

In addition to the definition (CLP Annex I, 2.15.1), organic peroxides may:

a. be flammable;

b. emit flammable gas when heated.

In general, organic peroxides do not have or have only weak oxidising properties.

The additional (subsidiary) labelling, as indicated in the list of classified organic peroxides included in the UN RTDG Model Regulations, Section 2.5.3.2.4, represents the additional hazardous properties.

Neither the burning properties nor the sensitivity to impact and friction form part of the classification procedure for organic peroxides in CLP. However, these properties may be of importance for the safe handling of organic peroxides (see Section 2.15.4.3.2, additional testing).

In addition, the following should be noted:

Explosive properties

The explosive properties do not have to be determined according to the CLP Annex I, Chapter 2.1, because explosive properties are incorporated in the decision logic for organic peroxides. Note that organic peroxides may have explosive properties when handled under higher confinement.

Flammable properties

The hazard statement for flammable properties for liquid organic peroxides should be based on the appropriate category for flammable liquids, as long as the flash point is relevant, (see Section 2.15.4.3.2). The translation table in Annex VII to CLP can be used for this.

2.15.4. Classification of substances and mixtures as organic peroxides

2.15.4.1. Identification of hazard information

The classification of an organic peroxide in one of the seven categories 'Types A to G' is dependent on its detonation, deflagration and thermal explosion properties, its response to heating under confinement, its explosive power and the concentration and the type of diluent added to desensitize the organic peroxide. Specifications of acceptable diluents that can be used safely are given in the UN RTDG Model Regulations, 2.5.3.5. The classification of an organic peroxide as Type A, B or C is dependent on the type of packaging in which the organic peroxide is tested as it affects the degree of confinement to which the organic peroxide is subjected. This has to be considered when handling the organic peroxide; stronger packaging may result in more violent reactions when the organic peroxide decomposes. This is why it is important that storage and transport is done in packaging, allowed for the type of organic peroxide, that conforms the requirements of the UN-packaging or IBC instruction (P520/IBC520) or tank instruction (T23).

The traditional aspects of explosive properties, such as detonation, deflagration and thermal explosion, are incorporated in the decision logic of CLP Figure 2.15.1. Consequently, explosive property determination as prescribed for the hazard class 'explosives' needs not to be conducted for organic peroxides.

A list of currently classified organic peroxides is included in the UN RTDG Model regulations, Section 2.5.3.2.4.

2.15.4.2. Classification criteria

In CLP, organic peroxides are not classified as oxidisers but they are a distinct hazard class.

Annex I: 2.15.2.1. Any organic peroxide shall be considered for classification in this class, unless it contains:

a) not more than 1,0 % available oxygen from the organic peroxides when containing not more than 1,0 % hydrogen peroxide; or

b) not more than 0,5% available oxygen from the organic peroxides when containing more than 1,0 % but not more than 7,0 % hydrogen peroxide.

[...]

In CLP decision logic Annex I, Figure 2.15.1, classification of organic peroxides is based on performance based testing both small scale tests and, where necessary, some larger scale test with the organic peroxide in its packaging. The concept of 'intrinsic properties' is, therefore, not applicable to this hazard class.

1 Organic peroxides are classified into one of the seven categories of 'Types A to G' according to
2 the classification criteria of CLP. The classification principles are given in decision logic Figure
3 2.15.1 of CLP and the Test Series A to H, as described in the Part II of the UN-MTC, should be
4 performed.

Annex I: 2.15.2.2. Organic peroxides shall be classified in one of the seven categories of 'Types A to G' for this class, according to the following principles:

(a) any organic peroxide which, as packaged, can detonate or deflagrate rapidly shall be defined as organic peroxide TYPE A;

(b) any organic peroxide possessing explosive properties and which, as packaged, neither detonates nor deflagrates rapidly, but is liable to undergo a thermal explosion in that package shall be defined as organic peroxide TYPE B;

(c) any organic peroxide possessing explosive properties when the substance or mixture as packaged cannot detonate or deflagrate rapidly or undergo a thermal explosion shall be defined as organic peroxide TYPE C;

(d) any organic peroxide which in laboratory testing:

(i) detonates partially, does not deflagrate rapidly and shows no violent effect when heated under confinement; or

(ii) does not detonate at all, deflagrates slowly and shows no violent effect when heated under confinement; or

(iii) does not detonate or deflagrate at all and shows a medium effect when heated under confinement;

shall be defined as organic peroxide TYPE D;

(e) any organic peroxide which, in laboratory testing, neither detonates nor deflagrates at all and shows low or no effect when heated under confinement shall be defined as organic peroxide TYPE E;

(f) any organic peroxide which, in laboratory testing, neither detonates in the cavitated state nor deflagrates at all and shows only a low or no effect when heated under confinement as well as low or no explosive power shall be defined as organic peroxide TYPE F;

(g) any organic peroxide which, in laboratory testing, neither detonates in the cavitated state nor deflagrates at all and shows no effect when heated under confinement nor any explosive power, provided that it is thermally stable, i.e. the SADT is 60 °C or higher for a 50 kg package⁽¹⁾, and, for liquid mixtures, a diluent having a boiling point of not less than 150 °C is used for desensitisation, shall be defined as organic peroxide TYPE G. If the organic peroxide is not thermally stable or a diluent having a boiling point less than 150 °C is used for desensitisation, the organic peroxide shall be defined as organic peroxide TYPE F.

Where the test is conducted in the package form and the packaging is changed, a further test shall be conducted where it is considered that the change in packaging will affect the outcome of the test.

⁽¹⁾ See UN RTDG, Manual of Test and Criteria, sub-sections 28.1, 28.2, 28.3 and Table 28.3.

5 A list of currently classified organic peroxides is included in the UN RTDG Model Regulations,
6 Section 2.5.3.2.4.

2.15.4.3. Testing and evaluation of hazard information

2.15.4.3.1. Thermal stability tests and temperature control

In addition to the classification tests given in decision logic Figure 2.15.1 of CLP, the thermal stability of the organic peroxide has to be assessed in order to determine the SADT. For the determination of the SADT, the testing method in UN-MTC, Part II, Section 28, may be used.

The SADT is defined as the lowest temperature at which self-accelerating decomposition of an organic peroxide may occur in the packaging as used in transport, handling and storage. The SADT is a measure of the combined effect of the ambient temperature, decomposition kinetics, package size and the heat transfer properties of the organic peroxide and its packaging.

There is no relation between the SADT of an organic peroxide and its classification in one of the seven categories 'Types A to G'. The SADT is used to derive safe handling, storage and transport temperatures (control temperature) and alarm temperature (emergency temperature).

Depending on its SADT an organic peroxide needs temperature control and the rules as given in CLP Annex I, 2.15.2.3, consist of the following two elements:

1. Criteria for temperature control:

The following organic peroxides need to be subjected to temperature control:

- a. Organic peroxide types B and C with a SADT ≤ 50 °C;
- b. Organic peroxide type D showing a medium effect when heated under confinement with a SADT ≤ 50 °C or showing a low or no effect when heated under confinement with a SADT ≤ 45 °C; and
- c. Organic peroxide types E and F with a SADT ≤ 45 °C.

2. Derivation of control and emergency temperatures:

Type of receptacle	SADT *	Control temperature	Emergency temperature
Single packagings and IBC's	20 °C or less	20 °C below SADT	10 °C below SADT
	over 20 °C to 35 °C	15 °C below SADT	10 °C below SADT
	over 35 °C	10 °C below SADT	5 °C below SADT
Tanks	< 50 °C	10 °C below SADT	5 °C below SADT

* i.e. the SADT of the organic peroxide as packaged for transport, handling and storage

It should be emphasized that the SADT is dependent on the nature of the organic peroxide itself, together with the volume and heat-loss characteristics of the packaging or vessel in which the organic peroxide is handled. The temperature at which self-accelerating decomposition occurs falls:

- as the size of the packaging or vessel increases; and
- with increasing efficiency of the insulation on the package or vessel.

The SADT is only valid for the organic peroxide as tested and when handled properly. Mixing the organic peroxide with other chemicals, or contact with incompatible materials (including incompatible packaging or vessel material) may reduce the thermal stability due to catalytic decomposition, and lower the SADT. This may increase the risk of decomposition and has to be avoided.

2.15.4.3.2. Additional considerations and testing

Explosive properties

The sensitivity of organic peroxides to impact (solids and liquids) and friction (solids only) may be of importance for the safe handling of the organic peroxide if they have pronounced explosive properties (e.g. they are liable to detonate, to deflagrate rapidly or show a violent effect when heated under confinement). Test methods to determine these properties are described in Test Series 3 of the UN-MTC (see Test 3 (a) (ii) and 3 (b) (i)). This information on the mechanical sensitivity should be included in the SDS.

Burning properties

In some national storage guidelines the burning rate is commonly used for classification for the purposes of storage and consequential storage requirements. Test methods are incorporated in these national storage regulations.

Flash point

The flash point for liquid organic peroxides is only relevant in the temperature range where the organic peroxide is thermally stable. Above the SADT of the organic peroxide determination of the flash point is not relevant because decomposition products are evolved.



NOTE: In case a flash point determination seems reasonable (expected flash point below the SADT) a test method using small amount of sample is recommended. In case the organic peroxide is diluted or dissolved, the diluent may determine the flash point.

Auto-ignition temperature

The determination of the auto ignition temperature is not relevant for organic peroxides. Available test methods are for non-decomposing vapour phases but the vapours of organic peroxides decompose during execution of the test and auto ignition of these organic peroxide vapours can never be excluded. This information should be included in the SDS.

Self-ignition temperature

Also the determination of the self-ignition temperature (applicable for solids) is not relevant. The thermal stability of organic peroxides is quantitatively given by the SADT.

Control and Emergency temperatures

The Control and Emergency temperatures are based on the SADT as in most cases determined by UN Test H.4. The Dewar vessel used in the UN Test H.4 is supposed to be representative for the organic peroxide handled in packages. For handling the organic peroxide in larger quantities (IBCs/tanks/vessels etc.) and/or in (thermally) insulated containers, the SADT has to be determined for that quantity with that degree of insulation. From that SADT the Control and Emergency temperatures can be derived (see also Section [2.15.4.3.1](#)).

2.15.4.3.3. Additional classification considerations

Currently the following properties are not incorporated in the classification of organic peroxides under the CLP:

- mechanical sensitivity i.e. impact and friction sensitivity (for handling purposes);
- burning properties (for storage purposes);
- flash point for liquids; and
- burning rate for solids.

Furthermore:

Annex I: 2.15.4.2. *Mixtures of already classified organic peroxides may be classified as the same type of organic peroxide as that of the most dangerous component. However, as two stable components can form a thermally less stable mixture, the SADT of the mixture shall be determined.*

Note: The sum of the individual parts can be more hazardous than the individual components.

1 Formulated commercial organic peroxides are classified according to their SADT.

2 **2.15.4.4. Decision logic**

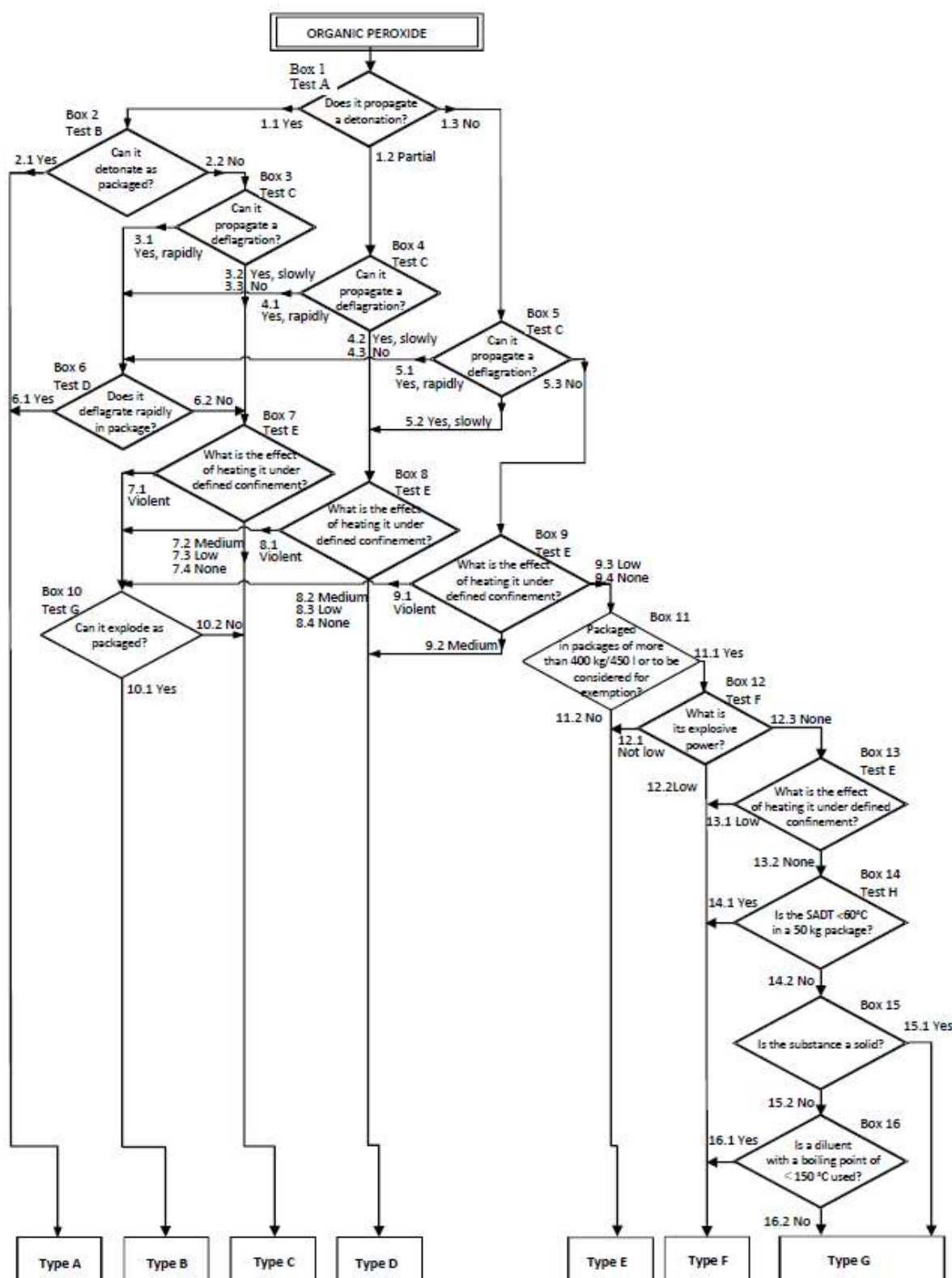
3 The decision logic for organic peroxides is applicable according to CLP.



NOTE: The person responsible for the classification of organic peroxides should be experienced in this field and be familiar with the criteria for classification.

4

1 **Figure 2.15.4—1 Decision logic 2.15 for organic peroxides**



2.15.5. Hazard communication for organic peroxides

2.15.5.1. Pictograms, signal words, hazard statements and precautionary statements

According to CLP the following label elements must be used for organic peroxide meeting the criteria for this hazard class:

Annex I: Table 2.15.1					
Label elements for organic peroxides					
Classification	Type A	Type B	Type C & D	Type E & F	Type G
GHS pictograms		 			There are no label elements allocated to this hazard category
Signal Word	Danger	Danger	Danger	Warning	
Hazard Statement	H240: Heating may cause an explosion	H241: Heating may cause a fire or explosion	H242: Heating may cause a fire	H242: Heating may cause a fire	
Precautionary statement Prevention	P210 P234 P235 P240 P280	P210 P234 P235 P240 P280	P210 P234 P235 P240 P280	P210 P234 P235 P240 P280	
Precautionary statement Response	P370 + P372 + P380 + P373	P370 + P380 + P375[+ P378] ¹	P370 + P378	P370 + P378	
Precautionary statement Storage	P403 P410 P411 P420	P403 P410 P411 P420	P403 P410 P411 P420	P403 P410 P411 P420	
Precautionary statement Disposal	P501	P501	P501	P501	

¹ See introduction to Annex I for details on the use of square brackets.

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

2.15.5.2. Additional labelling provisions for organic peroxides

Additional hazardous properties, resulting in additional (subsidiary) labelling, are indicated in the list of classified organic peroxides included in the UN RTDG Model Regulations, section 2.5.3.2.4.

2.15.6. Relation to transport classification

Division 5.2 within Class 5 of the UN RTDG Model Regulations covers organic peroxides. A list of currently classified organic peroxides is included in the UN RTDG Model Regulations, Section 2.5.3.2.4. This table includes organic peroxides Type B - Type F (and some formulations Type G, so-called exempted organic peroxides).

An exceptional case in this respect is a peroxyacetic acid formulation, as currently classified in the UN RTDG Model Regulations under UN 3149, with the following description: HYDROGEN PEROXIDE AND PEROXYACETIC ACID MIXTURE with acid(s), water and not more than 5 % peroxyacetic acid, STABILISED. In the classification procedure for organic peroxides, see decision logic in Section 2.15.4.4, this formulation will be assigned to organic peroxide Type G, and consequently no label elements are allocated. In view of the above, this formulation can be classified, also in accordance with CLP, as an Oxidising liquid, Category 2. See Annex VII for additional information on transport classification in relation to CLP classification.

2.15.7. Examples of classification for organic peroxides

2.15.7.1. Examples of substances and mixtures fulfilling the classification criteria

Substance to be classified: Example Peroxide

Molecular formula: n.a.

According to CLP Annex I, Section 2.15.2.1, the substance has an active oxygen content of 7.40 % and thus has to be considered for classification in the hazard class organic peroxides.

Test results and classification according to CLP decision logic 2.15.1 for organic peroxides and the UN-MTC, Part II, is as follows:

CLASSIFICATION TEST RESULTS	
1. Name of the organic peroxide:	Example Peroxide
2. General data	
2.1. Composition:	Example Peroxide, technically pure (97 %)
2.2. Molecular formula:	n.a.
2.3. Active oxygen content:	7.18 %
2.4. Physical form:	liquid
2.5. Colour:	colourless
2.6. Density (apparent):	900 kg/m ³

CLASSIFICATION TEST RESULTS	
3. Detonation (test series A)	
Box 1 of the decision logic:	Does the peroxide propagate a detonation?
3.1. Method:	UN Test A.1: BAM 50/60 steel tube test
3.2. Sample conditions:	peroxide assay 97 %
3.3. Observations:	fragmented part of the tube: 18 cm
3.4. Result:	No
3.6. Exit:	1.3
4. Deflagration (test series C)	
Box 5 of the decision logic:	Can the peroxide propagate a deflagration?
4.1. Method 1:	Time/pressure test (test C.1)
4.1.1. Sample conditions:	ambient temperature
4.1.2. Observations:	4000 ms
4.1.3. Result:	Yes, slowly
4.2. Method 2:	Deflagration test (test C.2)
4.2.1. Sample conditions:	temperature: 25 °C
4.2.2. Observations:	deflagration rate: 0.74 mm/s
4.2.3. Result:	Yes, slowly
4.3. Final result:	Yes, slowly
4.4. Exit:	5.2
5. Heating under confinement (test series E)	
Box 8 of the decision logic:	What is the effect of heating it under confinement?
5.1. Method 1:	Koenen test (test E.1)
5.1.1. Sample conditions:	■
5.1.2. Observations:	limiting diameter: 2.0 mm fragmentation type 'F'
5.1.3. Result:	Violent
5.2. Method 2:	Dutch pressure vessel test (test E.2)
5.2.1. Sample conditions:	■
5.2.2. Observations:	limiting diameter: 6.0 mm (with 10 g)

CLASSIFICATION TEST RESULTS	
5.2.3. Result:	Medium
5.3. Final result:	Violent
5.4. Exit:	8.1
6. Explosion test in package (test series G)	
Box 10 of the decision logic:	Can it explode as packaged?
6.1. Method:	Thermal explosion test in package (test G.1)
6.2. Sample conditions:	30 litre packaging,
6.3. Observations:	no fragmentation (N.F.)
6.4. Result:	No
6.5. Exit:	10.2
7. Thermal stability (outside of the decision logic)	
7.1. Method:	Heat accumulation storage test (test H.4)
7.2. Sample conditions:	mass 380 g. Half life time of cooling of Dewar vessel with 400 ml DMP:
	10.0 hrs. (representing substance in package)
7.3. Observations: self	accelerating decomposition at 35 °C
	no self accelerating decomposition at 30 °C
7.4. Result:	SADT 35 °C
8. General remarks:	The decision logic is given in Figure x ⁶
9. Final classification	
Hazard class:	Organic peroxide, Type C, liquid, temperature controlled
Label:	Flame (GHS02)
Signal word:	Danger
Hazard statement:	H242: Heating may cause a fire
Temperature control:	Needed based on SADT (35 °C, in package)
Control temperature*:	20 °C (in package)
Emergency temperature*:	25 °C (in package)

1 *see UN-MTC, table 28.2.

2

⁶ Not attached to this example.

2.15.7.2. Additional remarks

Explosive properties

As shown in Section [2.15.7.1](#) a substance and a mixture may have explosive properties when handled under greater confinement and where the packaging in which it was tested in UN Test G.1 (see point 6 of classification test results above) is changed. Such information should be given in the SDS.

The example in Section [2.15.7.1](#) shows a violent effect when heated under confinement (see point 5.3 of the above results). Consequently, also the impact sensitivity according to UN Test series 3, test 3 (a) (ii), BAM Fallhammer should be determined. For this example it amounts to 20 J. Such information should be given in the SDS.

Burning properties

For the example in Section [2.15.7.1](#) the burning properties as determined by the test method described in the storage guidelines, currently in place in France, Germany, Netherlands and Sweden, is 7.0 kg/min/m². Based on this figure and the classification as organic peroxide type C, the storage classification can be assigned in those countries.

Flash point

The example substance thermally decomposes before the temperature at which the vapour can be ignited is reached (see Section [2.15.4.3.2](#)) and consequently a flash point cannot be determined.

2.16. CORROSIVE TO METALS

2.16.1. Introduction

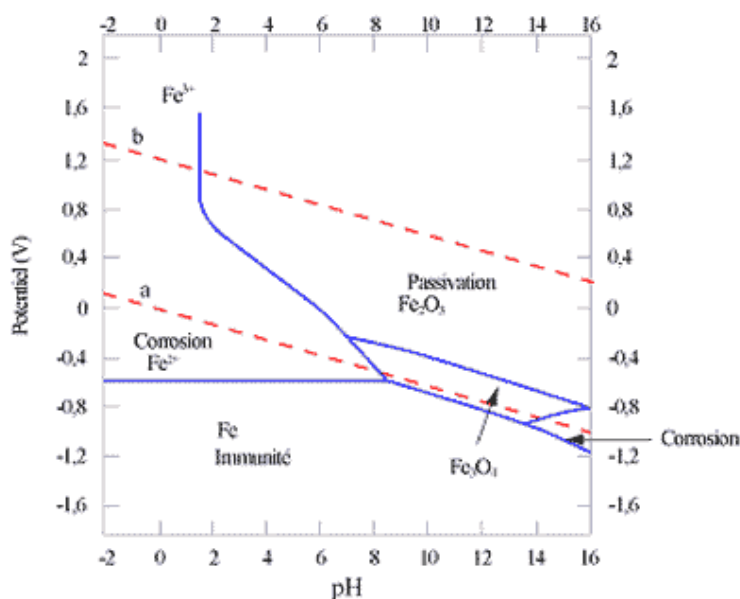
The criteria for 'Corrosive to metals' are found in Annex I, Section 2.16 of CLP and are identical to those in Chapter 2.16 of GHS.

The hazard class corrosive to metals is a physico-chemical property that is new in the EU classification scheme and appears for the first time in CLP. So far, only the health hazard corrosivity to skin was considered in the classification scheme. To some extent, both properties relate to each other and, in the context of transport of dangerous goods, have been considered for classification in class 8, despite the different nature of the hazard (material damage versus living tissue damage).

A substance or a mixture that is corrosive to metal under normal conditions is a substance or a mixture liable to undergo an irreversible electrochemical reaction with metals that leads to significant damage or, in some cases, even to full destruction of the metallic components. The corrosive to metal property is a quite complex property, since it is a substance (or mixture) related as well as a material (metal) related property. This means a corrosive substance or mixture leads to corroded material (metal), according to a number of external conditions. From the material side, many types of corrosion processes may occur, according to configurations, liquid or fluid media inducing the corrosion process, nature of metal, potential passivation occurring by oxide formation during corrosion.

From the substance or mixture side, many parameters may influence the corrosion properties of a substance or mixture, such as the nature of the chemical or the pH. From an electrochemistry point of view, corrosion conditions are often studied using Pourbaix diagrams, which plot the electrochemical potential (in Volt) that develops according to electrical charges transfer versus the pH-value. Such a diagram is shown for the case of iron and applies only for carbon steel corrosion (Jones, 1996).

Figure 2.16.1—1 Potential pH (also called Pourbaix) diagram for iron in water at 25 °C, indicating stable form of the Fe element and implicitly, corrosion domains



For the purposes of CLP, corrosion to metal will only be considered, by pure convention, for substances and mixtures that are liable to attack carbon steel or aluminium, two of the most common metals that may come in contact with chemical substances (containment material, reactor material). The classification scheme applied here must not be considered as a material

(metal) classification method for metals regarding resistance to corrosion. By no means steel or aluminium specimens that are treated to resist to corrosion, must be selected for testing.

2.16.2. Definitions and general considerations for the classification of substances and mixtures corrosive to metals

CLP comprises the following definition for substances and mixtures that are corrosive to metal.

Annex I: 2.16.1. Definition

A substance or a mixture that is corrosive to metals means a substance or a mixture which by chemical action will materially damage, or even destroy, metals.

2.16.3. Relation to other physical hazards

There is no direct relation to other physical hazards.

2.16.4. Classification of substances and mixtures as corrosive to metals

2.16.4.1. Identification of hazard information

Importance of the physical state of the test substance or mixture

There is no reference in the definition (CLP Annex I, 2.16.1) to the physical state of the substances or mixtures that needs consideration for potential classification in this hazard class. According to the test method to be employed for considering classification under this hazard class, we may state at least that gases are out of the scope of the corrosive to metal hazard class. Neither the corrosivity of gases nor the formation of corrosive gases is currently covered by CLP classes and are therefore **not** applicable here.

According to the classification criteria only substances and mixtures for which the application of the UN Test C.1 (described in part III, Section 37.4.1.1 of the UN-MTC) is relevant and needs to be considered. Application of classification criteria in the UN-MTC, Section 37.4 excludes solids, while 'liquids and solids that may become liquids (during transport)', have to be considered for such a classification.

The wording 'solids that may become liquids' was developed for UN RTDG Model Regulations classification purposes, and needs further explanation. Solids may become liquids by melting (due to increase in temperature). Solids having a melting point lower than 55 °C (which is the test temperature required in UN Test C.1) must then be taken into consideration. The other physical way to transform a solid into liquid is by dissolution in water or another solvent. Classification of solid substances that may become liquids by dissolution is subject to further expert judgement, and may need adaptation of the classification criteria or test protocol (see Section [2.16.4.4.2](#)). Interaction with liquids may come from air moisture or unintentional contact with water. Other solvent traces may result from the extraction process during manufacturing and these may induce corrosion in practice.

Substances and mixtures in a liquid state must be tested without any modification before testing. For other cases (solids that may become liquids), appropriate testing procedures require further work by the Committees of experts in charge of developing and updating the GHS at UN level. It needs to be further specified how such substances or mixtures must be prepared (transformed into liquids) to be able to determine their corrosivity to metals. As an example, it is thought that the quantity of solvent (water or any other solvent) to liquefy the test substance before testing would greatly influence results of the UN Test C.1 test and may not necessarily represent the real life situation of a product during transport, handling or use.

Non-testing data

Following parameters are helpful to evaluate corrosive properties before testing:

- melting points for solids;
- chemical nature of the substances and mixtures under evaluation (e.g. strong acids);
- pH values (liquids).

See also IR & CSA, Chapter R.7a: Endpoint specific guidance, Section R.7.1.2 (Melting point/freezing point).

Literature may also provide information on widely used substances and liquids 'compatibility tables', taking account of the corrosiveness of the products that may serve to decide whether testing must be conducted before assigning the corrosive to metals hazard class, on basis of expert judgement.

The following substances and mixtures should be considered for classification in this class:

- substances and mixtures having acidic or basic functional groups;
- substances or mixtures containing halogen;
- substances able to form complexes with metals and mixtures containing such substances.

2.16.4.2. Screening procedures and waiving of testing

Experience may have proven the corrosivity of given substances and mixtures. In such case no more testing is needed (see examples in Section 2.16.7).

Generally extreme pH-values point to a higher likelihood that the substance or mixture is corrosive. However, it cannot lead to immediate classification in the hazard class corrosive to metals. As a proof of that, Figure 2.16.1–1 shows that immunity zones (where steel does not corrode) still exist on the full spectrum of pH values as far as carbon steel is concerned.

Corrosivity is so complex that the evaluation of a mixture cannot be extrapolated from similar behaviour of constituents of a mixture. However, if one significant component of a mixture is corrosive to metals the mixture is likely to be corrosive to metals as well. Testing the actual mixture is therefore highly recommended. As already mentioned, solids are currently difficult to test according to the current CLP requirements, as the UN Test C.1 was designed for liquids.

Where an initial test on either steel or aluminium indicates the substance or mixture being tested is corrosive, the follow up test on the other metal is not required.

2.16.4.3. Classification criteria

Substances and mixtures of hazard class corrosive to metals are classified in a single hazard category on the basis of the outcome of the UN Test C.1 (UN-MTC, Part III, Section 37, paragraph 37.4).

Annex I: Table 2.16.1

Criteria for substances and mixtures corrosive to metals

Category	Criteria
1	Corrosion rate on either steel or aluminium surfaces exceeding 6,25 mm per year at a test temperature of 55 °C when tested on both materials.

2.16.4.4. Testing and evaluation of hazard information

2.16.4.4.1. General considerations

It is important to point out that the criteria of corrosion rate will never be applied in an absolute way, but by extrapolating the measured rate of corrosion over the test period to the annual assumed correlating corrosion rate. This exercise has to take account of the fact that the corrosion rate is not necessarily constant over time. Expert judgement may be required to consolidate the optimum test duration and to ascertain test results. However, the possibility of increasing the testing period from minimum one week to four weeks as well as the use of two different metals in the UN Test C.1 act as barriers against erroneous classification.

Whatever the result of the classification may be, the classification as corrosive to metals relates to steel and/or aluminium only and does not provide information with regard to the corrosivity potential to other metals than those tested.

Two types of corrosion phenomena need to be distinguished for classification of substances and mixtures in this hazard class, although not reported in CLP: the uniform corrosion attack and the localised corrosion (e.g. pitting corrosion, shallow pit corrosion).

[Table 2.16.4—1](#) (Section 37.4.1.4.1 of the UN- MTC) translates the corresponding minimum mass loss rates leading to classify the test substance or mixture as corrosive to metals for standard metal specimens (2 mm of thickness), according to time of exposure, for reasons of uniform corrosion process. In case of use of metal plates of a thickness that differs from the specified 2 mm (see comments in Section [2.4.2](#)), the values in tables [Table 2.16.4—1](#) and [Table 2.16.4—2](#) need adjustments due to the fact that the corrosion process depends on the surface of specimen.

Table 2.16.4—1 Minimum mass loss of specimens after different exposure times (corresponding to the criterion of 6.25 mm/year)

Exposure time	Mass loss
7 days	13.5 %
14 days	26.5 %
21 days	39.2 %
28 days	51.5 %

Table 2.16.4—2 (Section 37.4.1.4.2 of the UN-MTC) indicates the criteria leading to classification of the test substance or mixture as corrosive to metals for standard metal specimens, according to time of exposure, for reasons of localised corrosion process.

Table 2.16.4—2 Minimum intrusion depths after exposure times (corresponding to the criterion of localized corrosion of 6.25 mm/year)

Exposure time	Min. intrusion depth
7 days	120 µm
14 days	240 µm
21 days	360 µm
28 days	480 µm

It is not mentioned explicitly in the text that localised corrosion as well as uniform corrosion has also be taken into account. However, localised corrosion, that is entirely part of UN Test C.1 protocol, has actually to be taken into account. In addition, although the type of corrosion is not reflected in the classification result, this valuable information should be given in the SDS

2.16.4.4.2. Additional notes on best practice for testing

Competence required for testing

The overall evaluation of appropriate data for considering the corrosion properties of a substance or a mixture and in particular for testing it according to the mentioned criteria for this hazard class requires certain qualifications and experience. Expertise is often needed for this hazard class, which relates to a complex and multi-faceted hazardous phenomenon.

Selection of metal specimens

CLP refers to two types of metals (carbon steel and aluminium) meeting accurate specifications (technical characteristics of metal sheets and plate thickness). Thicker metal sheets, such as cast materials, of which the thickness is reduced by any form of mechanical treatment, may never be used. Mechanical reduction of sheet (metal) thickness could induce corrosion enhanced process due to cross section heterogeneity in metal grain and impurities. It is far better to use slightly different specifications of metal in the correct thickness or slightly different specimen plate thicknesses. It is recognised that it will not always be easy to obtain metal specimens with the profile as described above.

Regarding the type of aluminium or steel to be used for this test see UN-MTC, Sub-section 37.4.1.2.

Minimum corrosive media volume

In order to prevent any limitation on the corrosion process due to full consumption of the corrosive media before the end of the testing period, a minimum volume of substance or mixture (1.5 L, according to the UN-MTC) has to be used. (Note: volume/surface ratio of 10 mL/cm² is stated in DIN 50905, similar in ASTM G31-72.)

Adjustment of the test temperature

Corrosion processes are temperature dependent. In the context of CLP, the property corrosive to metals is assessed through testing metal specimens at a specified temperature of 55 °C ± 1 °C. In practice, it may be difficult with standard testing equipment to stay within the temperature window (55 °C ± 1 °C) of the gas phase, all over the test period. In such case, the test can be performed conservatively at a slightly higher temperature and somewhat lower accuracy (e.g. 57 °C ± 3 °C).

Selecting the appropriate test duration

The evaluation of the criterion of 6.25 mm/year is generally based on a test duration not exceeding 1 month. There is, however, the option to stop the test procedure already after 1 week (see table 1). For the decision on test duration, the non-linear behaviour of the corrosion process must be taken due account of. In borderline cases a non-appropriate test duration may result in either false positive or false negative results.

Specimen cleaning

Attention must be paid to the correct cleaning of the corroded residue before measurement of the corrosion characteristics. In case of adhesive corroded layer, the same cleaning process needs to be carried out on a non corroded sample to verify if the cleaning procedure is not significantly abrasive. For further information see UN-MTC, Sub-section 37.4.1.3.

Testing soluble solids

As said in Section 2.15.4.1, for solids that may become liquids through dissolution in water or in a solvent, the adequate testing procedure is more complex (not explicitly describe in the UN C.1 test protocol). In no case will simple dilution of the solid substance or mixture in any quantity of water lead to satisfactory testing of the substance or mixture for corrosion to metals.

For the specific case where the corrosion potential is linked to the presence of solvent traces (other than water), expert judgement is needed to determine if further testing must be performed (where the solid is put in interaction with the metallic part considered).

Example of equipment relevant for the performance UN Test C.1

Figure 2.16.4—1 Example of testing equipment available on the market to perform UN Test C.1



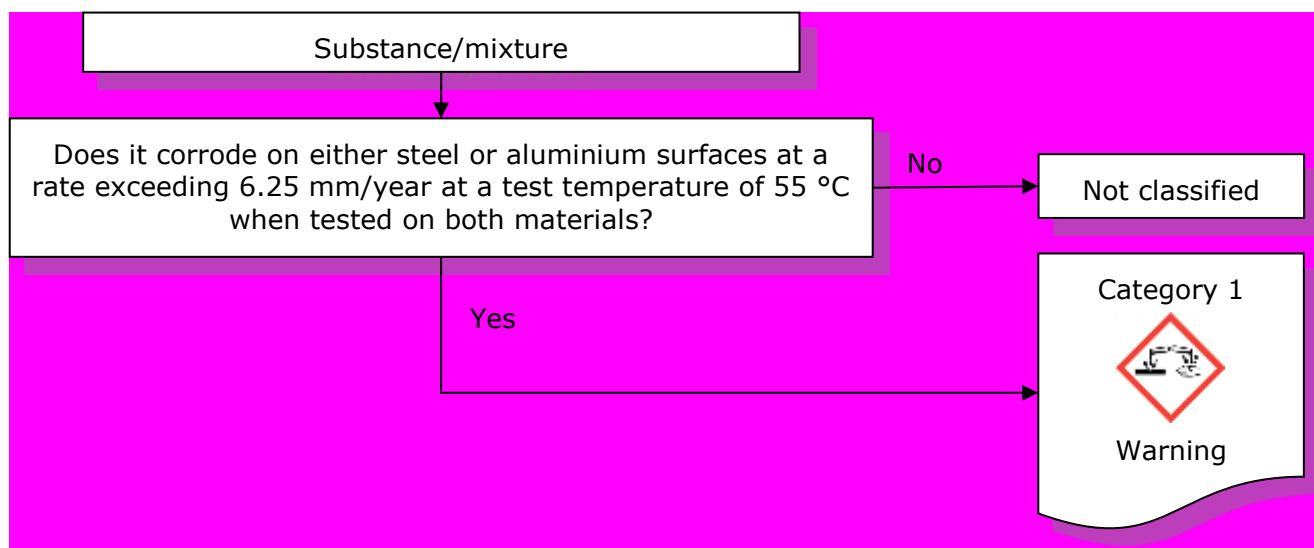
2.16.4.5. Decision logic

Classification of substances and mixtures corrosive to metals is done according to decision logics 2.16.4.1 as included in the GHS.



NOTE: The person responsible for the classification of substances and mixtures corrosive to metals should be experienced in this field and be familiar with the criteria for classification.

Figure 2.16.4–2 Decision logic for substances and mixtures corrosive to metals (Decision logic 2.16 of GHS)



2.16.5. Hazard communication for substances and mixtures corrosive to metals

2.16.5.1. Pictograms, signal words, hazard statements and precautionary statements

Table 2.16.2 of CLP Annex I provides the label elements for hazard class corrosive to metals. The hazard statement H290, using the wording 'may', reflects that classification under this hazard class does not cover all metals (testing only considers carbon steel and aluminium). Thus we may find examples of substances and mixtures that are classified in this hazard class corrosive to metals but will not induce corrosive action on other more corrosive resistant metals (e.g. platinum) than those serving as reference materials.

Label elements must be used for substances and mixtures meeting the criteria for classification in this hazard class in accordance with Table 2.16.2.

Annex I: 2.16.3. Table 2.16.2

Label elements for substances and mixtures corrosive to metals

Classification	Category 1
GHS Pictogram	
Signal Word	Warning
Hazard Statement	H290: May be corrosive to metals
Precautionary Statement, Prevention	P234
Precautionary Statement, Response	P390
Precautionary Statement, Storage	P406
Precautionary Statement, Disposal	
Note: Where a substance or mixture is classified as corrosive to metals but not corrosive to skin and/or eyes, the labelling provisions set out in section 1.3.6 shall be used.	

The wording of the Precautionary Statements is found in CLP Annex IV, Part 2.

Further, in section 1.3.6 of CLP Annex I a derogation from labelling requirements for substances or mixtures classified as corrosive to metals but not corrosive to skin and/or eyes is provided.

Annex I: 1.3.6 Substances or mixtures classified as corrosive to metals but not classified as skin corrosion or as serious eye damage (Category 1)

Substances or mixtures classified as corrosive to metals but not classified as skin corrosion or as serious eye damage (Category 1) which are in the finished state as packaged for consumer use do not require on the label the hazard pictogram GHS05.

2.16.6. Relation to transport classification

Class 8 of the UN RTDG Model Regulations covers substances and mixtures that are classified for corrosivity to skin, metals or both. Valuable information can be obtained from UN RTDG Model Regulations and the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI). Existing test results obtained in the context of the modal transport regulations (ADR, RID, ADN and IMDG Code, ICAO TI) may be applied since the UN Test C.1 serves as reference for testing in both classification systems. See Annex VII for additional information on transport classification in relation to CLP classification.

2.16.7. Examples of classification for substances and mixtures corrosive to metals

The following table lists some examples of substances and mixtures that should be classified or not in Class 2.16 (according to known UN Test C.1 results) in comparison with predicted results for skin corrosion hazard.

Table 2.16.7—1 Examples of classified and non classified substances and mixtures in Class 2.16

Note: 'Corroded' means corrosion attack in the sense of UN Test C.1;

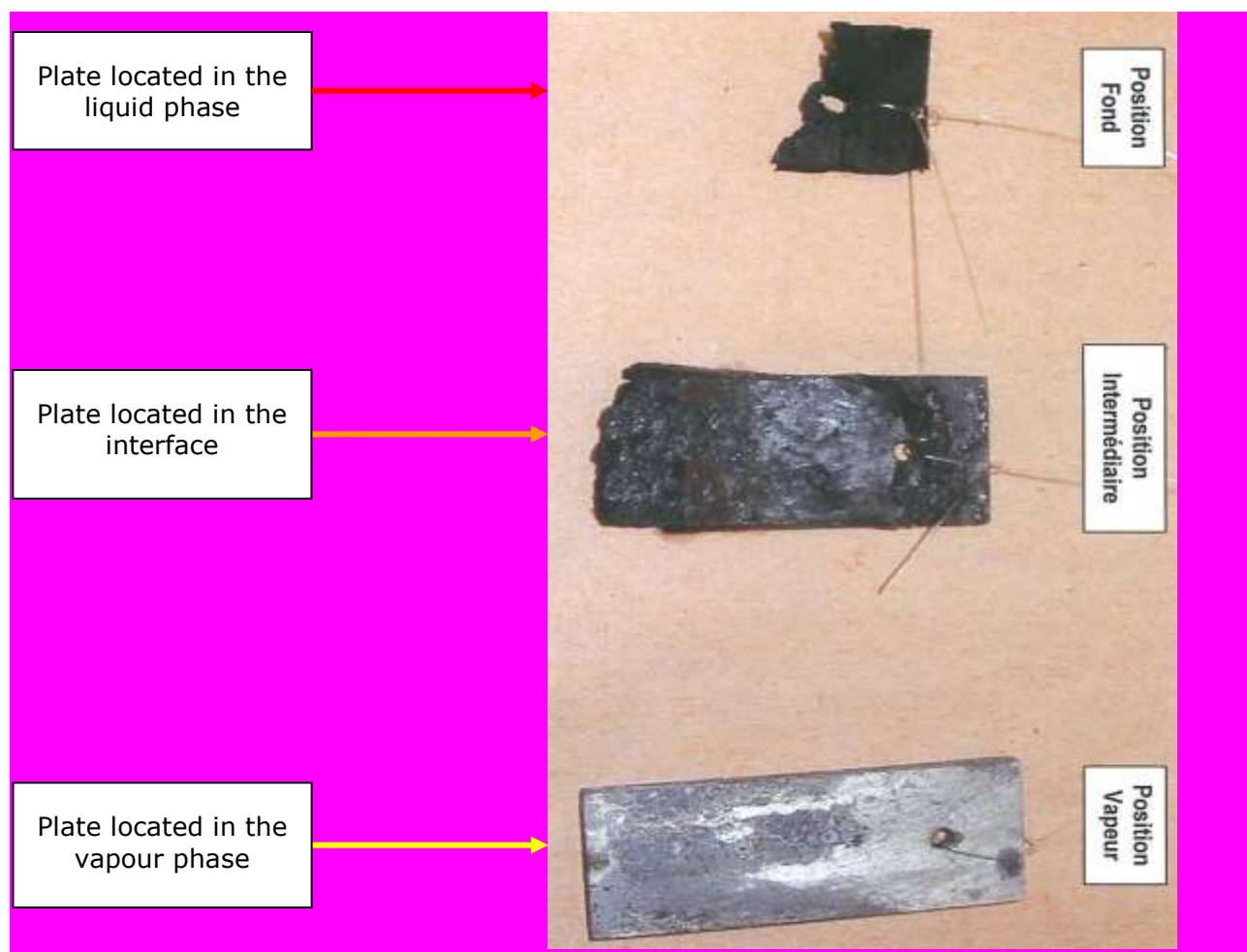
'Not corroded' means corrosion resistant in the sense of UN Test C.1;

'Positive' or 'Negative' are results from skin corrosion.

Substance or mixture	Steel	Aluminium	CLP Annex I, 2.16 classification	Skin (for comparison)
Hydrofluoric acid > 70 % (UN1790)	Not corroded	Corroded	Classified	Positive
Highly concentrated nitric acid (97 %) (UN2031)	Not corroded	Corroded	Classified	Positive
HNO ₃ red fuming (UN2032)	Not corroded	Not corroded	Not classified	Positive
Hydrochloric acid (diluted) (UN1789)	Corroded	Corroded	Classified	Negative
NaOH solutions (UN1824)	Not corroded	Corroded	Classified	Positive

2.16.7.1. Example of metal specimen plates after exposure to a corrosive mixture

Figure 2.16.7—1 Example of corroded metal plates after testing according to UN Test C.1 for a classified mixture



This example shows that the corrosion may develop at different rates according to the accurate position of the specimen related to the corroding mixture (sunk in the liquid, placed in the gas phase above liquid or at the liquid/gas interface).

2.16.8. References

ASTM G31-72(2004) *Standard Practice for Laboratory Immersion Corrosion Testing of Metals*.

Jones, D.A., *Principles and Prevention of Corrosion*, 2nd edition, 1996, Prentice Hall, Upper Saddle River, NJ. ISBN 0-13-359993-0 Page 50-52.

DIN 50905-1: 2007, *Corrosion of metals - Corrosion testing - Part 1: General guidance* (Korrosion der Metalle - Korrosionsuntersuchungen - Teil 1: Grundsätze).

3. PART 3: HEALTH HAZARDS

[See separate document for specific sections of Part 3 under consultation]

3.1. ACUTE TOXICITY

1 **3.2. SKIN CORROSION/IRRITATION**

2

1 **3.3. SERIOUS EYE DAMAGE/EYE IRRITATION**

2

3

1 **3.4. RESPIRATORY OR SKIN SENSITISATION**

2

1 **3.5. GERM CELL MUTAGENICITY**

2

1 **3.6. CARCINOGENICITY**

2

1 **3.7. REPRODUCTIVE TOXICITY**

2 **3.8. SPECIFIC TARGET ORGAN TOXICITY – SINGLE EXPOSURE (STOT-**
3 **SE)**

4 **3.9. SPECIFIC TARGET ORGAN TOXICITY – REPEATED EXPOSURE**
5 **(STOT-RE)**

6

4. PART 4: ENVIRONMENTAL HAZARDS

*[Please note, Part 4 is **not** under consultation]*

4.1. HAZARDOUS TO THE AQUATIC ENVIRONMENT

5. PART 5: ADDITIONAL HAZARDS

*[Please note, Part 5 is **not** under consultation]*

5.1. HAZARDOUS TO THE OZONE LAYER

ANNEXES

*[Please note, Annexes are **not** under consultation]*

I ANNEX I: AQUATIC TOXICITY

1 **II ANNEX II: RAPID DEGRADATION**

2

1 **III ANNEX III: BIOACCUMULATION**

2

IV ANNEX IV: METALS AND INORGANIC METAL COMPOUNDS

1 **V ANNEX V: COLLECTION OF INTERNET LINKS FOR THE**
2 **USERS OF THE GUIDANCE**
3
4
5

VI ANNEX VI: BACKGROUND DOCUMENT TO THE GUIDANCE FOR SETTING SPECIFIC CONCENTRATION LIMITS FOR SUBSTANCES CLASSIFIED FOR REPRODUCTIVE TOXICITY ACCORDING TO REGULATION (EC) NO 1272/2008

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